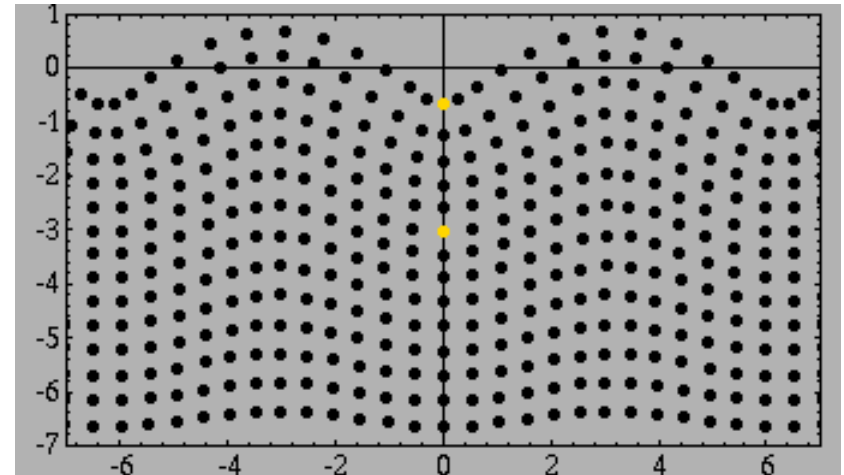


United International University

Waves and Oscillation

Md. Asaduzzaman Asad
Institute of Natural Sciences



Part I CSE/EEE (Waves & Oscillations)

Simple harmonic oscillators (SHM)

Damped oscillators (DHM)

Driven oscillators (FHM)

Resonance

Wave Equation

Part II CSE (Electricity & Magnetism)

Electric Force

Electric Field

Electric Potential

Capacitor and Capacitance

Current, Resistance & EMF

Magnetic Field

Inductor and Inductance

Part II EEE (Mechanics)

Vectors and Vectors Algebra

Work, Energy and Power

Conservation of Energy and Momentum

Elastic and Inelastic Collisions

Rotational Dynamics

Part III CSE (Quantum Physics)

Introduction to Quantum Mechanics
Particle Properties of Wave
Wave Properties of Particles
Schrodinger Equation & Application
Atomic Physics

Part III EEE (Physical Optics)

Theories of Light

Interference of Light

Diffraction of Light

Polarization of Light

What to do in this course:

1. Read the relevant sections in the textbook
... the course notes will guide you.
2. Do all homework and problem sets.
3. Get help early ... from Course Teacher.
4. there are no shortcuts ... put effort in to understand things ...

Problem-solving and homework

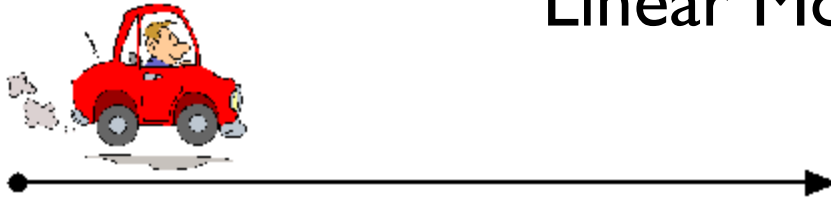
Each Chapter you will be given a take-home problem set to complete and hand in for marks ... In addition to this, you need to work through the problems in *Book*, in your own time, at home.

You will not be asked to hand these in for marks. Get help from your friends, the course Teacher ...

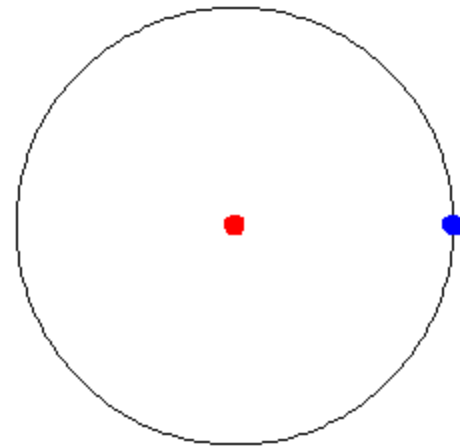
Do not take shortcuts. Mastering these problems is a fundamental aspect of this course.

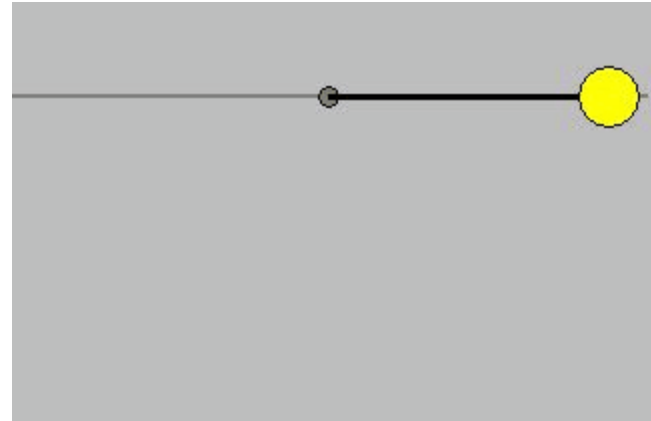
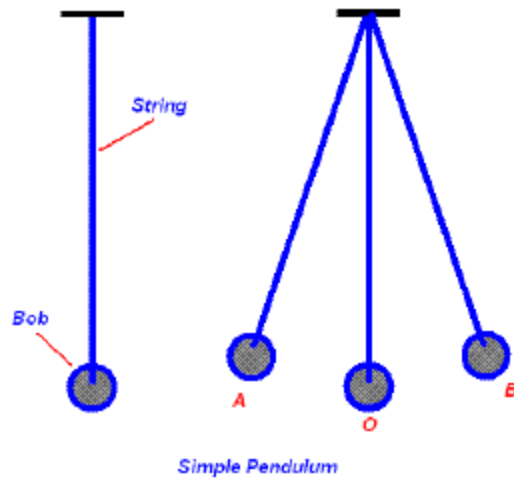
Different Kind of Motions

Linear Motion



Uniform Circular Motion

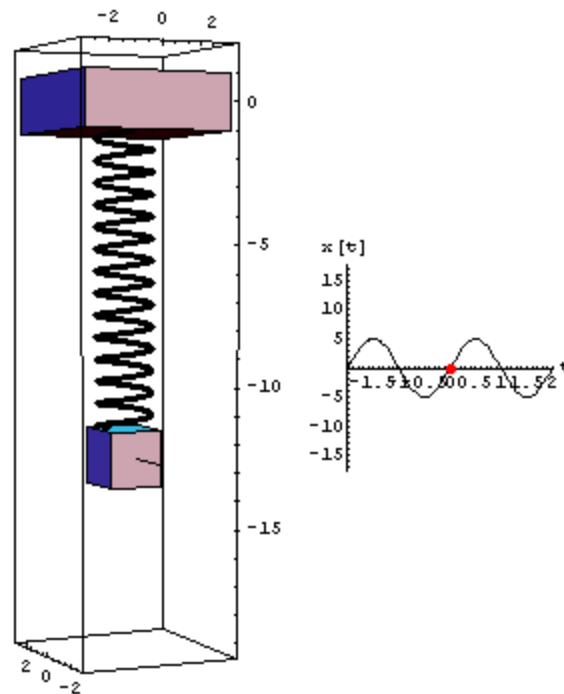




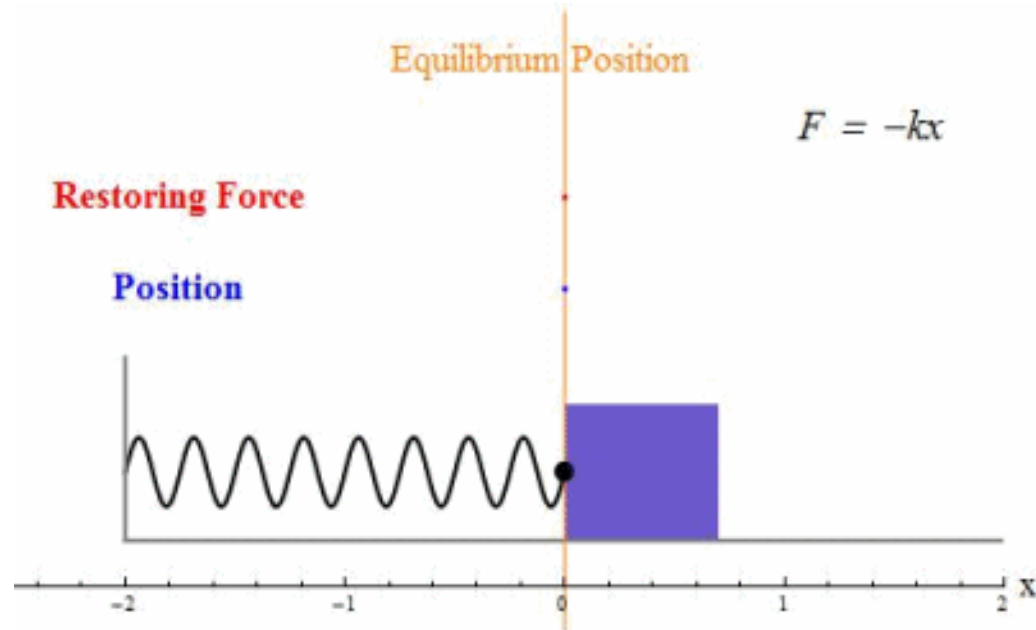
Oscillatory Motion (Simple Pendulum)



Oscillatory Motion (Spring Mass)



Oscillatory Motion (Spring Mass)



Oscillatory Motion (Spring Mass)

Oscillatory Phenomena

... observed in many physical systems ...
from the very small...(e.g. dipole resonance in nuclei)... to the very large (earthquake waves, stars,...)

Mechanical systems to lasers
..... from violin strings to electrical systems

Oscillatory Motion

Motion which is periodic in time, that is, motion that repeats itself in time.

Examples:

- Power line oscillates when the wind blows past it
- Earthquake oscillations move buildings

- 
- Block attached to a spring
 - Motion of a swing
 - Motion of a pendulum
 - Vibrations of a stringed musical instrument
 - Motion of a cantilever
 - Oscillations of houses, bridges, ...
 - All clocks use simple harmonic motion

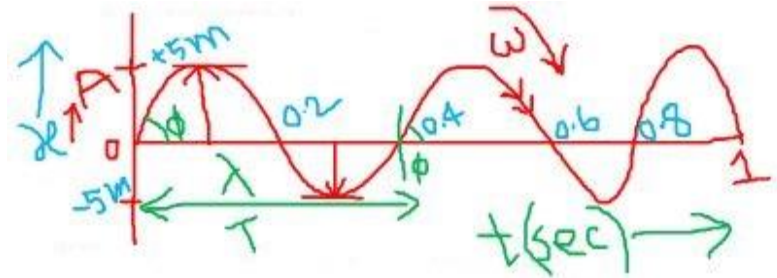
Periodic Motion: Many kind of motion repeat over and over, such as, the vibrations of quartz crystal in a watch, swinging pendulum in a clock and back-and-forth motion of a piston in an engine. This kind of motions are called periodic motion.

Example: Clock motion, motion of Moon, a rocking chair, a bouncing ball, a vibrating tuning fork, the Earth in its orbit around the Sun, Simple pendulum motion, etc.

Amplitude: The amplitude of the motion, denoted by A , is the *maximum magnitude of displacement* from the equilibrium position. It is always positive.

Mathematically,

$$\therefore A = \sqrt{\frac{2E}{k}} \quad \therefore A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0}\right)^2}$$



Period: The period T , is the time required for one (complete) oscillation. Mathematically, $T = \frac{1}{f} = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$

Frequency: The frequency, f , is the number of (complete) cycles in a unit time (in a linear path). $f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$

Angular Frequency: The angular frequency, ω , is the number of (complete) cycles in a unit time (in an angular path).

$$\omega = 2\pi f = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

Phase constant: The initial angle possessed by a particle for which a wave or motion generated. Mathematically,

$$\delta = \sin^{-1} \left(\frac{x_0}{A} \right) = \cos^{-1} \left(\frac{x_0}{A} \right) = \tan^{-1} \left(-\frac{v_0}{\omega_0 x_0} \right) = \sin^{-1} \left(-\frac{v_0}{\omega_0 A} \right)$$

Wavelength: The distance required for one (complete) oscillation. Mathematically, $v = f\lambda \Rightarrow \lambda = \frac{v}{f} = \frac{2\pi}{k}$

Phase angle: The angle possessed by a particle at instantaneous time for which a wave or motion generated.

$$\theta = (\omega t + \delta) = (\omega t + \theta_0) = \varphi$$

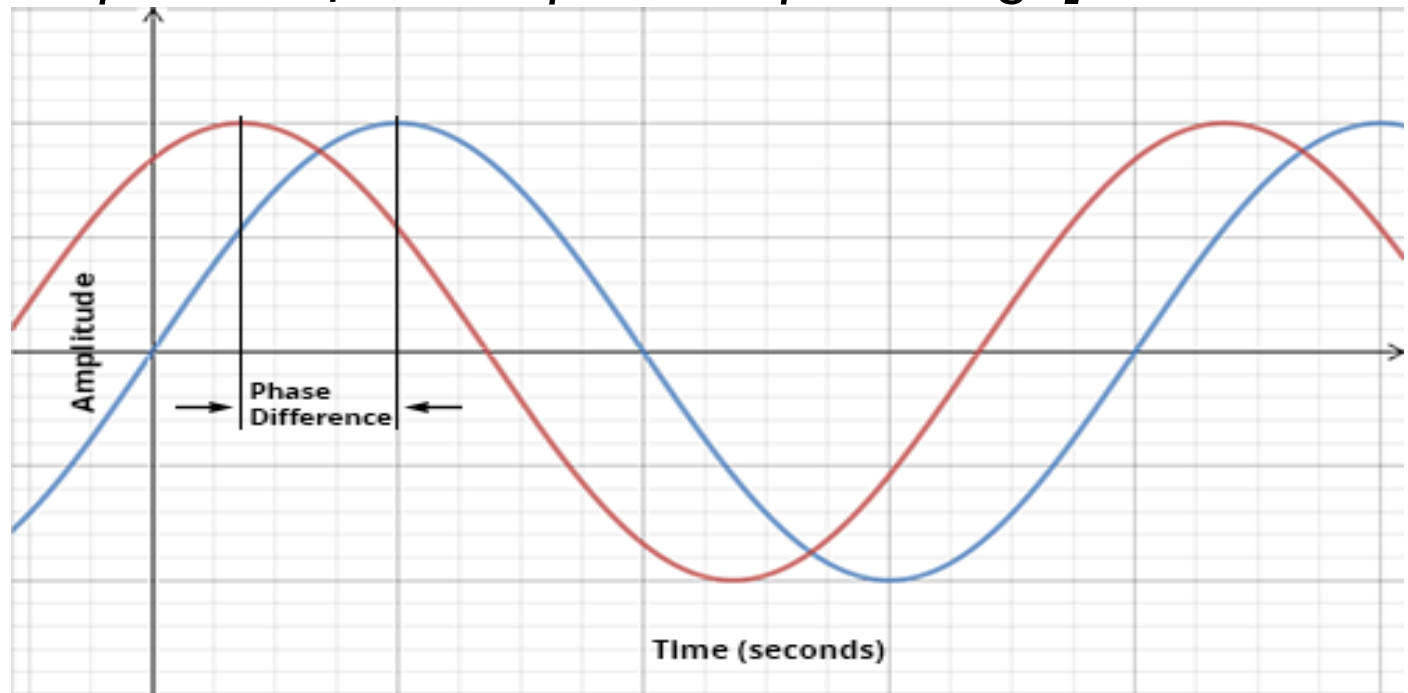
Let the signal F be a periodic function of one real variable, and T be its period [that is, the smallest positive real number such that $F(t+T)=F(t)$ for all t]. Then the *phase of F at any argument t* is

$$\varphi(t) = 2\pi \left[\frac{t - t_0}{T} \right] \qquad \theta = \omega t$$

Phase angle: The phase angle refers to the angular component of a periodic wave.

It is a complex entity. A periodic wave is represented as: $A\angle\theta$, v.z., $V = V_0\angle\theta$ Where $A=V_0$ is the magnitude and θ is the phase angle. *It is measured using degrees or radians.*

So, we can conclude that "Phase angle specifies the instantaneous location or instantaneous timing of a point within a wave cycle of a repetitive waveform" [Instantaneous phase position or instantaneous time in a position of a wave produces phase angle]



Simple Harmonic Motion (SHM) : The simplest kind of oscillation occurs when the restoring force F_x is directly proportional to the displacement from the equilibrium x .

When the restoring force is directly proportional to the displacement from the equilibrium as given by equation

$$F_x = -kx$$

The oscillation is called Simple Harmonic Motion (SHM).
Restoring force is a force that acts to bring a body to its equilibrium position.

Simply, the force responsible for restoring original size and shape is called the restoring force. Example: Spring force, elastic force., etc.

Simple Harmonic Motion (SHM) :

Any motion that repeats at regular interval of time is called simple harmonic motion. This motion is a sinusoidal function of time.

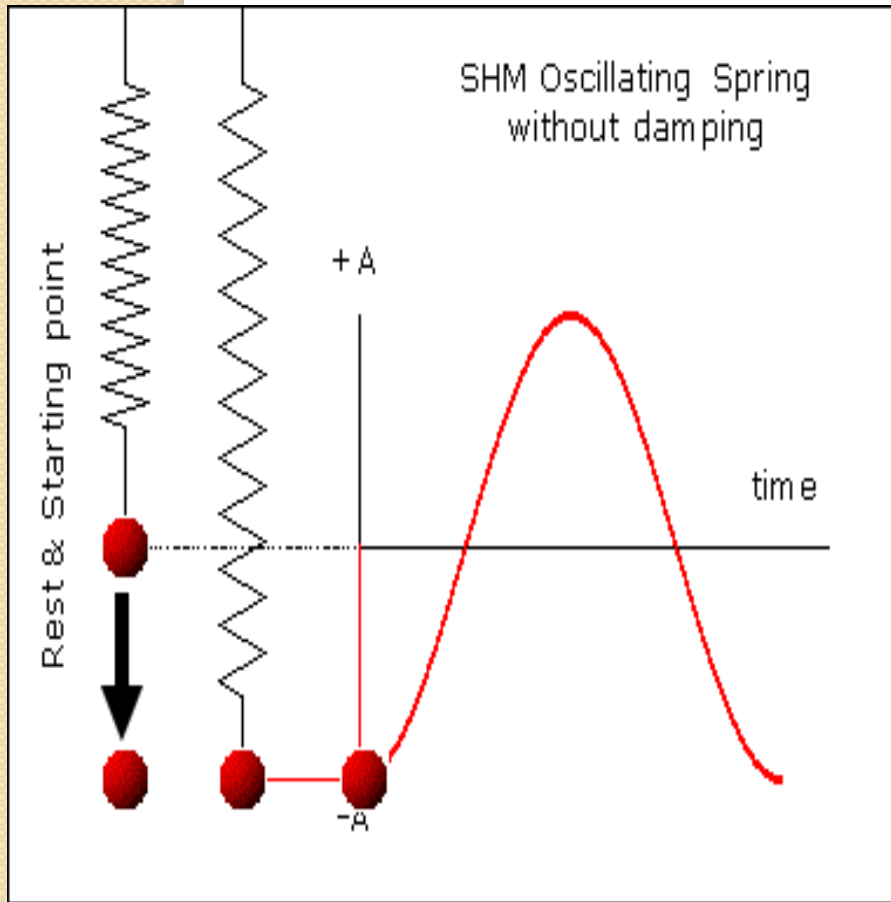
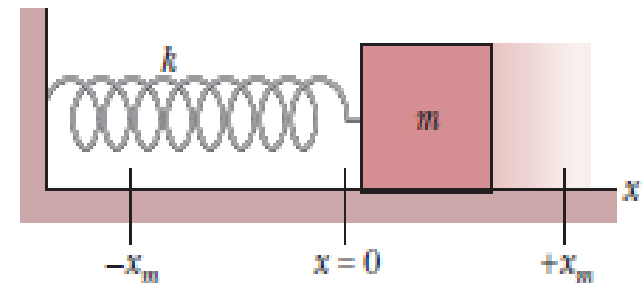


Figure shows that a particle is moving back and forth repeatedly along the x-axis between extreme points $+x_m = +A$ and $-x_m = -A$. This motion is a simple harmonic motion (SHM).

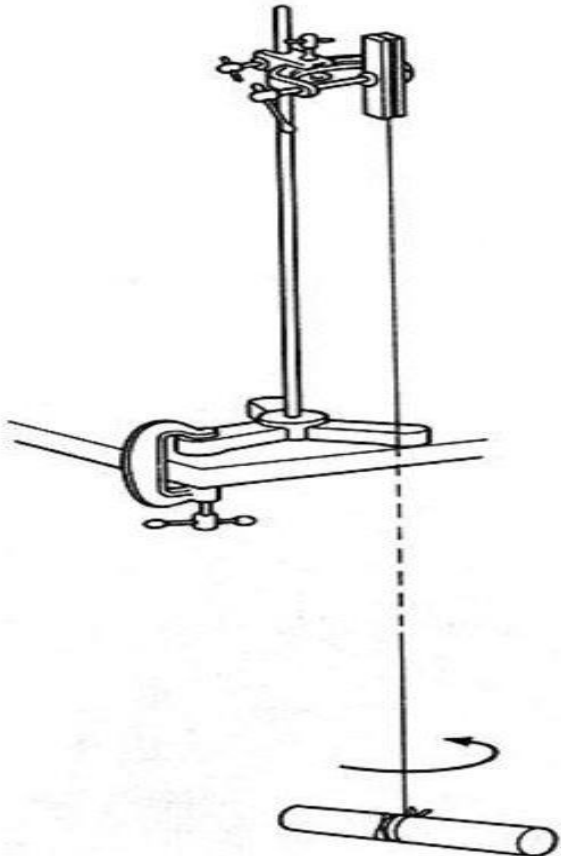
There is no loss of energy in this motion, so continues for indefinite time.



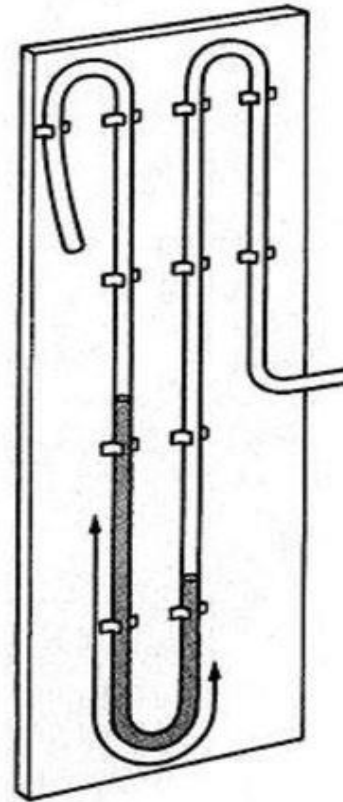
Simple Harmonic Motion (SHM) :

The motion under the influence of an elastic restoring force and in absence of all friction, the motion would continue once it had been started. This motion is called SHM.

Example: Oscillating simple pendulum, vibration of the eardrum, motion of a spring, oscillation of a liquid column in a U-tube, spring-mass system, torsional pendulum, wig-wag, etc.



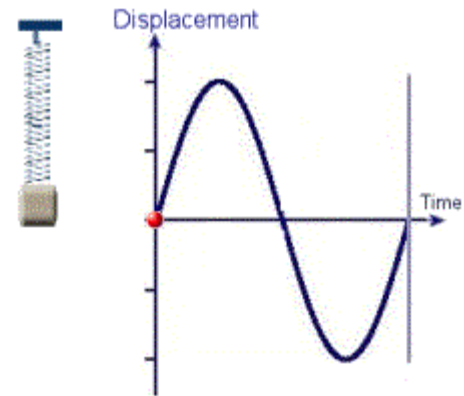
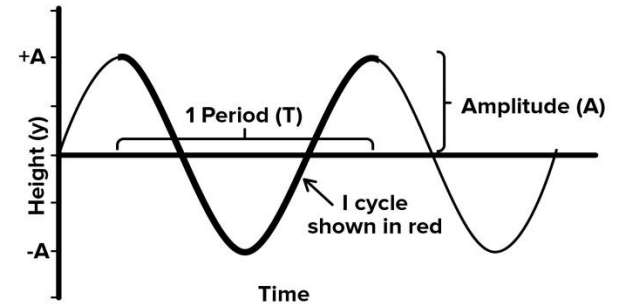
Torsional Pendulum



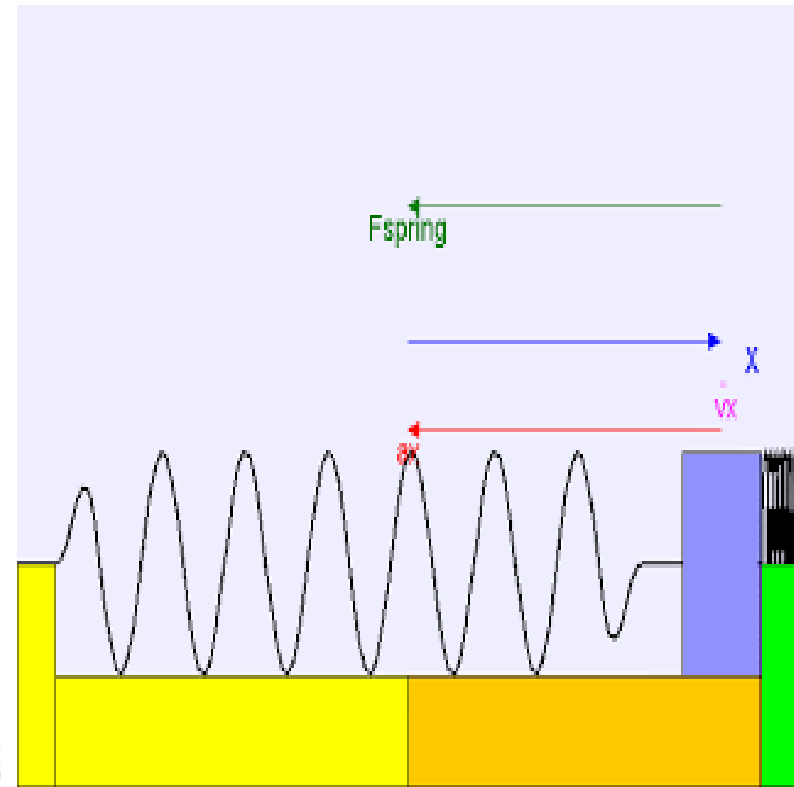
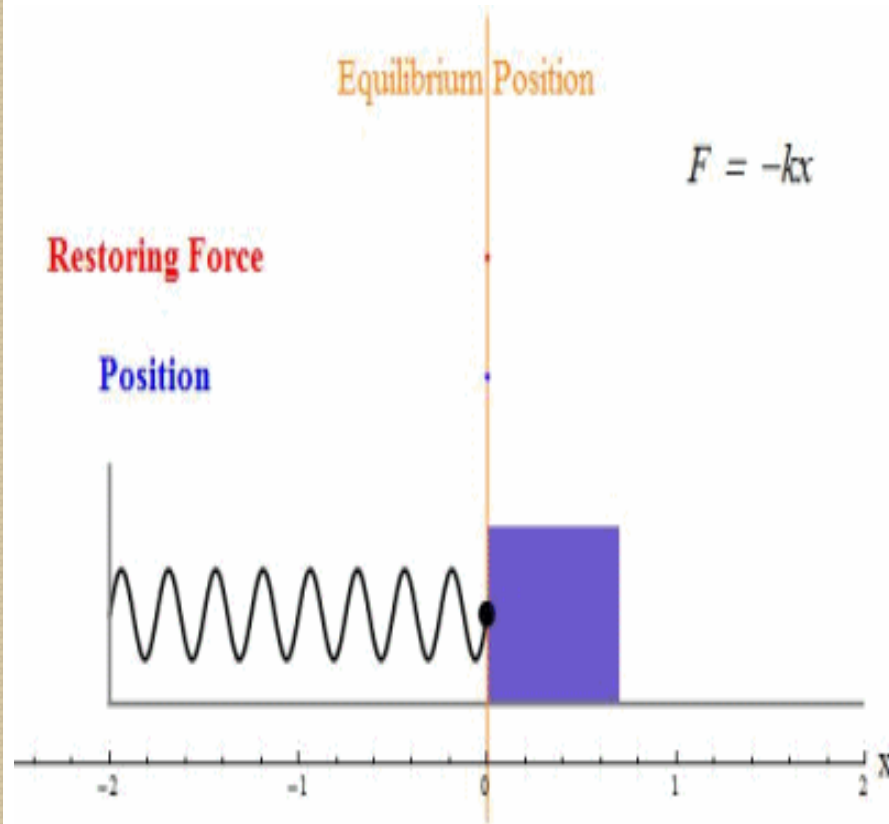
Oscillating water column

Characteristics Of SHM

- The particle's acceleration in simple harmonic motion is directly proportional to its displacement.
- Acceleration is opposite to the displacement.
- Motion is directed towards its mean location/mean position or motion is directed to the equilibrium position.
- Motion must be periodic
- The particle is subjected to a restoring force that is proportional to its displacement from the equilibrium position.
- At particular interval of time motion becomes opposite.
- Motion is always tending to a straight line.
- The particle's total energy is always constant.
- Net force can be described by Hooke's law.
- There is no damping (slowing down due to friction or other non-conservative forces).
- It oscillates with equal displacement on either side of the equilibrium position.
- Restoring force magnitude increases as the displacement increases.
- At equilibrium position the restoring force is zero. The particle oscillates back and forth around this position.

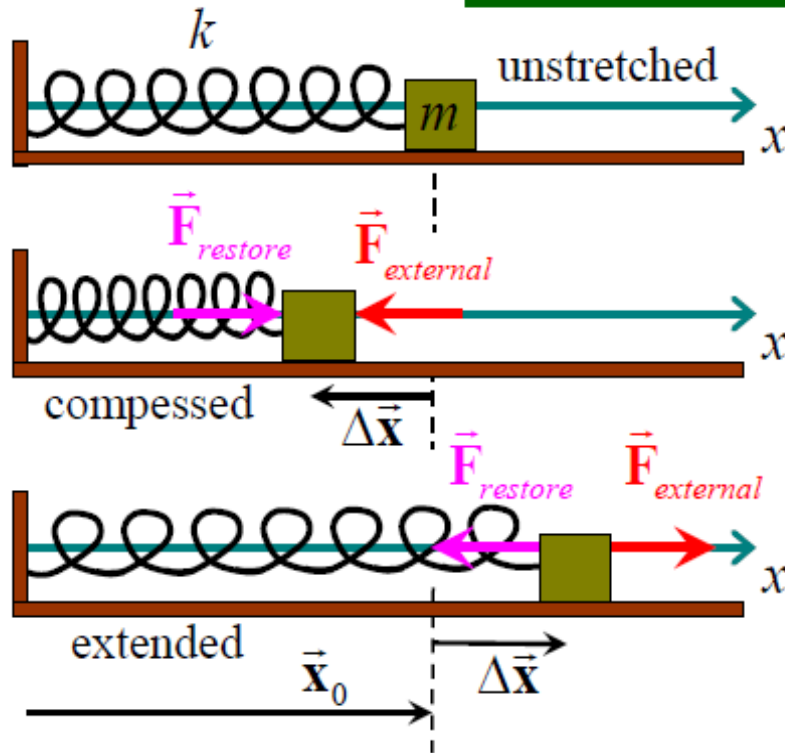


Simple Harmonic Motion (SHM):



□ *Derive the differential equation of SHM and find its solution (Mechanical system, Momentum Approach):*

Mass-spring oscillator



Hooke's Law:

Restoring force,

$$\vec{F}_{\text{restore}} = -k\Delta\vec{x}$$

where $\Delta\vec{x} = \vec{x} - \vec{x}_0$

and k is the “spring constant”
[N m⁻¹]

Start with the momentum principle: $\frac{d\vec{p}}{dt} = \vec{F}_{\text{net}}$

For horizontal forces on the mass: $\frac{dp_x}{dt} = -kx$

$$\therefore \frac{d(mv_x)}{dt} = -kx \quad \text{or} \quad \frac{d}{dt} \left(m \frac{dx}{dt} \right) = -kx$$

$$\therefore \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

SHM with Mechanical System (Momentum Approach):

Mass-spring oscillator ...2

$$\frac{d^2x(t)}{dt^2} = -\frac{k}{m}x(t)$$

... a second order differential equation
... we know that if we displace a mass-spring system from its rest position and then release it, it will perform SHM ...

Guess a trial solution: $x(t) = A \cos(\omega t + \phi)$

$$\text{then } \frac{d^2x}{dt^2} = -A\omega^2 \cos(\omega t + \phi)$$

and substitute into our DE: $-A\omega^2 \cos(\omega t + \phi) = -A \frac{k}{m} \cos(\omega t + \phi)$

... which is true provided $\omega^2 = \frac{k}{m}$

Therefore our solution is $x(t) = A \cos(\omega t + \phi)$ where $\omega = \sqrt{\frac{k}{m_{10}}}$

SHM with Mechanical System (Momentum Approach):

Mass-spring oscillator ...3

We will write a particular value of ω as ω_0 , as the **natural angular frequency** of the oscillator – the frequency that it “wants” to oscillate at.

$$\text{Mass-spring system: } \omega_0 = \sqrt{\frac{k}{m}} \quad \text{and} \quad T = 2\pi\sqrt{\frac{m}{k}}$$

$$\text{System parameters: } m, k \longrightarrow \omega_0$$

$$\text{Initial conditions: } \longrightarrow A, \phi$$

(Note that ω_0 is independent of A)

$$\text{Frequency, } f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

❑ *Derive the differential equation of SHM and find its solution (Mechanical system, Hook's Law Approach) :*

Alternative Method-1

[SHM with Mechanical System (Hook's Law Approach)]

Simple harmonic Oscillator:

According to Hook's law

$$F \propto -x$$

$$\Rightarrow F = -kx \quad ; [k \text{ is spring constant}]$$

$$\Rightarrow ma = -kx$$

$$\Rightarrow m \frac{d^2x}{dt^2} = -kx$$

$$\Rightarrow \boxed{m \frac{d^2x}{dt^2} + kx = 0} \rightarrow \text{Second order differential eqn of SHM}$$

----- (1)

Simple harmonic Oscillator : Alternative Method-1

SHM with Mechanical System (Hook's Law Approach):

$$E_K = \frac{1}{2}mv^2 \rightarrow \text{Kinetic energy}$$
$$E_P = \frac{1}{2}kx^2 \rightarrow \text{Potential energy}$$
$$E \rightarrow \text{Total Energy}$$

$$\Rightarrow E_K + E_P = \text{constant}$$
$$\therefore E_K + E_P = E$$

.....(2)

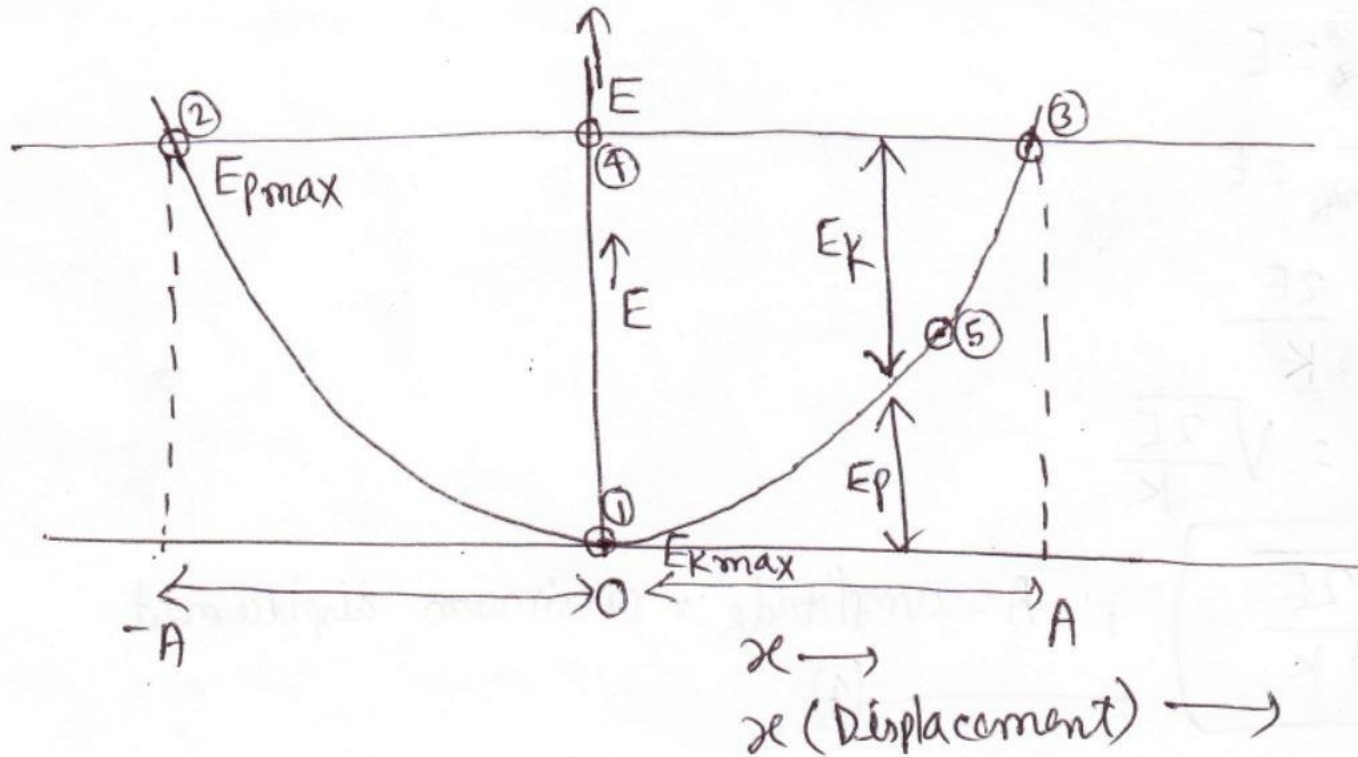


Fig: Total energy vs Displacement graph

Simple harmonic Oscillator : Alternative Method-1

SHM with Mechanical System (Hook's Law Approach):

for maximum kinetic energy:

$$\therefore V_{\max} = \sqrt{\frac{2E}{m}} \quad (3)$$

for maximum potential energy:

$$x_{\max} = \sqrt{\frac{2E}{K}}$$

$$A = \sqrt{\frac{2E}{K}} \quad (4)$$

Velocity at any instant:

$$V = \sqrt{\frac{K}{m}} \sqrt{A^2 - x^2}$$
$$\therefore V = \omega \sqrt{A^2 - x^2} \quad (5)$$

Simple harmonic Oscillator : Alternative Method-1

SHM with Mechanical System (Hook's Law Approach):

$$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{K}{m}} \sqrt{A^2 - x^2}$$

$$\Rightarrow \frac{dx}{\sqrt{A^2 - x^2}} = \sqrt{\frac{K}{m}} dt$$

$$\Rightarrow \int \frac{dx}{\sqrt{A^2 - x^2}} = \int \sqrt{\frac{K}{m}} dt$$

$$\Rightarrow \sin^{-1}\left(\frac{x}{A}\right) = \sqrt{\frac{K}{m}} t + C \quad ; \quad C \rightarrow \text{constant} \quad (6)$$

When $t=0$, $x=x_0$

$$\therefore \sin^{-1}\left(\frac{x_0}{A}\right) = 0 + C$$

Simple harmonic Oscillator : Alternative Method-1

SHM with Mechanical System (Hook's Law Approach):

$$\therefore C = \sin^{-1} \left(\frac{x_0}{A} \right)$$

Put the value of C in eqn (6),

$$\sin^{-1} \frac{x}{A} = \sqrt{\frac{k}{m}} t + \sin^{-1} \frac{x_0}{A}$$

$$\Rightarrow \sin^{-1} \frac{x}{A} = \sqrt{\frac{k}{m}} t + \sin^{-1} \theta_0 \quad ; \left[\frac{x_0}{A} = \theta_0 \right]$$

$$\Rightarrow \sin \left(\sin^{-1} \frac{x}{A} \right) = \sin \left(\sqrt{\frac{k}{m}} t + \theta_0 \right) \quad ; \left[\begin{array}{l} \text{for a small angle} \\ \sin \theta_0 = \sin^{-1} \theta_0 \\ = \theta_0 \end{array} \right]$$

$$\Rightarrow \frac{x}{A} = \sin \left(\sqrt{\frac{k}{m}} t + \theta_0 \right)$$

$$\Rightarrow \boxed{x = A \sin \left(\sqrt{\frac{k}{m}} t + \theta_0 \right)} \\ = A \sin (\omega t + \theta_0)$$

$$\therefore x(t) = A \cos(\omega t + \phi_0)$$

$\rightarrow$ Eqn of SHM

$$\therefore x = A \sin(\omega t + \phi_0)$$

$$\therefore x(t) = A \sin(\omega t + \phi_0)$$

□ *Derive the differential equation of SHM and find its solution (Mechanical system, Newton's 2nd Law Approach):*

Alternative Method-2

[SHM with Mechanical System (Newton's 2nd Law Approach)]

In the case of spring mass system the relation between the restoring force F and the displacement x is

$$F \propto x$$

or, $F = -kx$ (1)

where k is the proportionality constant and generally called the spring constant or force constant. But $-kx$ is called the restoring force working opposite to the displacement. Using Newton's 2nd law of motion we can write

$$F = ma = m \frac{d^2x}{dt^2}$$

Equation (1) can now be recast as

$$m \frac{d^2x}{dt^2} = -kx$$

Alternative Method-2

SHM with Mechanical System (Newton's 2nd Law Approach):

$$\begin{aligned}\frac{d^2x}{dt^2} + \frac{k}{m}x &= 0 \\ \frac{d^2x}{dt^2} + \omega^2x &= 0\end{aligned}\tag{2}$$

This is the differential equation for a simple harmonic motion.

Solution of the differential equation for SHM

Let the trial solution be

$$\begin{aligned}x &= e^{\alpha t} \\ \text{So, } \frac{dx}{dt} &= \alpha e^{\alpha t} \quad \text{and} \quad \frac{d^2x}{dt^2} = \alpha^2 e^{\alpha t}\end{aligned}$$

Equation (2) can be rewritten as

$$\begin{aligned}\alpha^2 e^{\alpha t} + \omega^2 e^{\alpha t} &= 0 \\ \therefore (\alpha^2 + \omega^2) e^{\alpha t} &= 0\end{aligned}$$

Alternative Method-2

SHM with Mechanical System (Newton's 2nd Law Approach):

Now $e^{\alpha t} \neq 0$ because it is the solution. Therefore

$$\begin{aligned}\alpha^2 + \omega^2 &= 0 \\ \text{or, } \alpha^2 &= -\omega^2 = (i\omega)^2 \quad [\text{here, } i^2 = -1] \\ \therefore \alpha &= \pm i\omega\end{aligned}$$

Two independent solution are of the form $e^{i\omega t}$ and $e^{-i\omega t}$. So, the general solution may be written as

$$\begin{aligned}x &= B_1 e^{i\omega t} + B_2 e^{-i\omega t} \\ &= B_1 (\cos \omega t + i \sin \omega t) + B_2 (\cos \omega t - i \sin \omega t) \\ &= (B_1 + B_2) \cos \omega t + i(B_1 - B_2) \sin \omega t\end{aligned} \tag{3}$$

Let us replace constants B_1 and B_2 by other two constants A and ϕ such that $B_1 + B_2 = A \cos \phi$ and $i(B_1 - B_2) = A \sin \phi$.

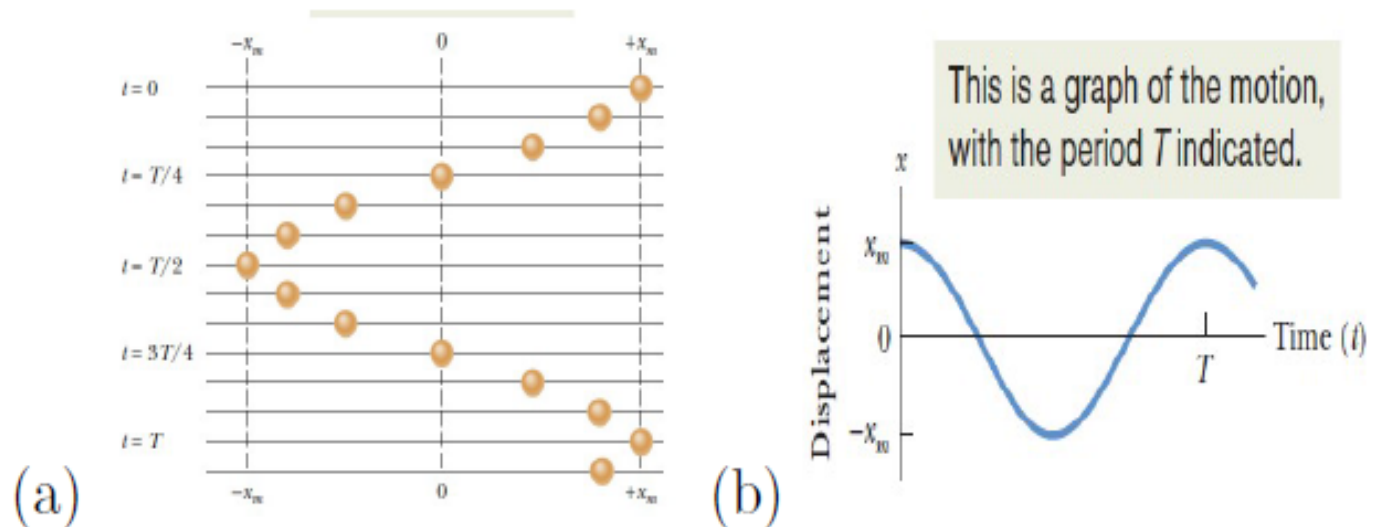
Alternative Method-2

SHM with Mechanical System (Newton's 2nd Law Approach):

Then

$$\begin{aligned}x &= A \cos \omega t \cos \phi + A \sin \omega t \sin \phi \\x(t) &= A \cos(\omega t - \phi)\end{aligned}\quad (4)$$

Here, $A (= x_m)$ denotes the amplitude of the oscillation and ϕ denotes the phase constant. This is the equation for the displacement of the oscillation.



Alternative Method-2

SHM with Mechanical System (Newton's 2nd Law Approach):

Time period of oscillation

After a time period T the object will come to its initial position. So,

$$\begin{aligned} A \cos(\omega t - \phi) &= A \cos(\omega[t + T] - \phi) \\ \text{or, } A \cos(2\pi + [\omega t - \phi]) &= A \cos(\omega[t + T] - \phi) \\ \text{or, } \omega(t + T) - \phi &= 2\pi + \omega t - \phi \\ \therefore T &= \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}} \quad (5) \end{aligned}$$

$$\text{and angular frequency } \omega = \frac{2\pi}{T} = 2\pi \nu. \quad (6)$$

□ **Graphical Representation and Mathematical Proof of maximum displacement, maximum velocity, and maximum acceleration of SHM:**

Velocity of the body undergoing SHM **Alternative Method**

The velocity of the oscillating body is

$$\begin{aligned}v(t) &= \frac{dx(t)}{dt} = \frac{d}{dt}(A \cos(\omega t - \phi)) \\V(t) &= -A\omega \sin(\omega t - \phi) \\&= -\omega x(t)\end{aligned}\tag{7}$$

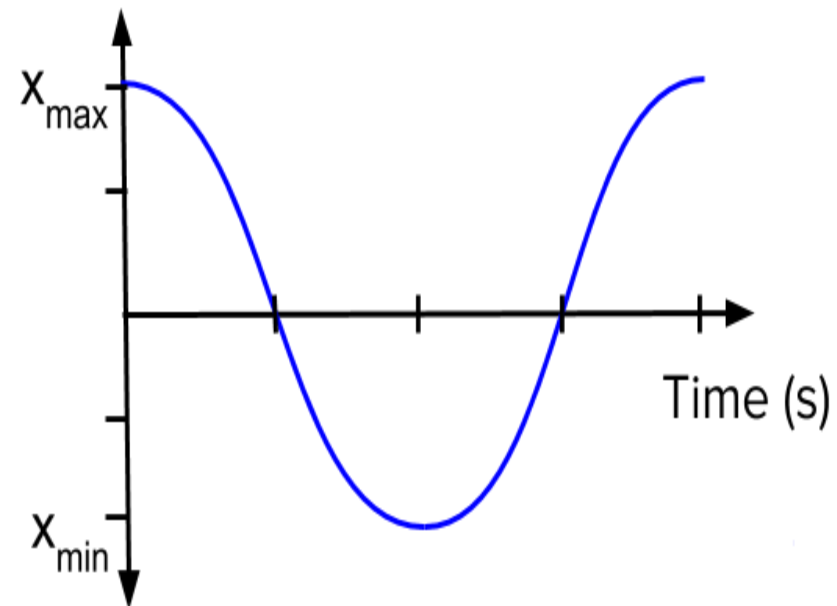
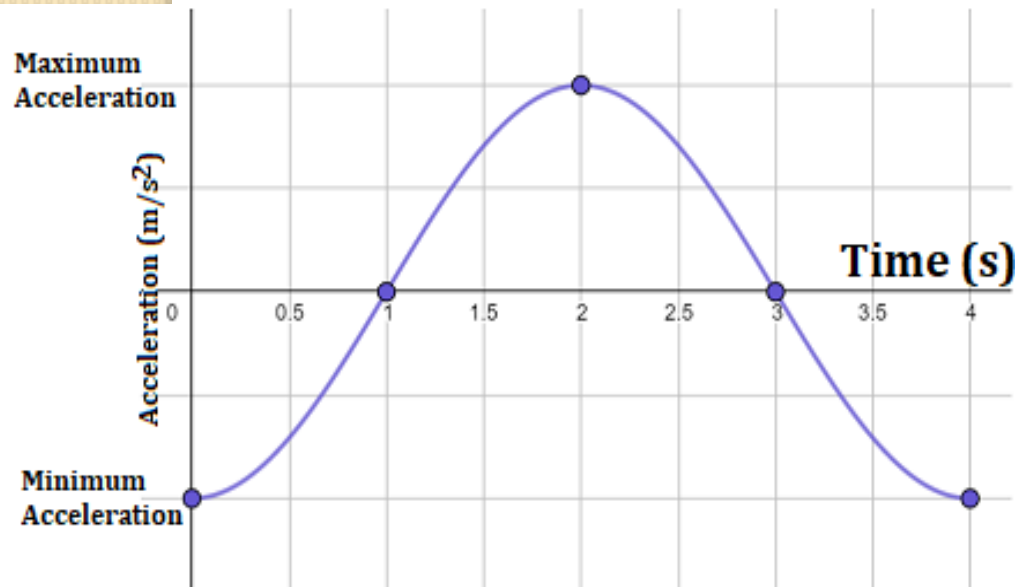
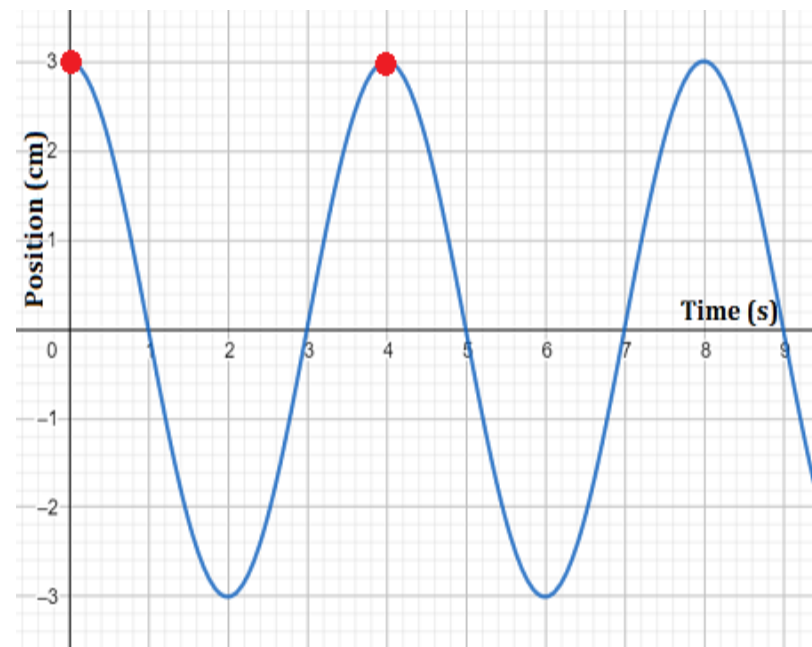
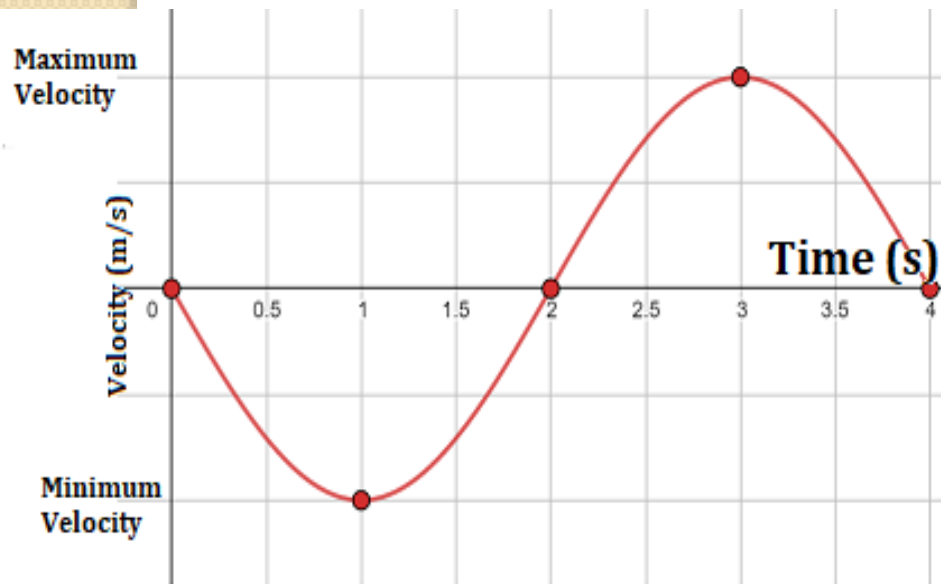
where $A\omega$ is the amplitude or maximum velocity. Therefore

$$\begin{aligned}v^2 &= A^2 \omega^2 \sin^2(\omega t - \phi) \\&= \omega^2 (A^2 - A^2 \cos^2(\omega t - \phi)) \\&= \omega^2 (A^2 - x(t)^2) \\\therefore v(t) &= \omega \sqrt{A^2 - x(t)^2}\end{aligned}\tag{8}$$

The maximum velocity of oscillation is

$$v_{max} = \omega A \quad \text{and} \quad v_{min} = -\omega A$$

Alternative Method



□ Total Energy of SHM calculation and its Graphical Representation: Alternative Method (Energy in Mechanical System)

Acceleration of the body undergoing SHM

The acceleration of the oscillating body is

$$\begin{aligned}a(t) &= \frac{dv(t)}{dt} = -A\omega^2 \cos(\omega t - \phi) \\&= -\omega^2 [A \sin(\omega t - \phi)] \\&= -\omega^2 x(t)\end{aligned}\tag{9}$$

where $A\omega^2$ is the amplitude or maximum acceleration.

Kinetic Energy, Potential energy and total energy

The kinetic energy of the oscillating body is

$$\begin{aligned}K &= \frac{1}{2}mv(t)^2 \\&= \frac{1}{2}mA^2\omega^2 \sin^2(\omega t - \phi) \quad \left[\because \omega^2 = \frac{k}{m}\right] \\&= \frac{1}{2}kA^2 \sin^2(\omega t - \phi)\end{aligned}\tag{10}$$

Total Energy of SHM calculation and its Graphical Representation: Alternative Method (Energy in Mechanical System)

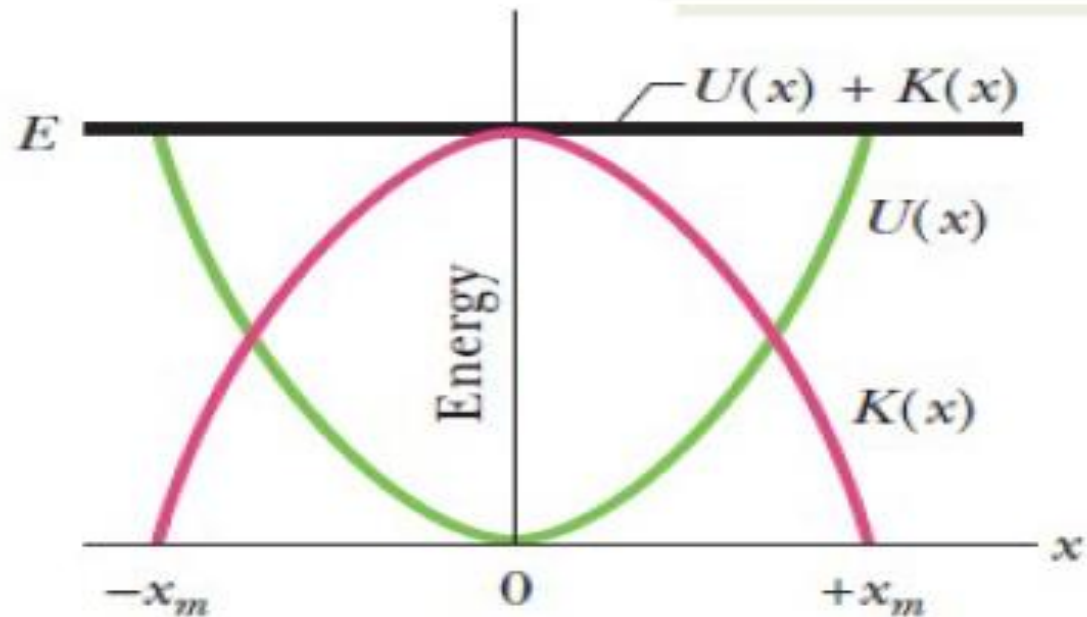
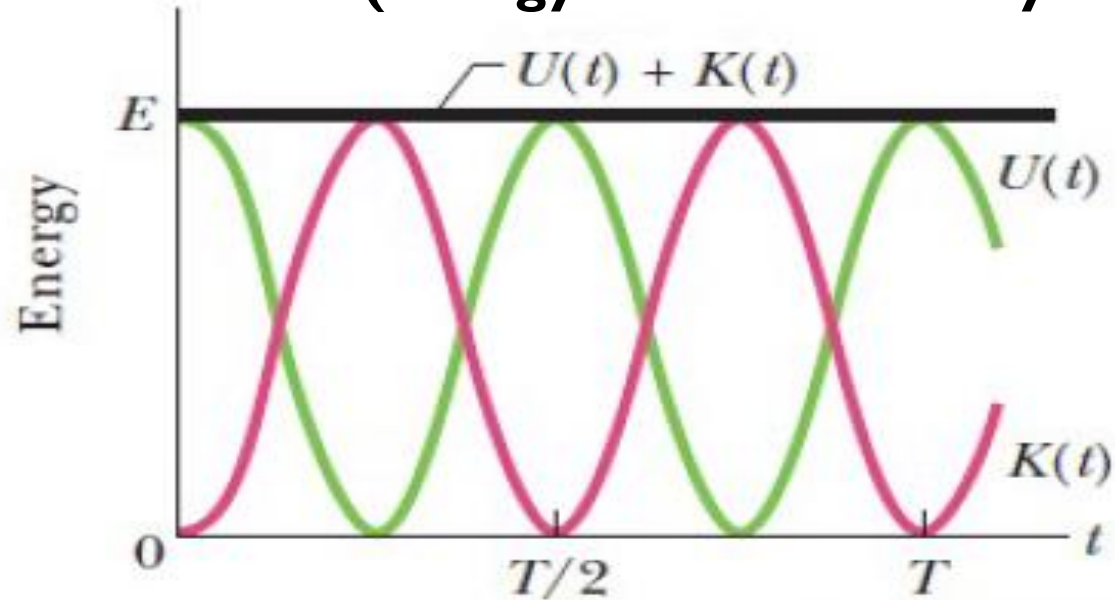
The potential energy is

$$\begin{aligned} U &= - \int_0^x F(x) dx = k \int_0^x x dx \\ &= \frac{1}{2} k x^2 \\ &= \frac{1}{2} k A^2 \cos^2(\omega t - \phi) \end{aligned} \quad (11)$$

The Total energy of an oscillator is

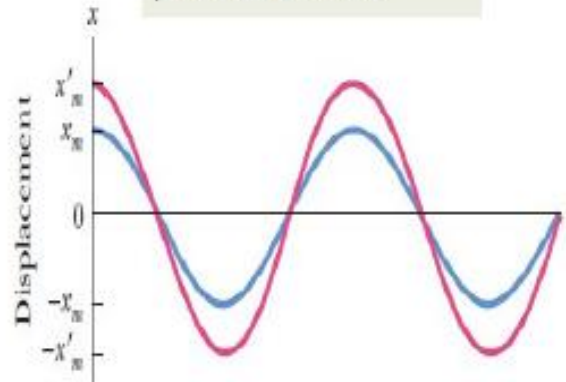
$$\begin{aligned} E &= K + U \\ &= \frac{1}{2} k A^2 \sin^2(\omega t - \phi) + \frac{1}{2} k A^2 \cos^2(\omega t - \phi) \\ &= \frac{1}{2} k A^2 [\sin^2(\omega t - \phi) + \cos^2(\omega t - \phi)] \\ &= \frac{1}{2} k A^2 \end{aligned} \quad (12)$$

Total Energy of SHM calculation and its Graphical Representation: Alternative Method (Energy in Mechanical System)

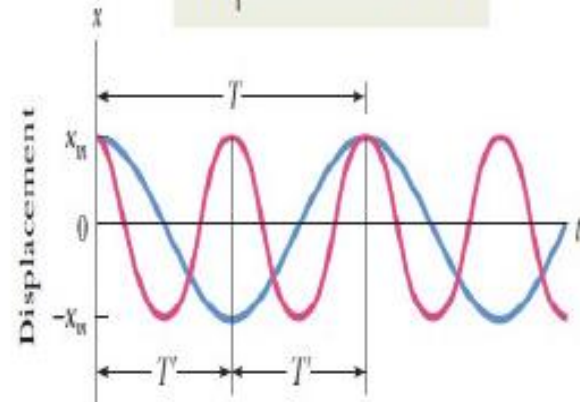


Physical interpretation of oscillations:

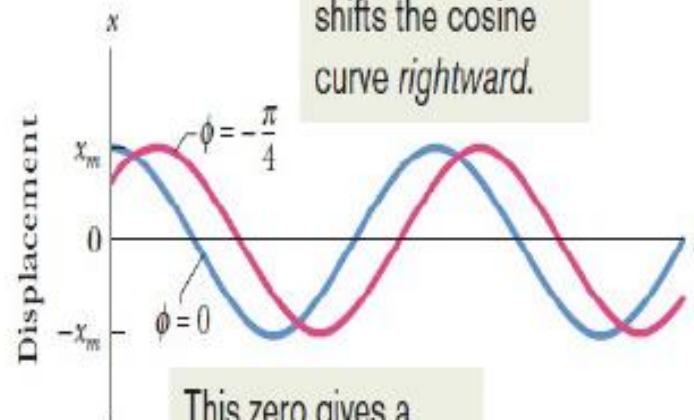
The amplitudes are different, but the frequency and period are the same.



The amplitudes are the same, but the frequencies and periods are different.



This *negative* value shifts the cosine curve *rightward*.



This zero gives a regular cosine curve.

Graphical Representation of SHM

Let P be a particle moving on the circumference of a circle of radius a . The foot of perpendicular vibrates on the diameter YY'.

The displacement graph is a sine curve represented by $ABCDE$ (Fig. 1.3).

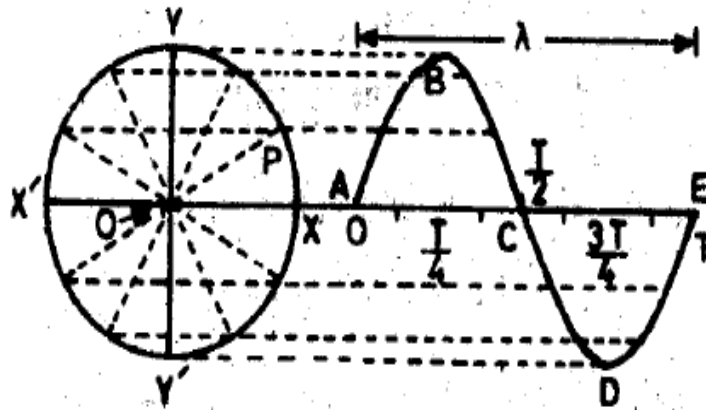
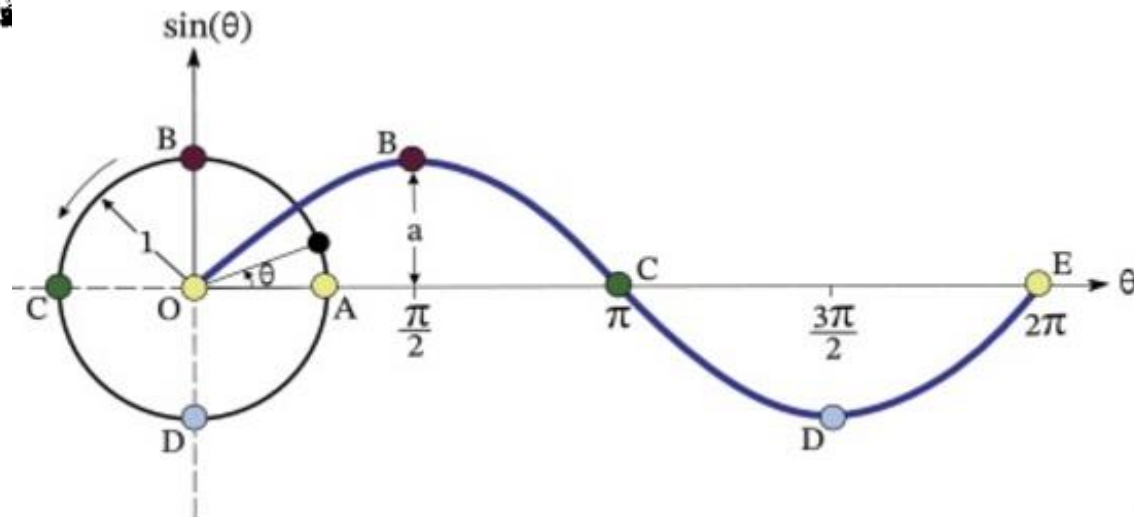


Fig. 1.3. Displa

$$y = a \sin \omega t = a \sin 2\pi \frac{t}{T}$$



Graphical Representation of SHM

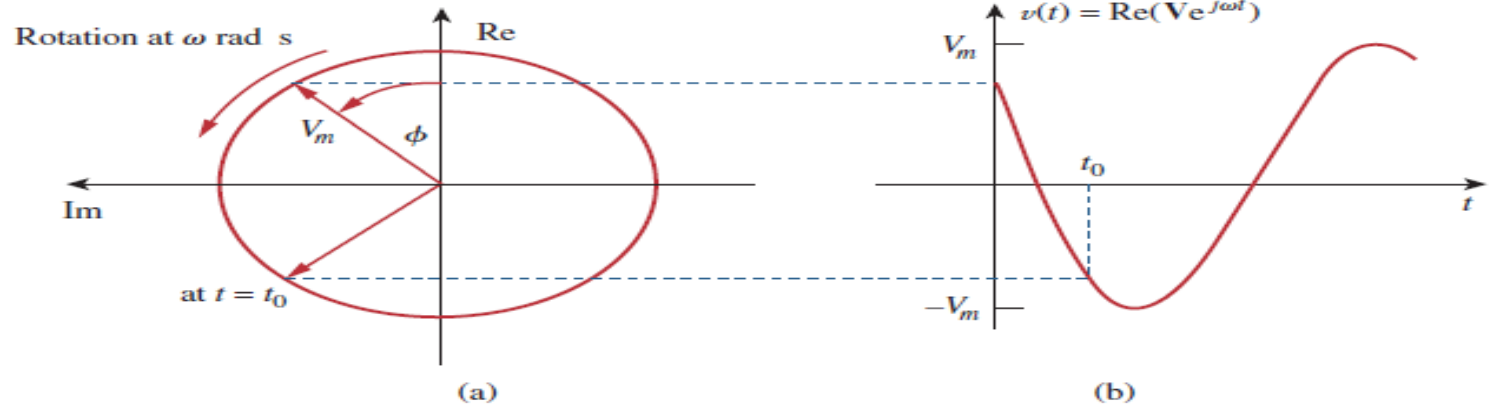
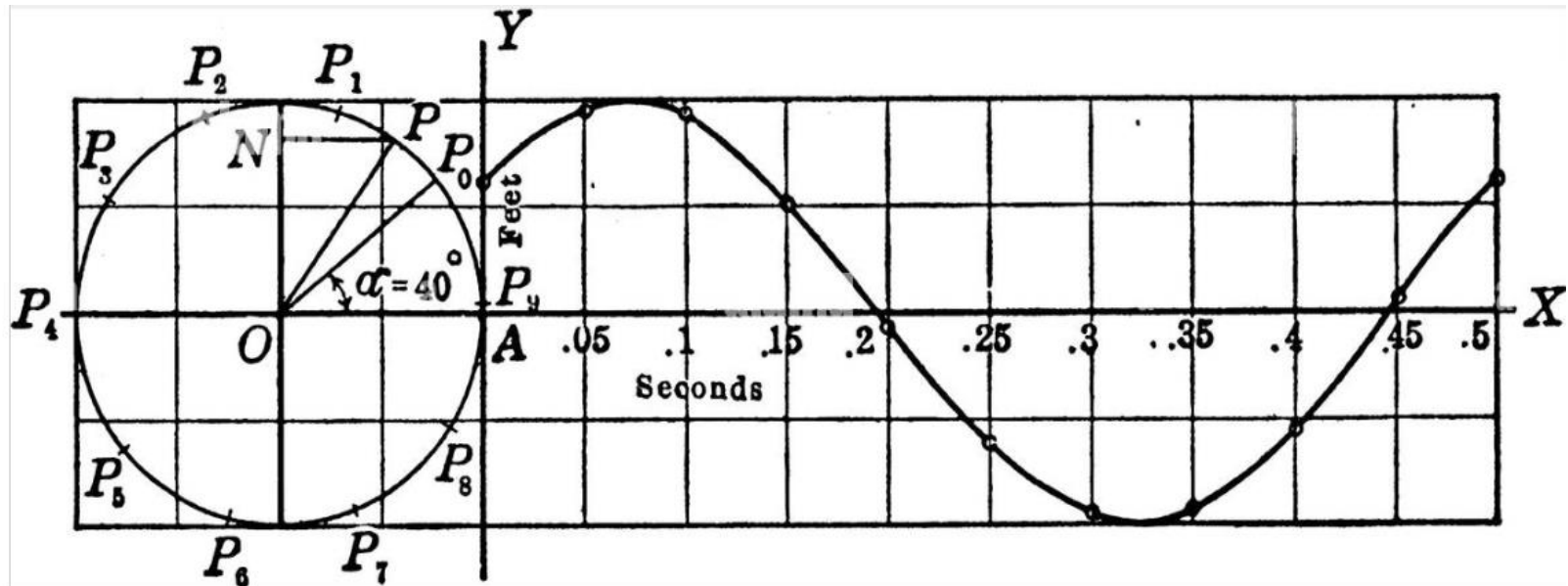
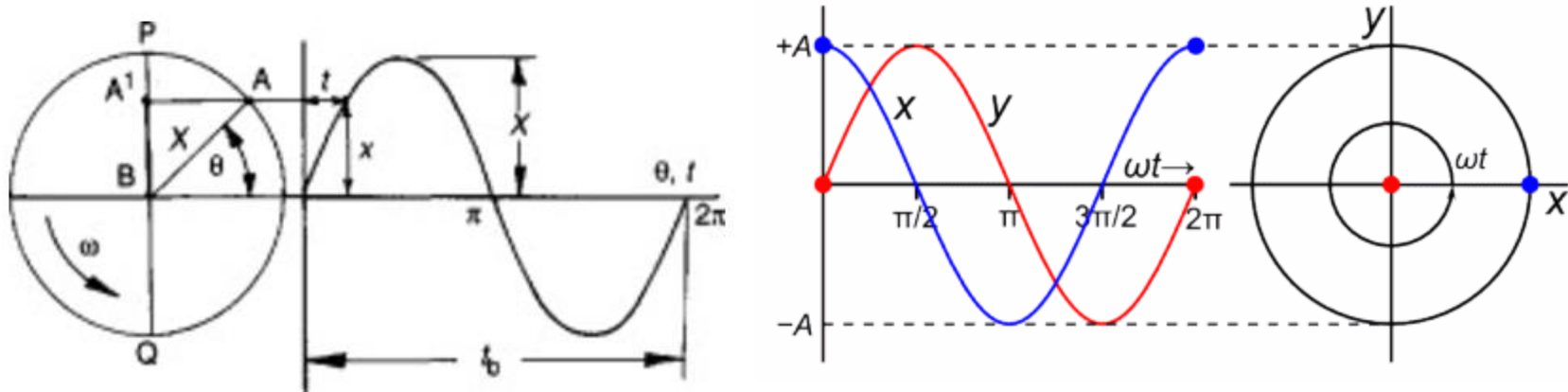


Figure 9.7

Representation of $V e^{j\omega t}$: (a) sinor rotating counterclockwise, (b) its projection on the real axis, as a function of time.



Graphical Representation of SHM



Referring to the figure, point A rotates with constant angular velocity ω at radius AB. The projection of A on to PQ, i.e. A', moves with simple harmonic motion. If A'B is plotted to a base of the angle of rotation θ , a so-called 'sine curve' is produced. The base of the graph can also represent time. The time for one complete rotation is the 'periodic time' t_p . If $AB = X$ and $A'B = x$, then $x = f(x) = X \sin \omega t$, where $\omega = \theta/t$.

Periodic time, $t_p = 2\pi/\omega$

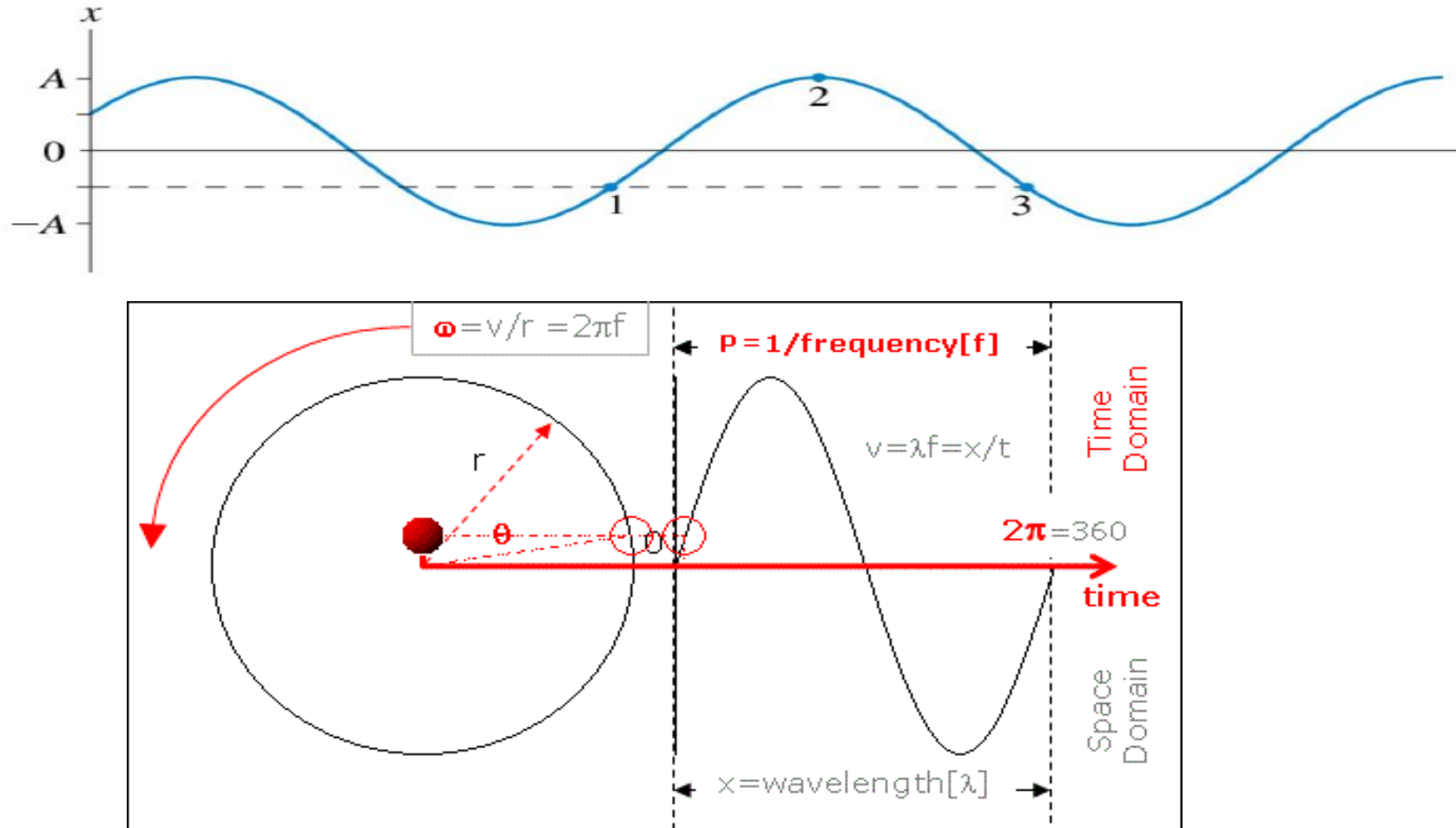
Frequency, $f = \frac{1}{t_p} = \omega/2\pi$

$1^r = 1^c = 57.2958^\circ = 57.3^\circ \approx 57.3^\circ = 57.3^\circ \text{ degree} = 1 \text{ rad} = 1 \text{ radian}$

$1^\circ = 1^{deg} = 1 \text{ degree} = 0.0174533^r = 0.0174533^r = 0.0174533 \text{ radian}$
 $\approx 0.017 \text{ radian} \approx 0.017 \text{ rad}$

Graphical Representation of SHM

Example-0.1: The figure shows below a position-versus-time graph for a particle in SHM. What is the phase constant? What is the phase at point 2?



Example-0.2: An object in simple harmonic motion has an amplitude of 4.0 cm, a frequency of 2.0 Hz, and a phase constant of $2\pi/3$ rad. Draw a position graph showing two cycles of motion.

□ ***Acceleration and Displacement relation of SHM : $a = -\omega^2 x$***

$$x(t) = A \cos(\omega_0 t + \phi)$$

$$v(t) = \frac{dx(t)}{dt} = -A \omega_0 \sin(\omega_0 t + \phi)$$

$$a(t) = \frac{d^2 x(t)}{dt^2} = \frac{dv(t)}{dt} = -A \omega_0^2 \cos(\omega_0 t + \phi)$$

$$= -\omega_0^2 \times [A \cos(\omega_0 t + \phi)]$$

$$= -\omega_0^2 \times x(t)$$

$$= -\omega_0^2 x(t) = -\omega^2 x(t)$$

$$\therefore a = -\omega^2 x$$

□ *Graphical Representation and Mathematical Proof of Phase Difference between displacement, velocity, and acceleration of a SHM* : $\delta(a, x) = \pi$, $\delta(v, x) = \frac{\pi}{2}$, and $\delta(v, a) = \frac{\pi}{2}$.

Simple harmonic oscillator

$$x(t) = A \cos(\omega_0 t + \phi)$$

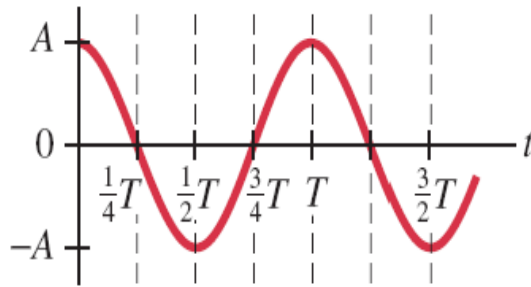
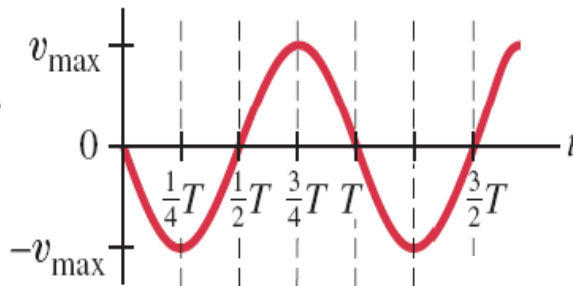
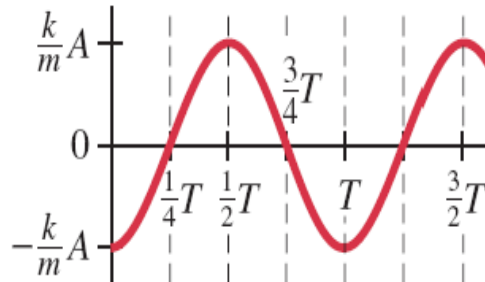
$$v(t) = \frac{dx(t)}{dt} = -A\omega_0 \sin(\omega_0 t + \phi)$$

$$a(t) = \frac{d^2x(t)}{dt^2} = \frac{dv(t)}{dt} = -A\omega_0^2 \cos(\omega_0 t + \phi)$$

$$\begin{aligned} \dots \text{ acceleration} &= -(\text{constant}) \cdot (\text{displacement}) \\ &= -A\omega_0^2 \cos(\omega_0 t + \phi) \\ &= A\omega_0^2 \cos(\omega_0 t + \phi + \pi) \end{aligned}$$

Phase difference between acceleration and displacement is π

Phase difference between v and x (and v & a) is $\frac{\pi}{2}$

Displacement x Velocity v Acceleration a 

The velocity and acceleration for simple harmonic motion can be found by differentiating the displacement:

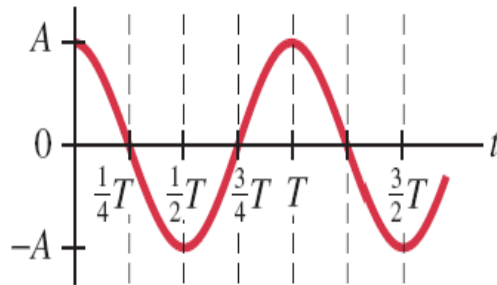
$$x(t) = A \cos(\omega_o t + \phi)$$

$$\therefore x = A \cos(\omega t + \phi)$$

$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

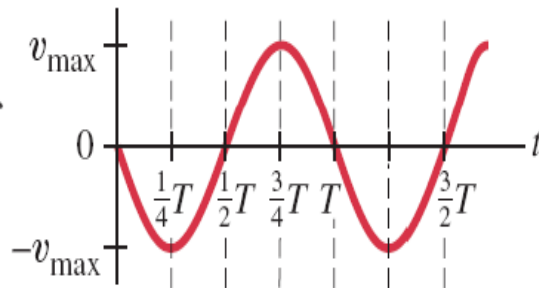
$$a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = -\omega^2 A \cos(\omega t + \phi).$$

Displacement x



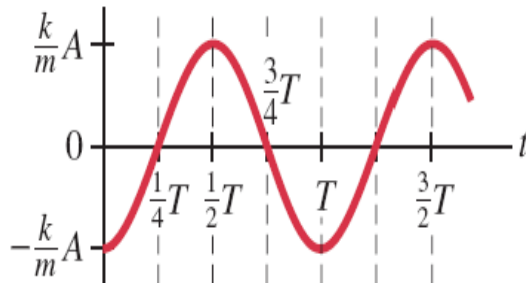
The phase difference between acceleration and displacement is, $\delta = \delta(a, x) = \delta(x, a)$
 $= \{\pi + (\omega t + \phi)\} - \{\omega t + \phi\}$
 $= \pi$

Velocity v



Similarly, the phase difference between velocity and displacement is, $\delta = \delta(v, x) = \delta(x, v)$
 $= \{\frac{\pi}{2} + (\omega t + \phi)\} - \{\omega t + \phi\}$
 $= \frac{\pi}{2}$

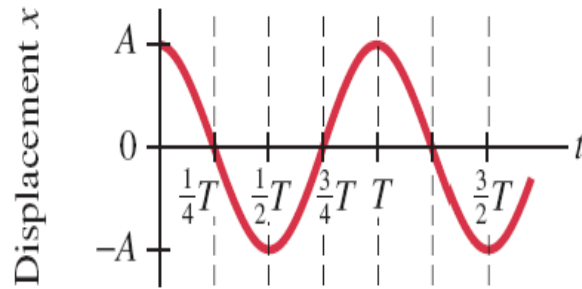
Acceleration a



Similarly, the phase difference between velocity and acceleration is, $\delta = \delta(v, a) = \delta(a, v) = \frac{\pi}{2}$

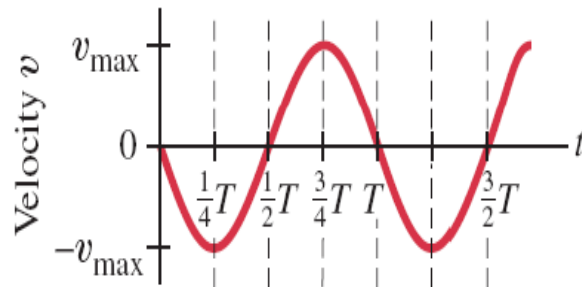
So that, Phase difference between acceleration and displacement is $\delta(a, x) = \pi$, Phase difference between velocity and displacement is $\delta(v, x) = \frac{\pi}{2}$, and Phase difference between velocity and acceleration is $\delta(v, a) = \frac{\pi}{2}$.

□ **Graphical Representation and Mathematical Proof of maximum displacement, maximum velocity, and maximum acceleration of SHM:**
Magnitude of amplitud $A_a > A_v > A_x$



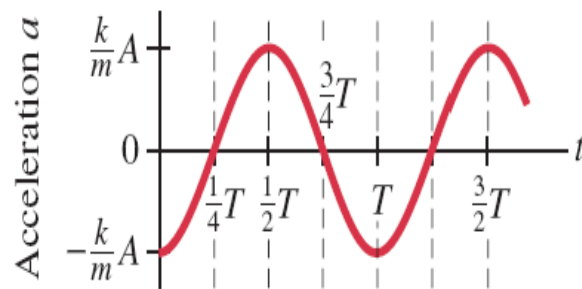
$$x = A \cos(\omega t + \varphi),$$

$$v = -\omega A \sin(\omega t + \varphi)$$



$$= \omega A \cos(\omega t + \varphi + \frac{\pi}{2}),$$

$$a = -\omega^2 A \cos(\omega t + \varphi)$$



$$= -\omega^2 A \sin(\omega t + \varphi + \frac{\pi}{2}).$$

$$= \omega^2 A \cos[\pi + (\omega t + \phi)]$$

$A_a = -A\omega^2 = |\pm A\omega^2| = A\omega^2 = a_{max}$ **=Acceleration amplitude**
=Amplitude of the acceleration Equation=Maximum Acceleration

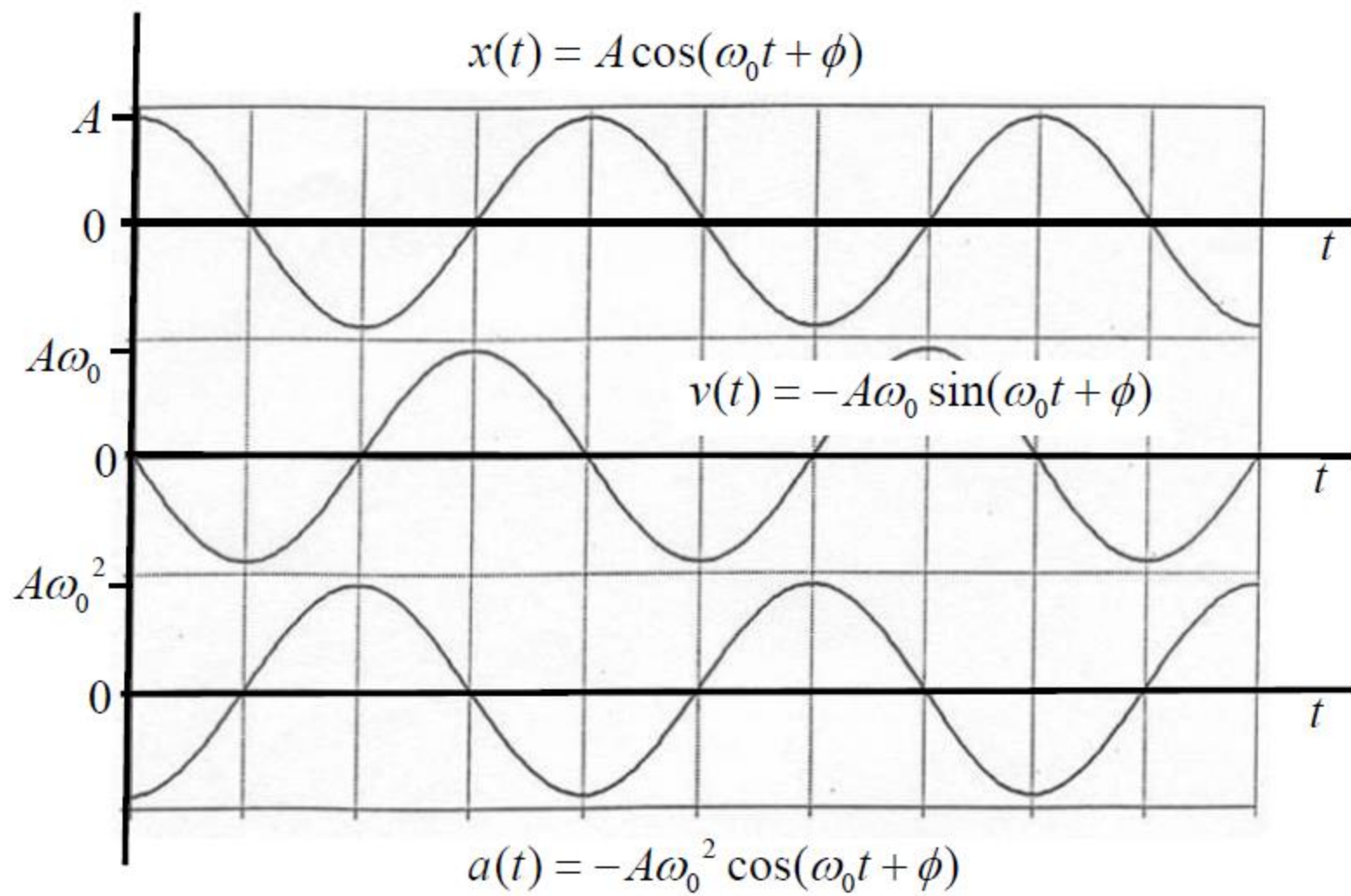
$A_v = -A\omega = |\pm A\omega| = A\omega = v_{max}$ **=Velocity amplitude**
=Amplitude of the velocity Equation=Maximum Velocity

$A_x = A_y = -A = A = |\pm A| = A = x_{max} = y_{max} = x_m = y_m = x_o = y_o = A_o$
=Displacement amplitude=Amplitude=Amplitude of the displacement
Equation=Maximum Displacement

The magnitude of the amplitude, therefore

$$\therefore A_a > A_v > A_x$$

$$\therefore A_a > A_v > A$$



Phase, Phase shift, and Phase difference

TEXTBOOK SAYS: "Phase is a variable that depends on position and time."

$$\varphi(x, t) = kx \pm \omega t + \varphi$$

$$\varphi(x, t) = \left(\frac{2\pi}{\lambda}\right)x \pm 2\pi ft + \varphi$$

$$\theta = \omega t + \delta = \omega t + \theta_o = \varphi$$

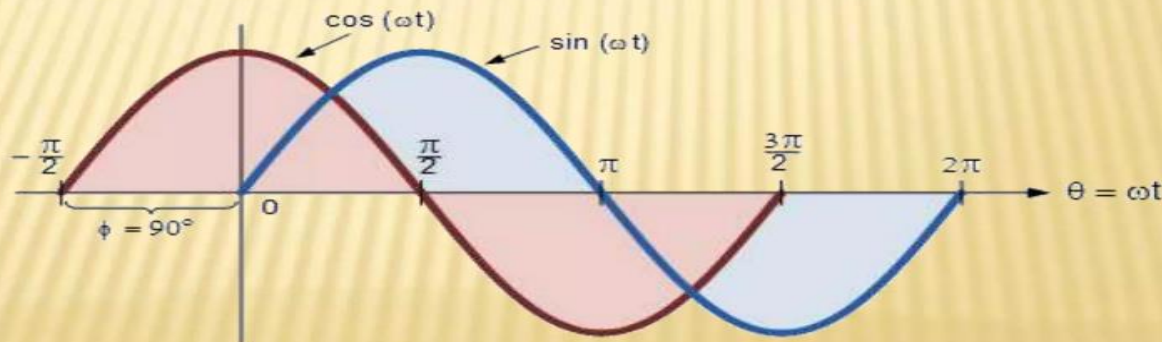
$$\theta = \varphi = \tan^{-1}\left(-\frac{v}{\omega x}\right)$$

$$\delta = \sin^{-1}\left(\frac{x_0}{A}\right) = \cos^{-1}\left(\frac{x_0}{A}\right) = \tan^{-1}\left(-\frac{v_0}{\omega_0 x_0}\right) = \sin^{-1}\left(-\frac{v_0}{\omega_0 A}\right)$$

The phase or phase angle of a sine signal is often noted with Φ or θ and measured in radians (rad) or degrees ($^\circ$) and can vary between **$-\pi$ and $+\pi$ rad** or **-180° and $+180^\circ$** .

The phase of periodic wave describes where the wave is in its cycle. δ is a phase constant.

The phase of a wave is dependant on position and time and allow us flexibility to describe the graph with a variety of different notations



Phase, Phase shift, and Phase difference

Phase Difference

- The phase of periodic wave describes where the wave is in its cycle.
- Phase difference is used to describe the phase position of one wave relative to another.
- At a glance, the **difference of phase position of the waves is called phase difference.**
Usually expressed by δ or Δ in angle.

Phase, Phase shift, and Phase difference

- **PHILIP SAYS:** The phase difference, $\Delta\phi$, can be thought of as the value between two different phases.

$$\Delta\phi = \phi_2 - \phi_1$$

- If we plug in the two phases we identified earlier, we can determine the phase constant.

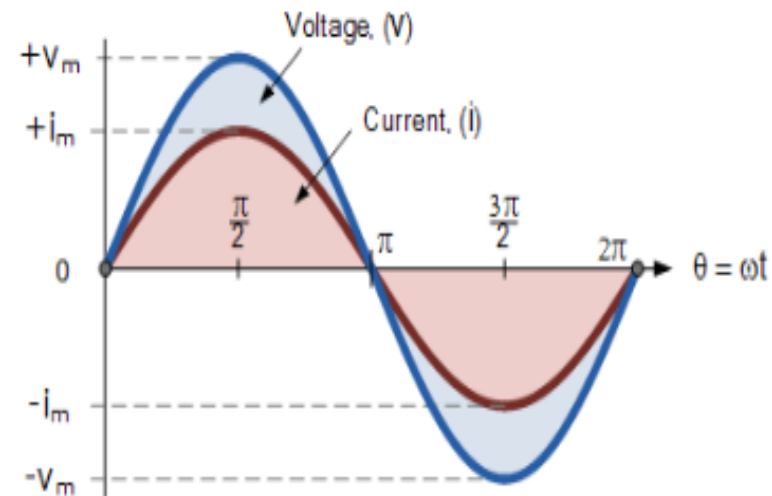
$$\begin{aligned}\Delta\phi &= (kx_0 + \omega t_2 + \phi) - (kx_0 + \omega t_1 + \phi) \\ &= \omega t_2 + \phi - \omega t_1 + \phi\end{aligned}$$

This represents an instance where two sinusoidal waves with different initial displacements are observed at one position over time, and we would like to know what the difference between their phases are.

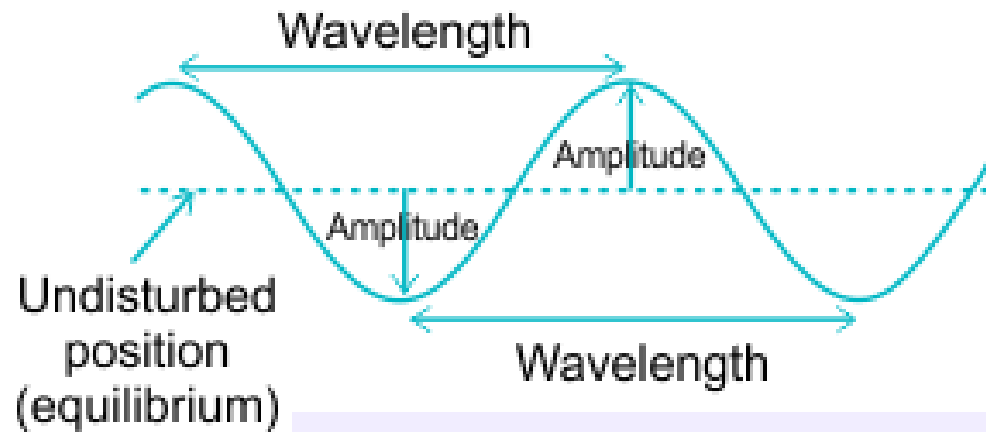
Phases are of two types- in phase and out of phase.

In Phase, waves are the waves which are 0° or 360° apart.

Out-of-phase waves are the waves that are 180° apart.

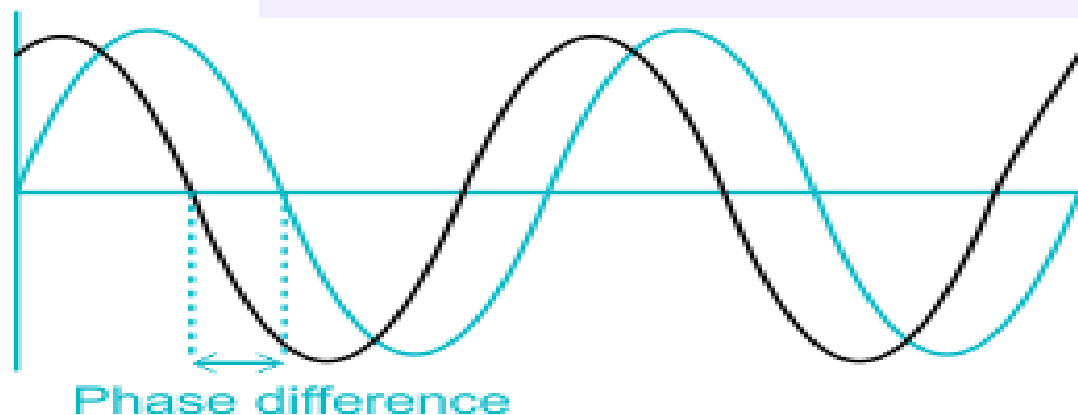


Phase Difference, Path Difference, and Time Difference Relation



The relation between
phase difference and
path difference

$$\frac{\Delta x}{\lambda} = \frac{\Delta \phi}{2\pi}$$



Phase Difference, Path Difference, and Time Difference Relation

Phase difference of 2π radian is equivalent to a path difference λ and a time difference of period T .

$$\text{Phase difference} = \frac{2\pi}{\lambda} \times \text{path difference}$$

$$\phi = \frac{2\pi}{\lambda} \times x \Rightarrow x = \frac{\lambda}{2\pi} \times \phi$$

$$\text{Phase difference} = \frac{2\pi}{T} \times \text{time difference}$$

$$\phi = \frac{2\pi}{T} \times t \Rightarrow t = \frac{T}{2\pi} \times \phi$$

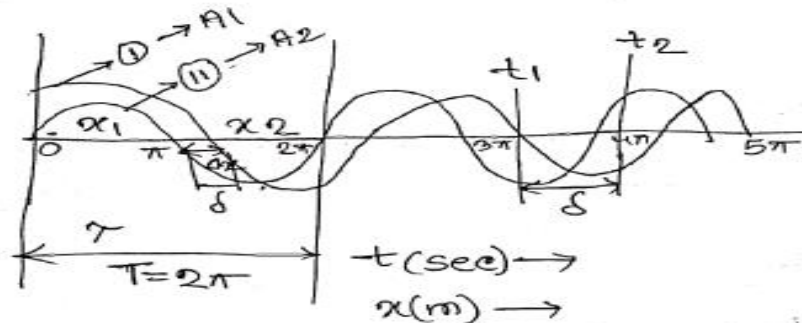
$$\text{Time difference} = \frac{T}{\lambda} \times \text{path difference}$$

$$t = \frac{T}{\lambda} \times x \Rightarrow x = \frac{\lambda}{T} \times t$$

Phase Difference Calculation Process

Suppose, $x = A \sin(\omega t + \delta)$

$\delta \rightarrow \Delta \rightarrow$ Phase difference $= \delta_2 - \delta_1$



Wave 1 - cosine wave
Wave 2 - sine wave

Fig-1

If λ distance covered then angle is 2π

$$1 \text{ --- } \frac{2\pi}{\lambda}$$

$$\therefore x \text{ --- } \frac{2\pi}{\lambda} \times x$$

$$\boxed{\delta = \frac{2\pi}{\lambda} \times \Delta x}$$

$\Delta x = x_2 - x_1$
= phase position difference
Phase difference

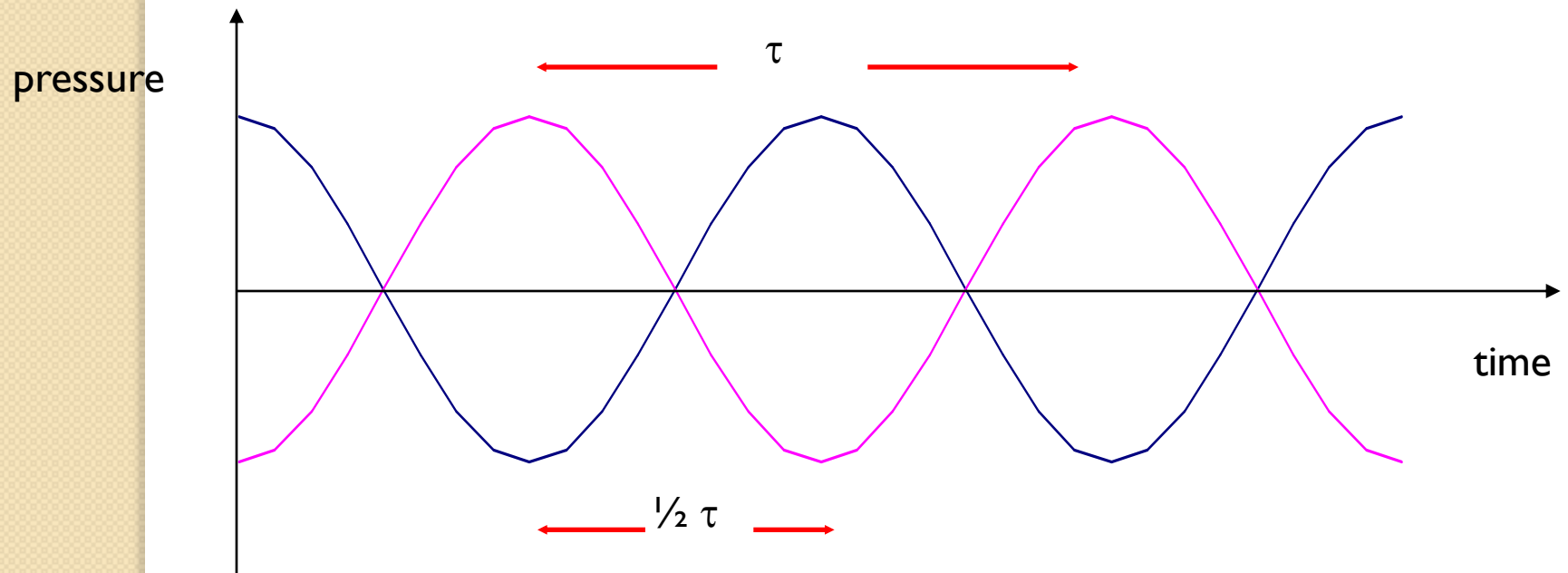
If T times covered, then angle is 2π

$$\text{If } 1 \text{ --- } \frac{2\pi}{T}$$

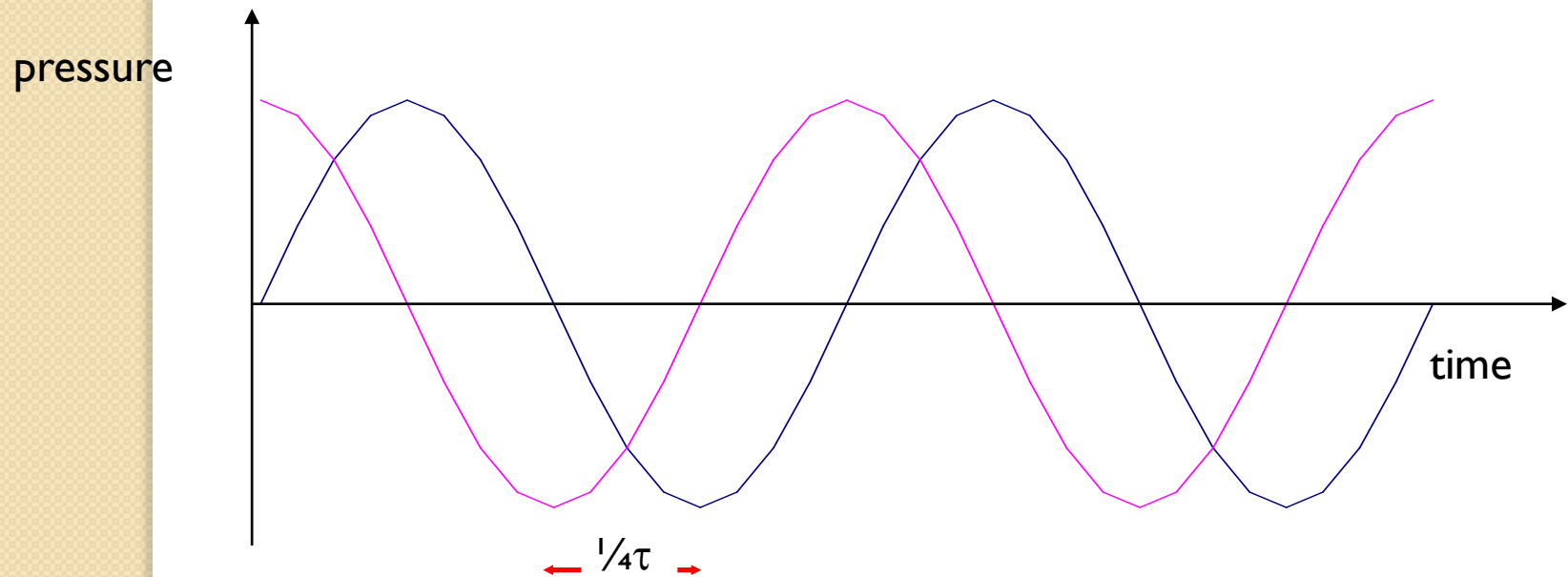
$$\therefore t \text{ --- } \frac{2\pi}{T} \times t$$

Phase difference $\therefore \boxed{\delta = \frac{2\pi}{T} \times \Delta t}$ $\Delta t = t_2 - t_1$
= time difference

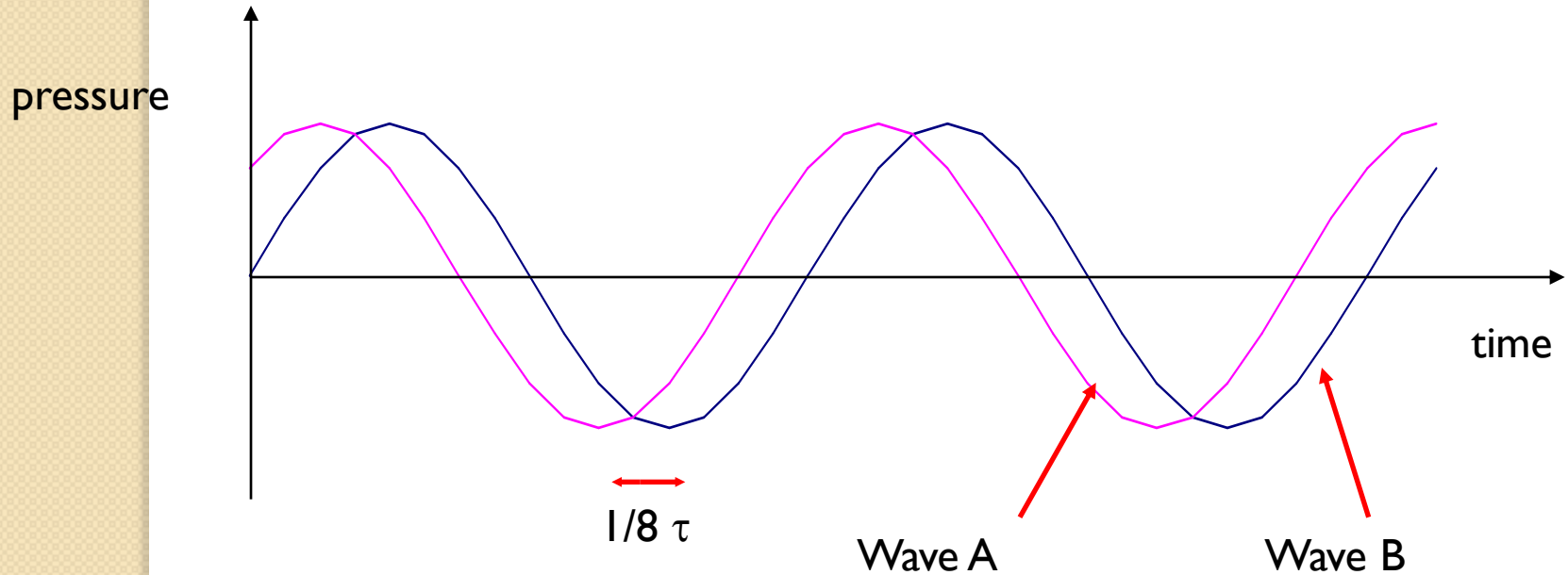
Phase Difference 180°



Phase Difference 90°



Phase Difference 45°



Phase, Phase shift, and Phase difference

$$y = A \sin(\omega t + \frac{\pi}{4}) \dots \dots \dots (1)$$

From Eqⁿ (1), the graph will start from $\omega t = -\frac{\pi}{4}$

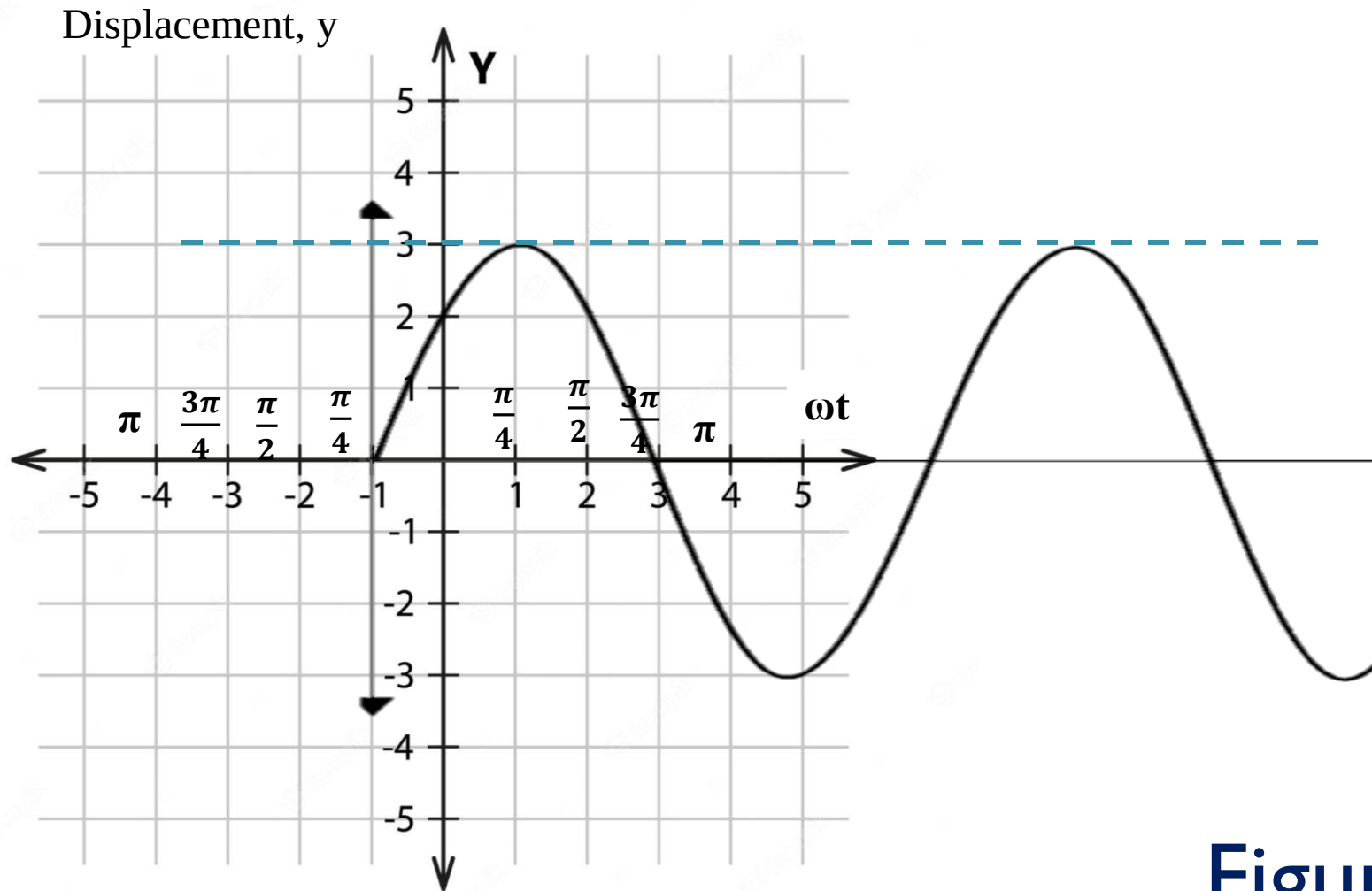


Figure: 01

Phase, Phase shift, and Phase difference

Displacement y vs. Angle ωt

$$\omega t = -\frac{\pi}{4}$$

$$\phi = \frac{\pi}{4}$$

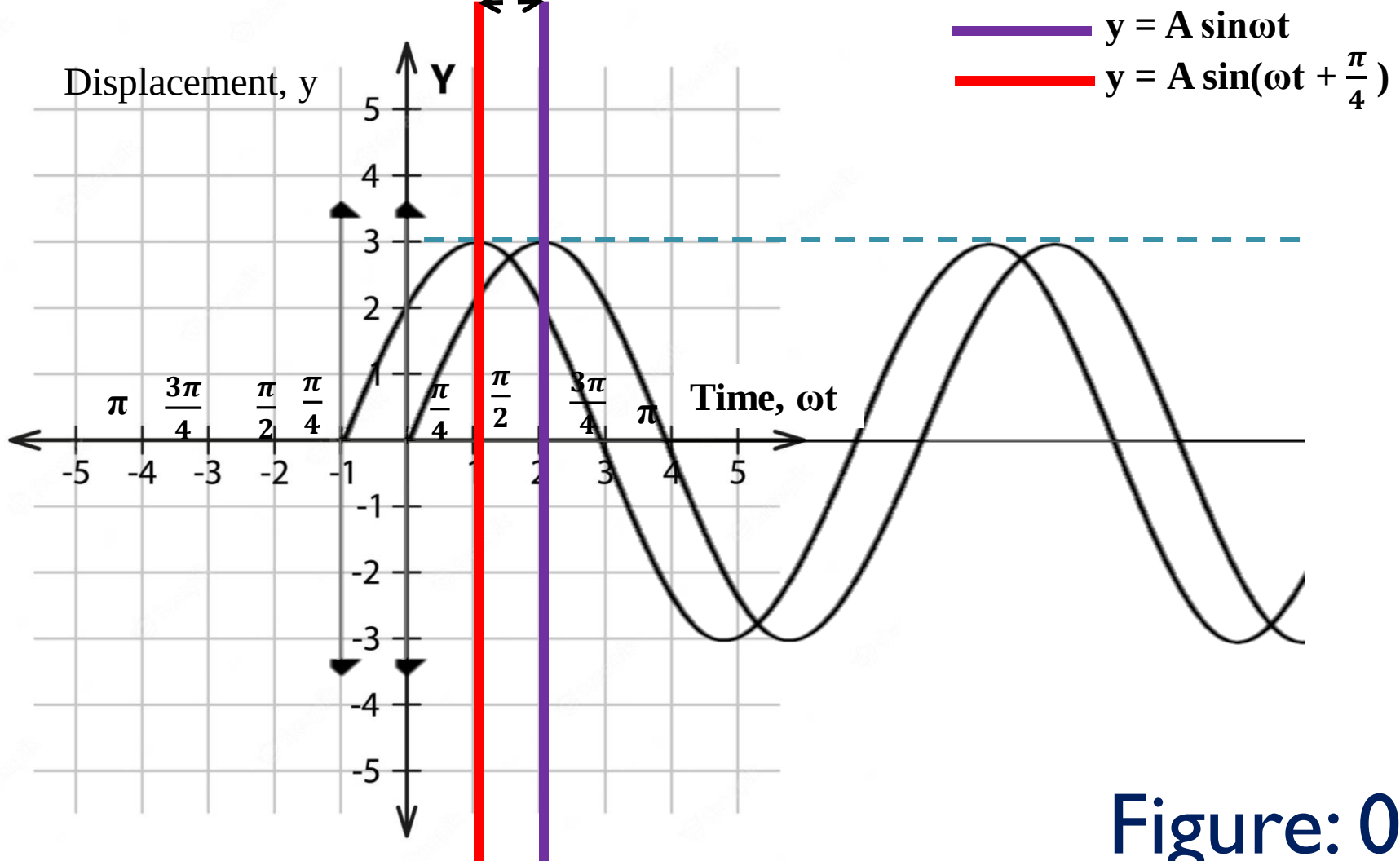


Figure: 01

Phase, Phase shift, and Phase difference

Displacement y vs. Time t

$$\omega t = -\frac{\pi}{4} \quad \phi = \frac{\pi}{4}$$

Displacement, y

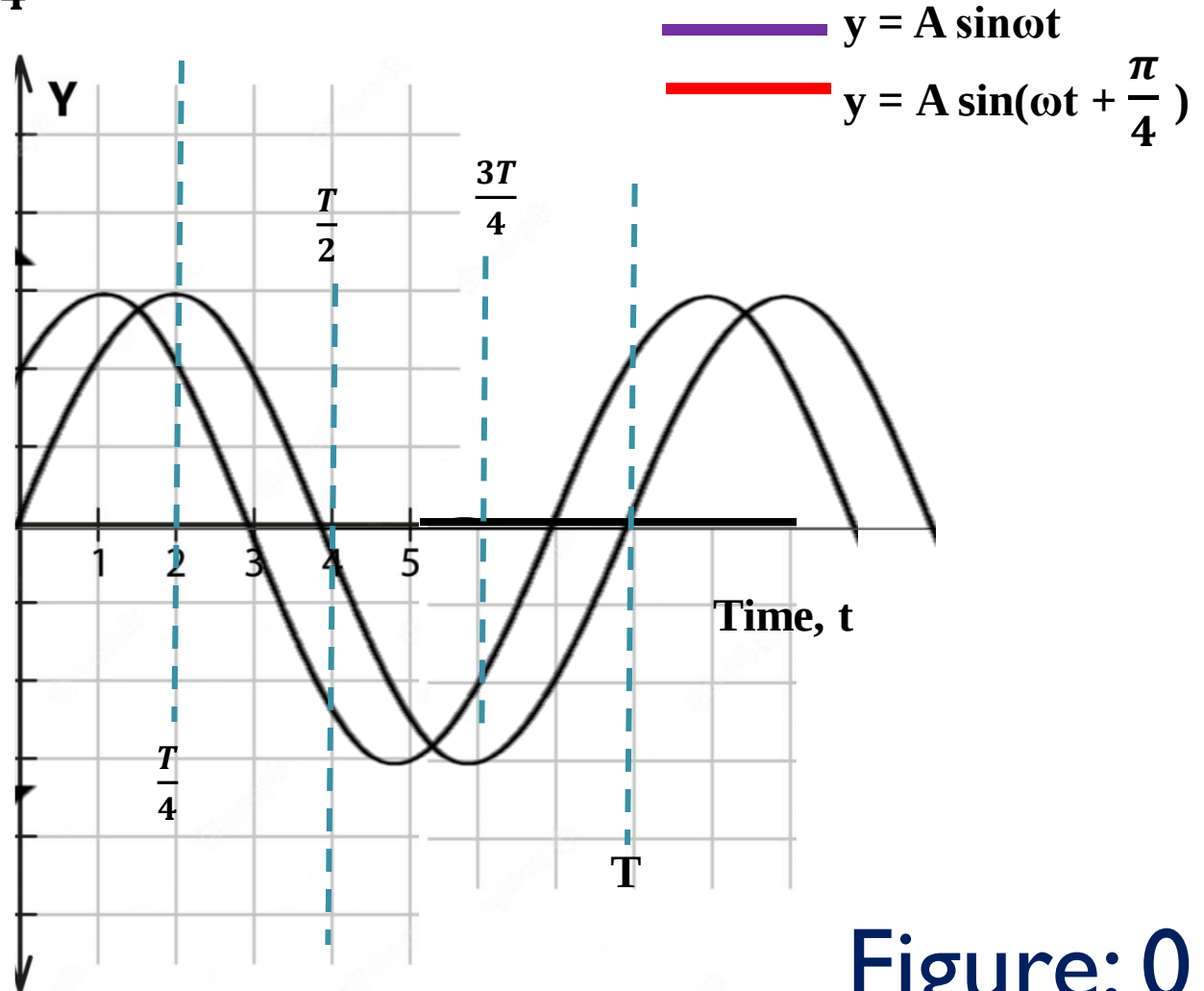


Figure: 01

Phase, Phase shift, and Phase difference

$$y = A \sin(\omega t - \frac{\pi}{4}) \dots \dots \dots (2)$$

From Eqⁿ (2), the graph will start from $\omega t = -(-\frac{\pi}{4}) = \frac{\pi}{4}$

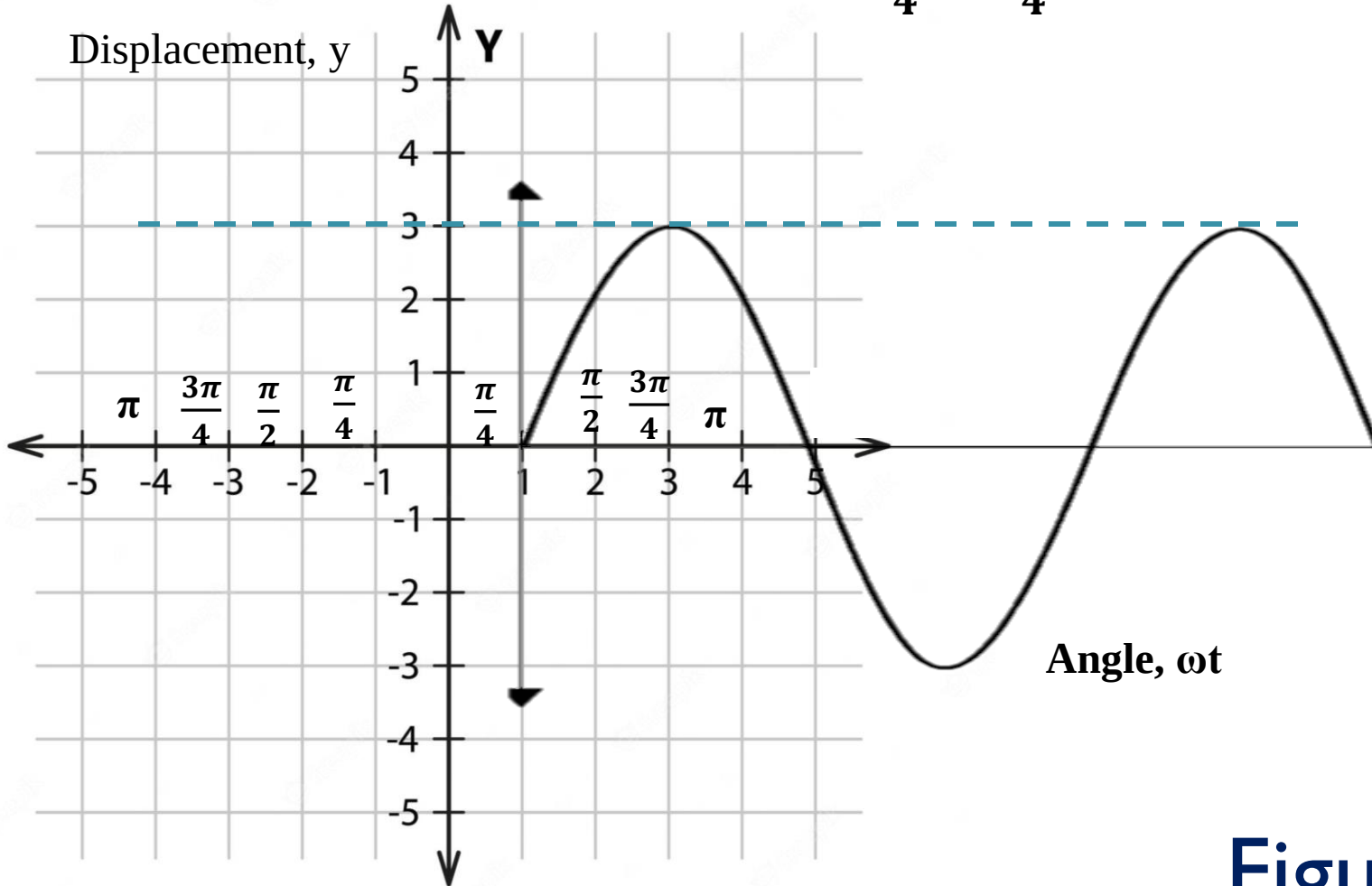


Figure: 02

Phase, Phase shift, and Phase difference

Displacement y vs. Angle ωt

$$\omega t = \frac{\pi}{4}$$

$$\phi = -\frac{\pi}{4}$$

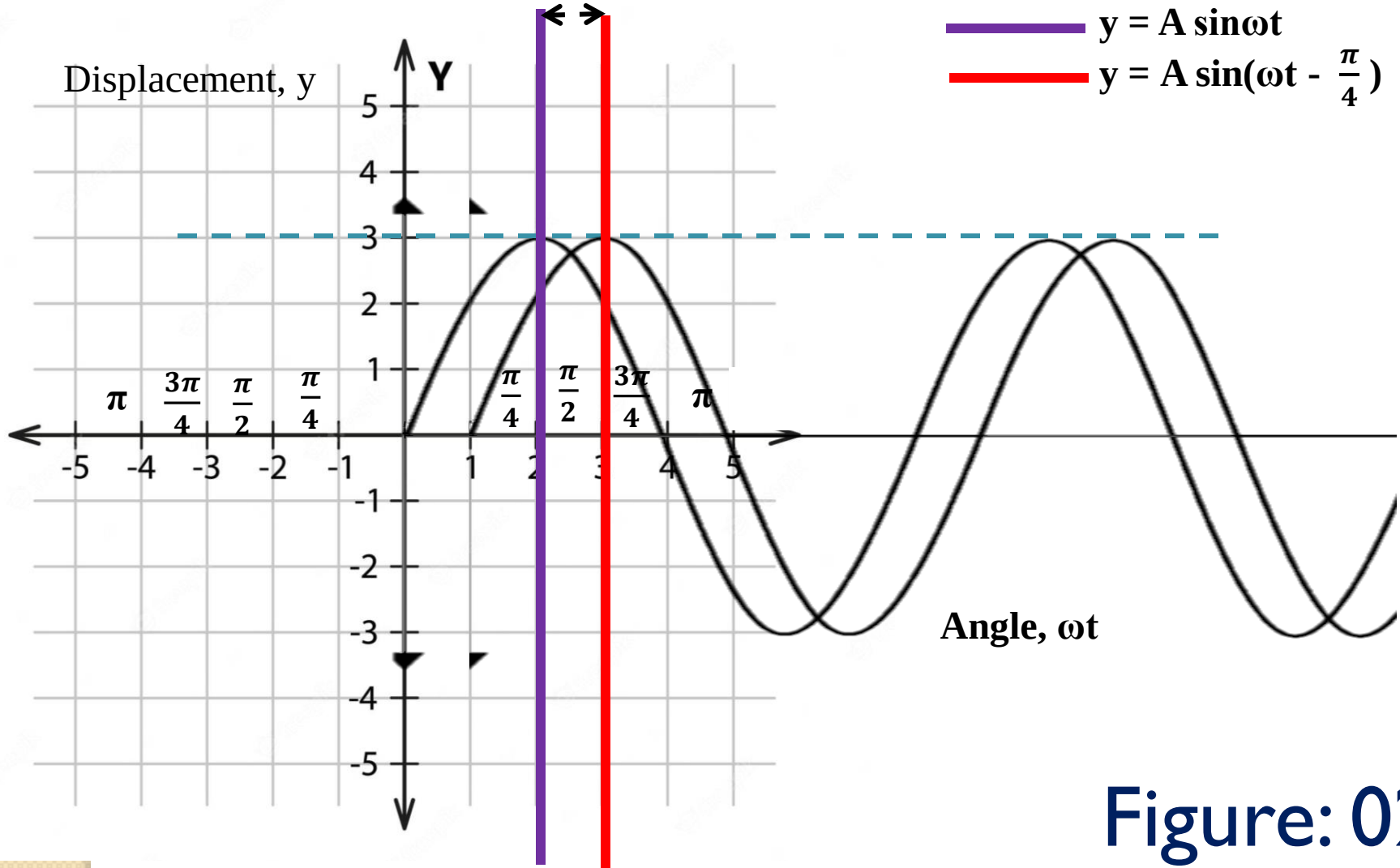


Figure: 02

Phase, Phase shift, and Phase difference

Displacement y vs. Time t

$$\omega t = \frac{\pi}{4}$$

$$\phi = -\frac{\pi}{4}$$

Displacement, y

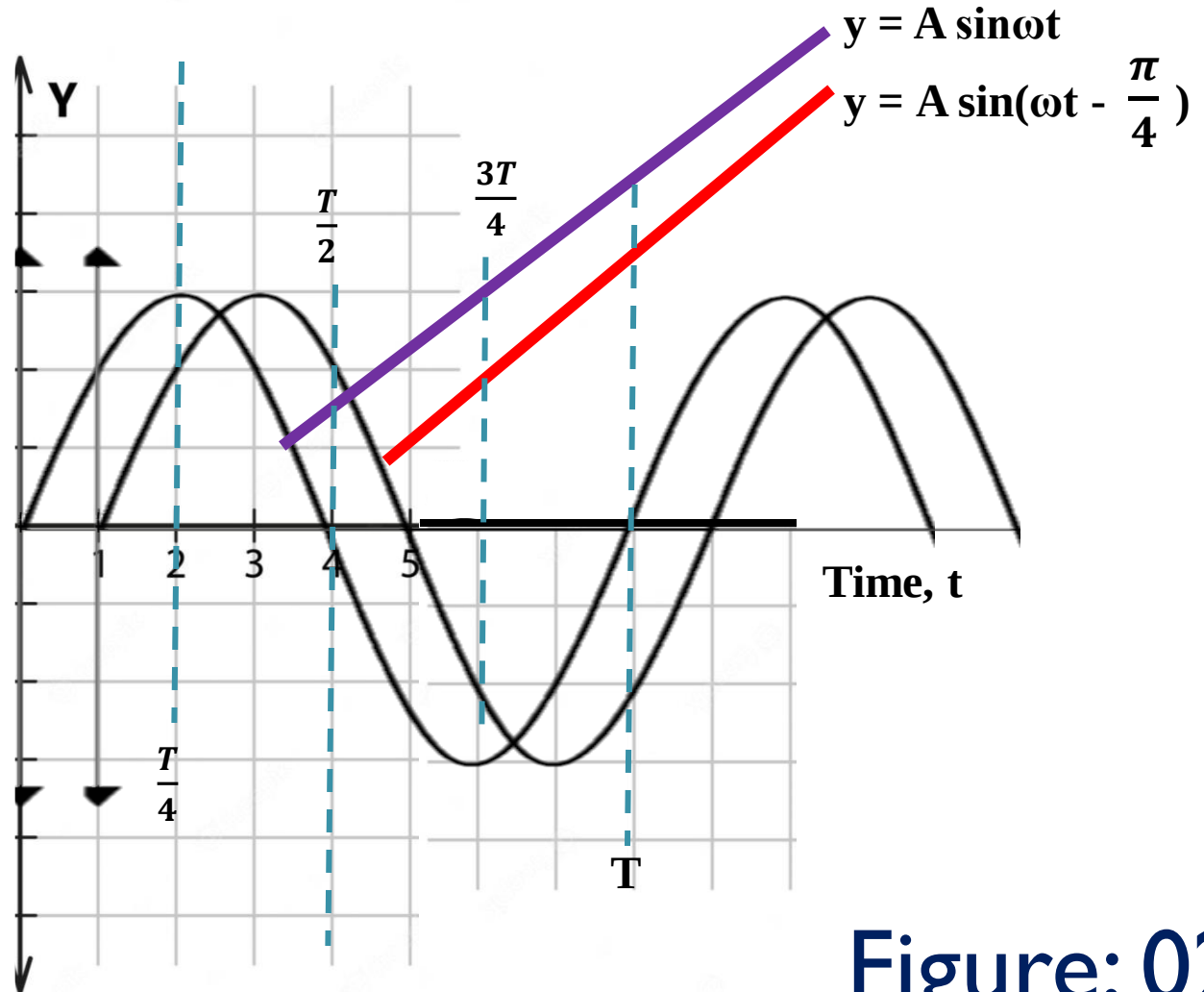


Figure: 02

Phase, Phase shift, and Phase difference

$$y = A \sin(\omega t + \frac{\pi}{2}) \dots \dots \dots (3)$$

From Eqⁿ (3), the graph will start from $\omega t = -(\frac{\pi}{2})$

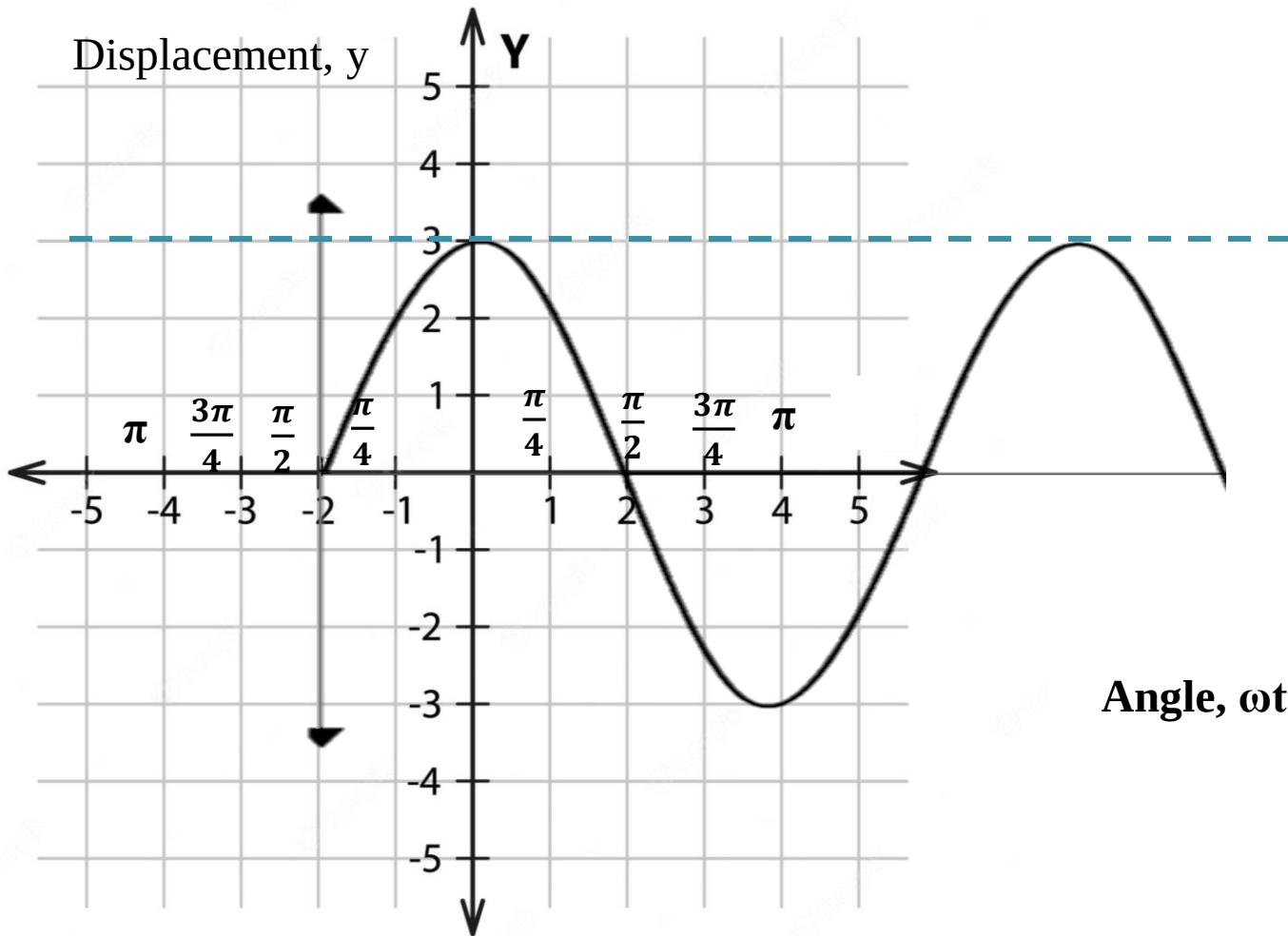


Figure: 03

Phase, Phase shift, and Phase difference

Displacement y vs. Angle ωt

$$\omega t = -\frac{\pi}{2}$$

$$\phi = \frac{\pi}{2}$$

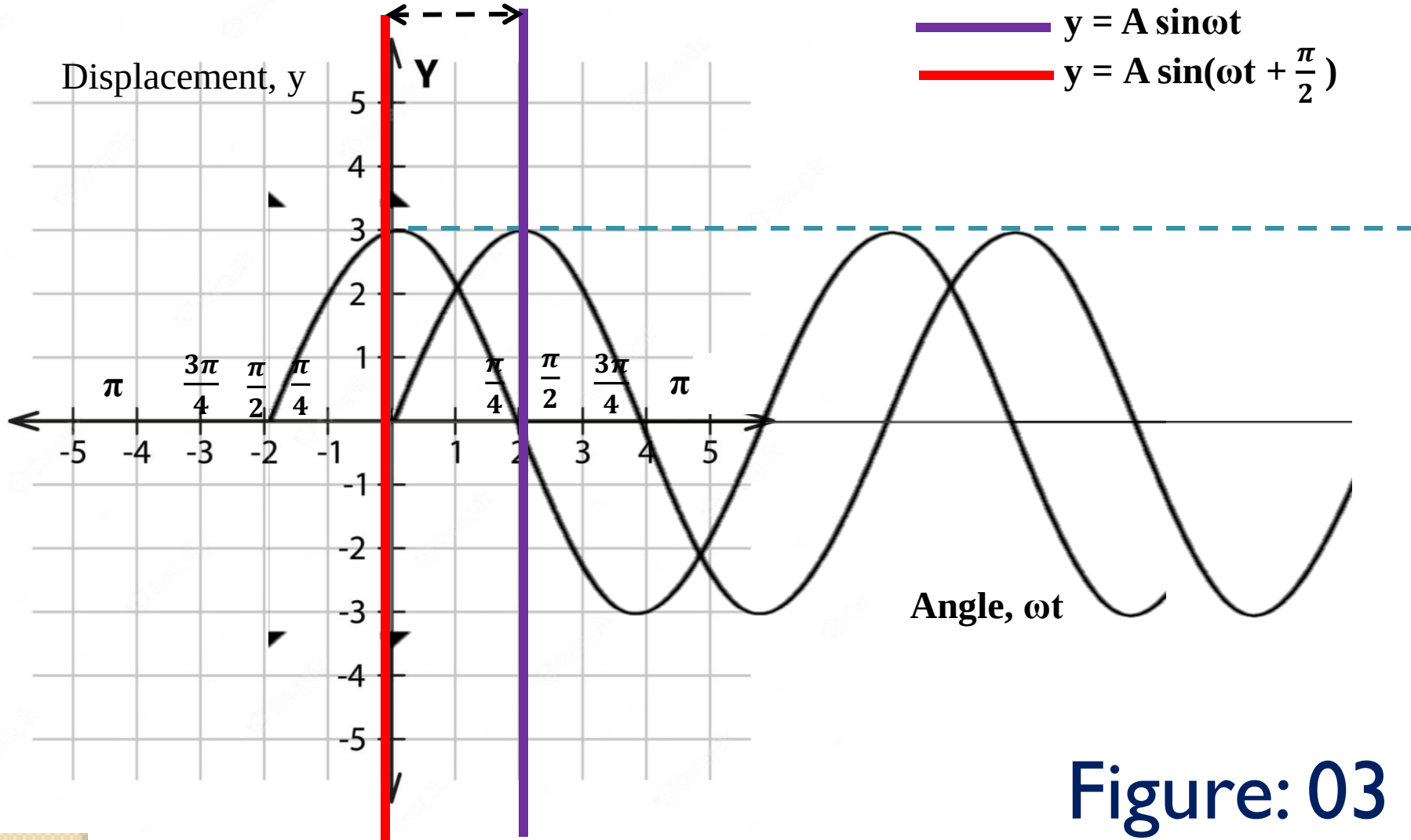


Figure: 03

Phase, Phase shift, and Phase difference

Displacement y vs. Time t

$$\omega t = -\frac{\pi}{2} \quad \phi = \frac{\pi}{2}$$

Displacement, y

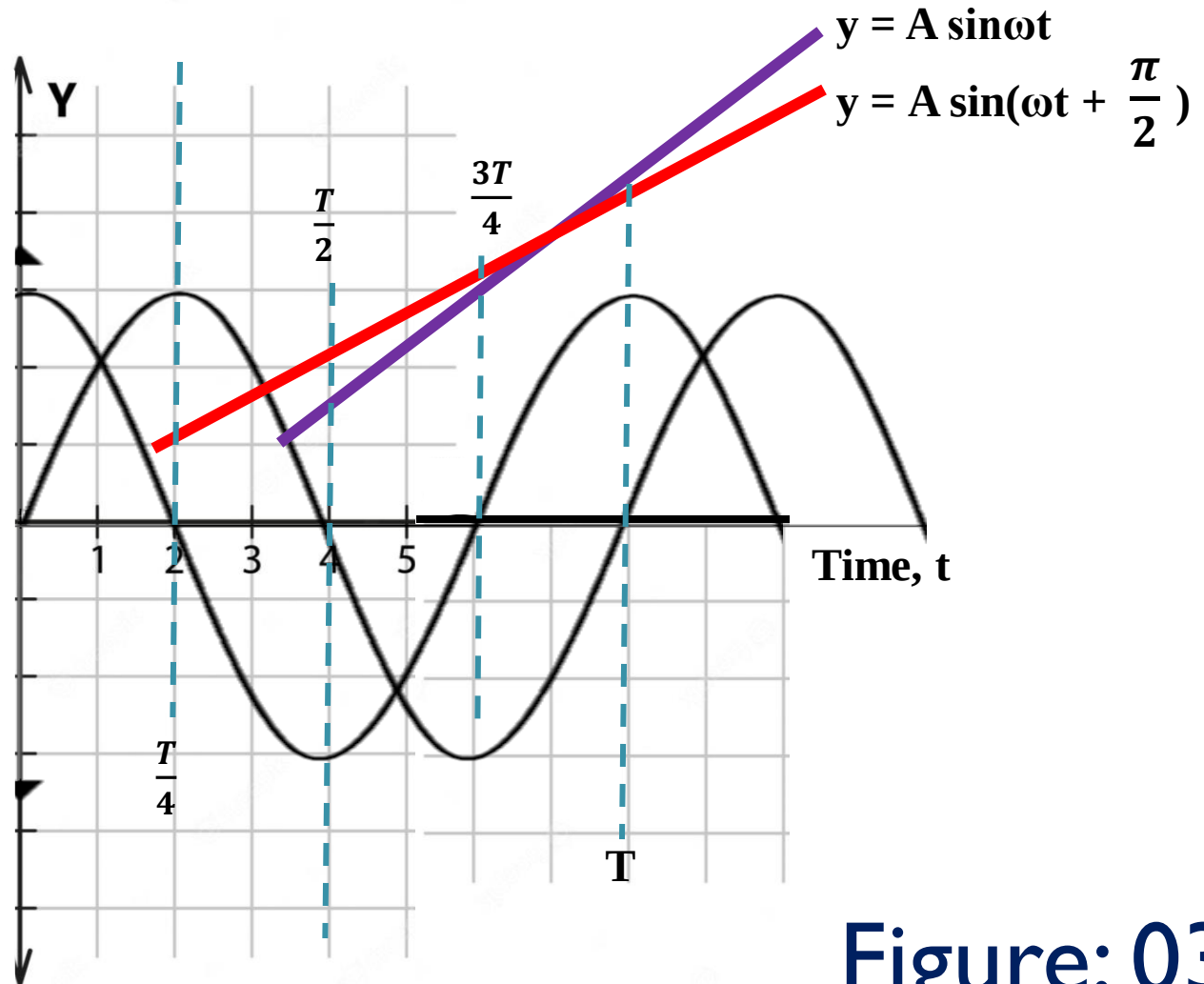


Figure: 03

Phase, Phase shift, and Phase difference

Now if we consider the particle with $y = A \sin(\omega t + \frac{\pi}{3})$(4)

Graph for y vs. ωt and the graph will start from $\omega t = -(\frac{\pi}{3})$ and the graph for eqn. (4) is

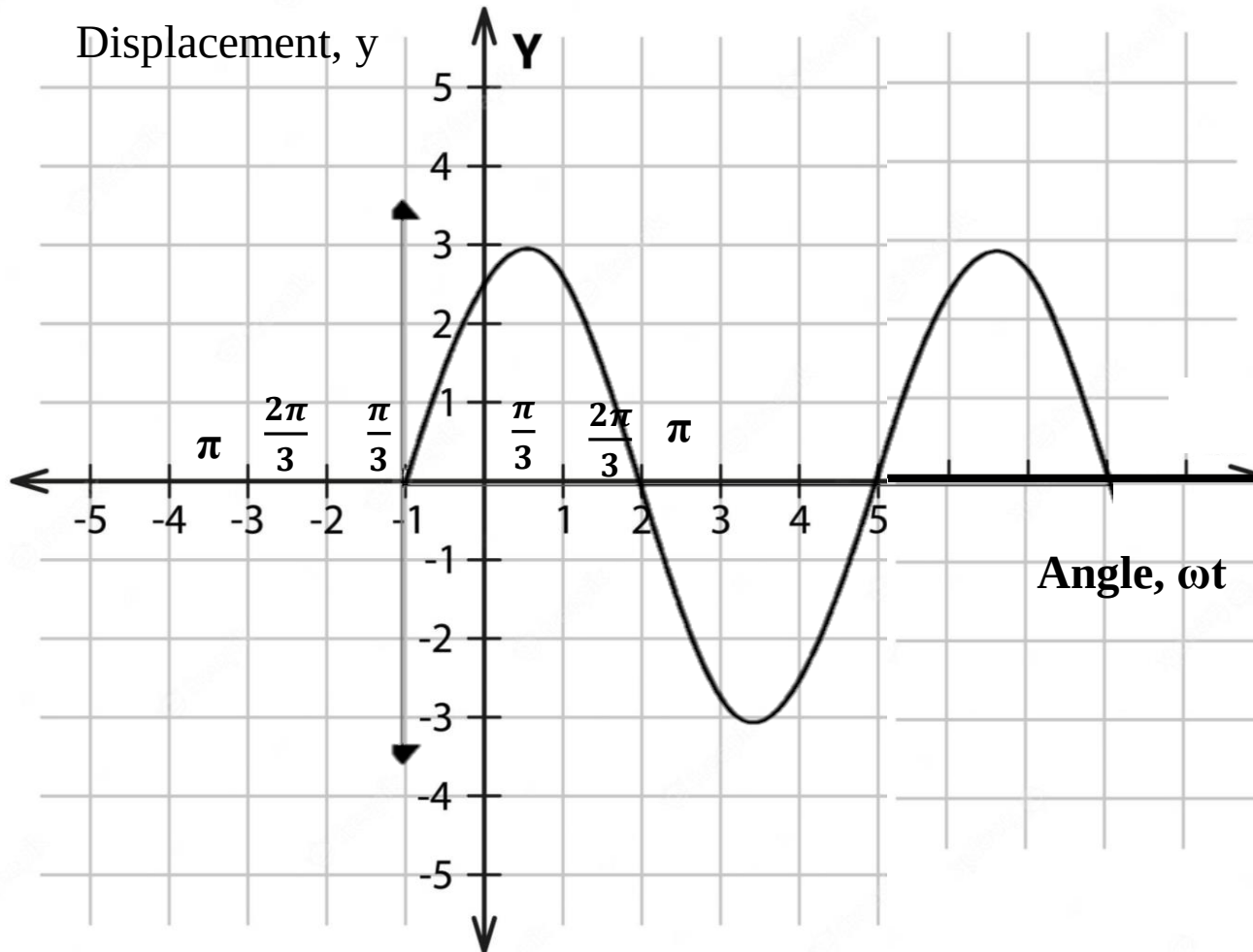


Figure: 04

Phase, Phase shift, and Phase difference

Displacement y vs. Angle ωt

$$\omega t = -\frac{\pi}{3}$$

$$\phi = \frac{\pi}{3}$$

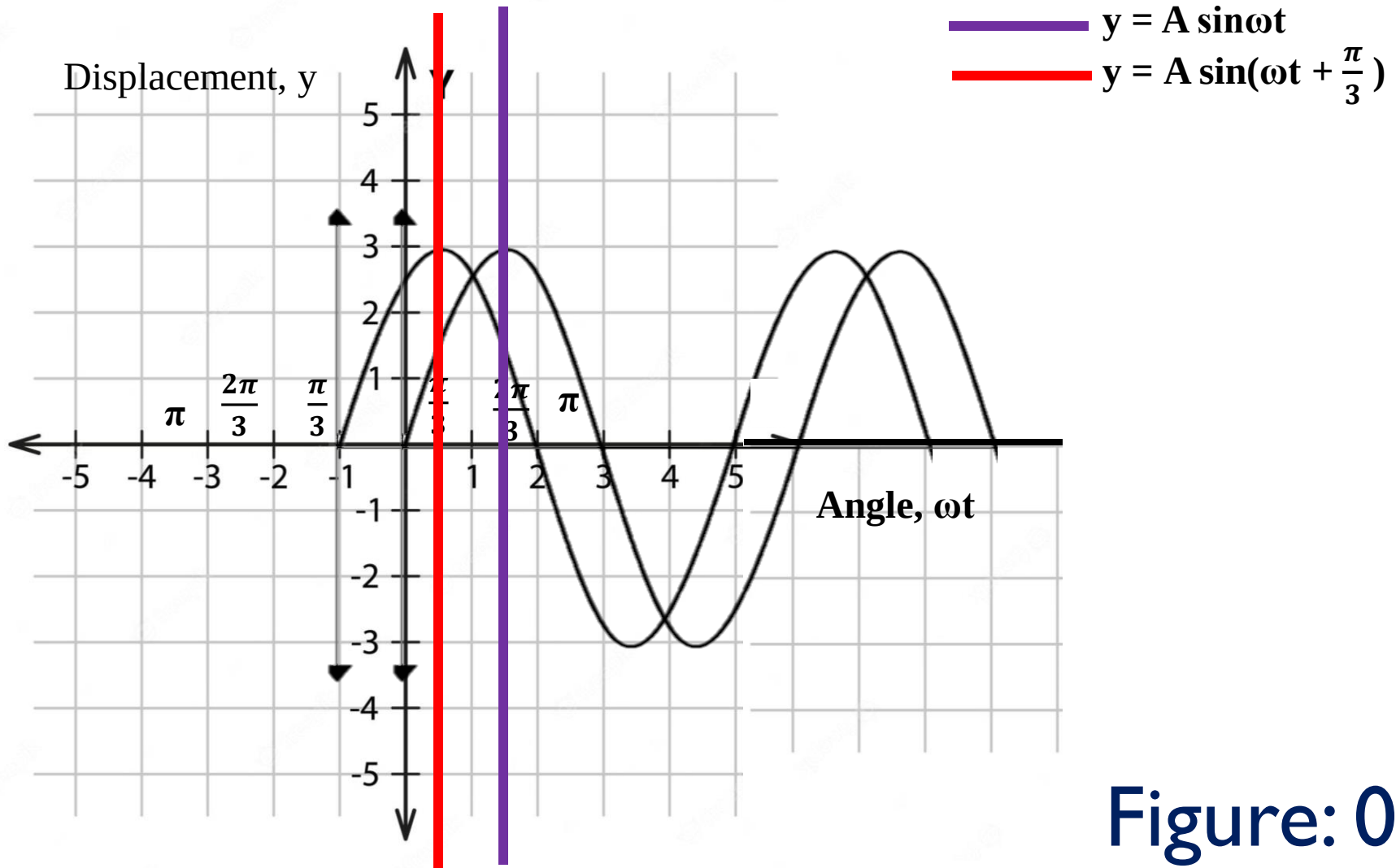


Figure: 04

Phase, Phase shift, and Phase difference

Displacement y vs. time t

$$\omega t = -\frac{\pi}{3}$$

$$\phi = \frac{\pi}{3}$$

Displacement, y

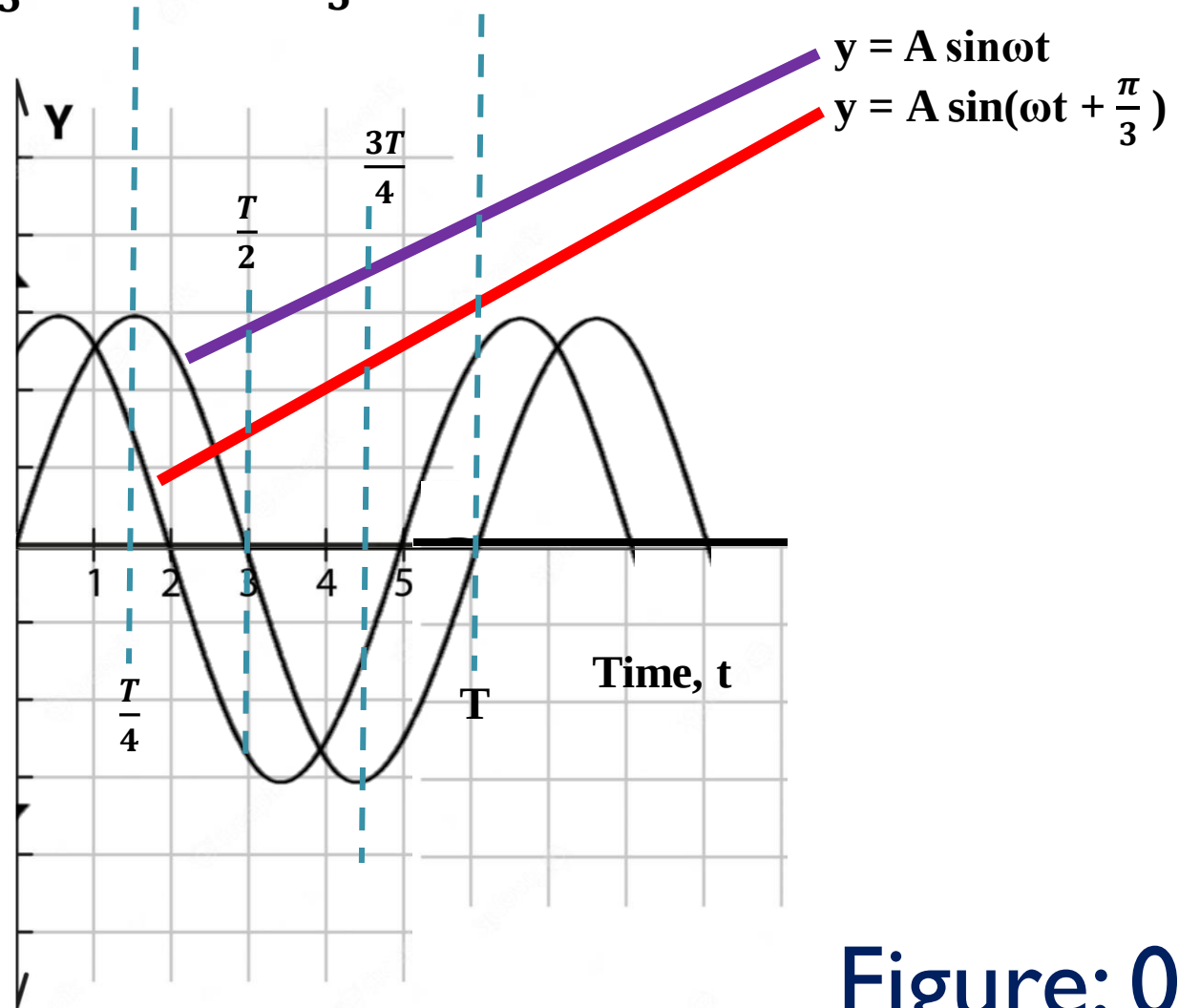


Figure: 04

Phase, Phase shift, and Phase difference

Example-0.3 : Some Practice Problems

$$(i) y = A \sin(\omega t - \frac{\pi}{3})$$

$$(ii) y = A \sin(\omega t - \frac{\pi}{2})$$

$$(iii) y = A \sin(\omega t + \pi)$$

$$(iv) y = A \sin(\omega t - \pi)$$

$$(v) y = A \sin(\omega t + \frac{3\pi}{4})$$

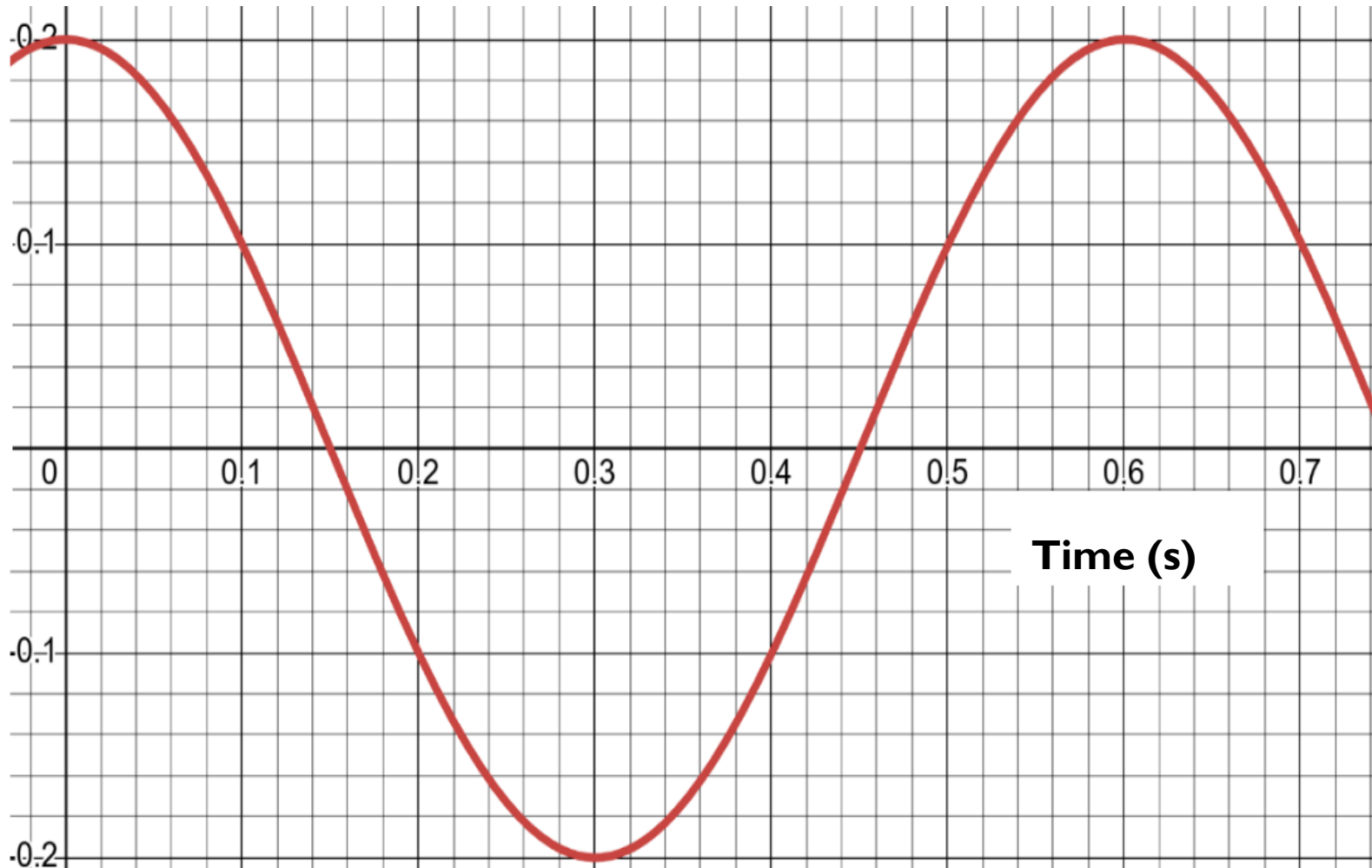
$$(vi) y = A \sin(\omega t - \frac{3\pi}{4})$$

$$(vii) y = A \sin(\omega t + \frac{3\pi}{2})$$

$$(viii) y = A \sin(\omega t + 2\pi)$$

Phase, Phase shift, and Phase difference

Example-0.4 :



Graph-1

Phase, Phase shift, and Phase difference

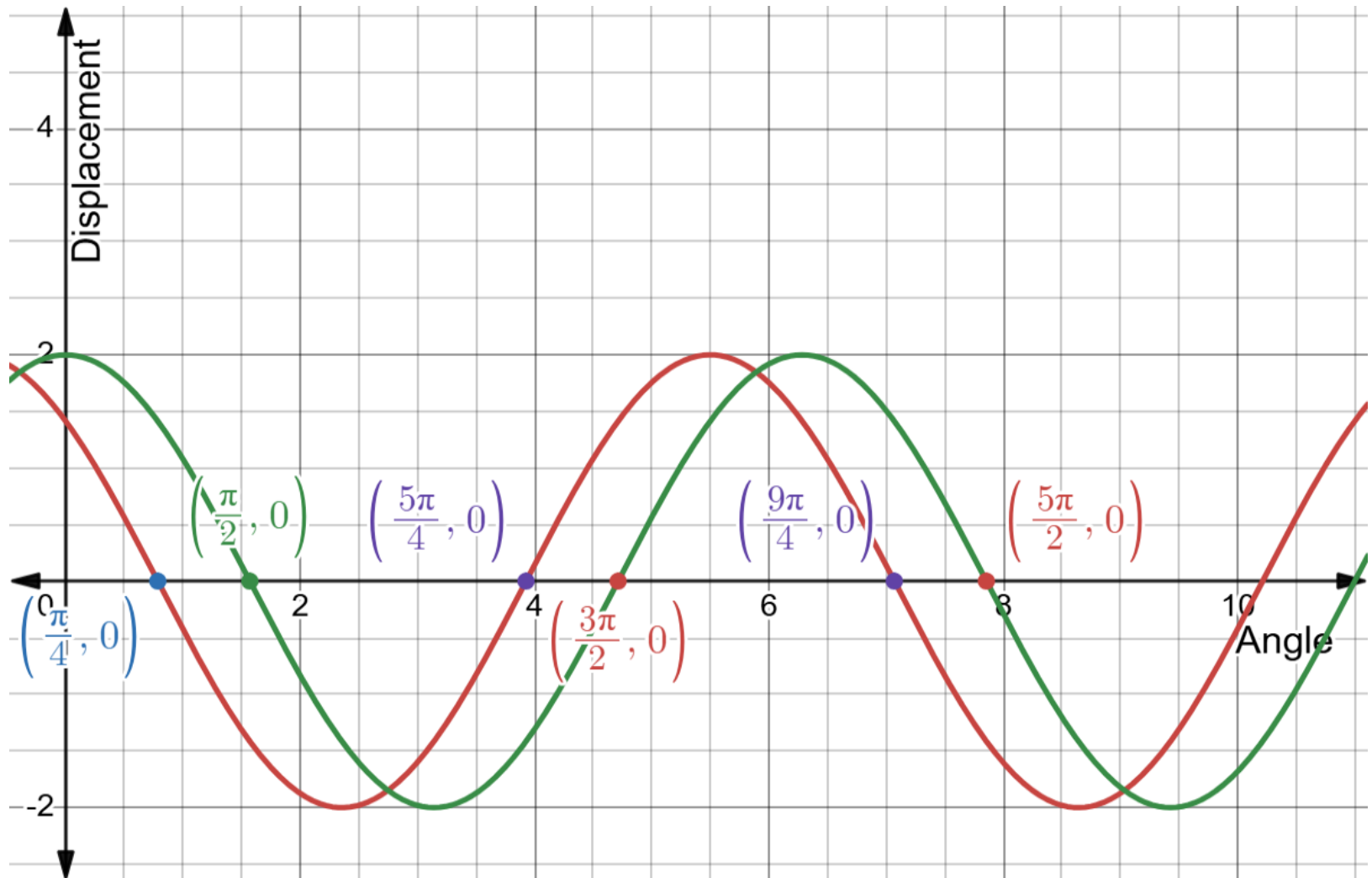
Example-0.4 : Position vs. time graph for a mass spring system oscillating in Simple Harmonic Oscillation is illustrated in Graph-1. The mass of the mass-spring system is 0.25 kg. Find out the (i) equation of displacement vs. time, (ii) velocity at $t = 2$ sec, (iii) acceleration at $t = 3$ sec, (iv) potential energy at $t = 0.25$ sec, (v) kinetic energy at $t = 4$ sec, (vi) total energy, (vii) maximum displacement, (viii) maximum velocity, (ix) maximum acceleration, (x) phase constant, (xi) angular frequency, (xii) frequency, (xiii) Time period, (xiv) velocity at $x=0.1$ m, (xv) equation of velocity vs. time, (xvi) velocity graph, (xvii) equation of acceleration vs. time, (xviii) acceleration graph, (xix) the kinetic energy at $x=14$ cm from the equilibrium point, (xx) the potential energy at $x=10$ cm from the equilibrium point, (xxi) mathematically calculate the position where the kinetic energy is 0, (xxii) mathematically calculate the position where the potential energy is 0, (xxiii) the position where the kinetic energy equal to potential energy, (xxiv) If figure represents the velocity graph, write down the velocity equation (Say, Graph-2), (xxv) Find out the acceleration equation and acceleration graph from the velocity equation (Graph-2), (xxvi) Find out the displacement equation and displacement graph from velocity equation (Graph-2),

Phase, Phase shift, and Phase difference

Example-0.4 : (xxvii) Find out the displacement equation and displacement graph from acceleration equation or acceleration graph (Graph-2), (xxviii) Find out the % Difference and % Accuracy of your obtained calculative answer with graphical answer (Graph-1) for phase constant, (xxix) If the graph mentioned in the figure has a phase constant $+\frac{\pi}{4}$, design the displacement equation and draw the corresponding displacement graph (Graph-3), (xxx) Find out the velocity equation and velocity graph of Graph-3, (xxxi) Find out the acceleration equation and acceleration graph of Graph-3, (xxxii) Compare the frequency of the two graphs (Graph-1 vs Graph-3), which one is greater? (xxxiii) If the angular frequency of the Graph-3 is double than that of Graph-1, compare the frequency of both ($f_1:f_2$)? (xxxiv) Compare the maximum velocity of the two waves (Graph-1 and Graph-3)? (xxxv) Compare the maximum acceleration of the two waves (Graph-1 and Graph-3)? (xxxvi) Compare the time period of two waves (Graph-1 and Graph-3)? (xxxvii) Find out the % Difference and % Accuracy of your obtained calculative answer with graphical answer (Graph-3) for phase constant, (xxxviii) Draw the Potential Energy, Kinetic Energy, and Total Energy, vs Time graph of Graph-1 for $x=0.1\text{m}$., and (xxxix) Draw the Potential Energy, Kinetic Energy, and Total Energy, vs Time graph of Graph-3 for $x=0.1\text{m}$.

Phase, Phase shift, and Phase difference

Example-0.5 :

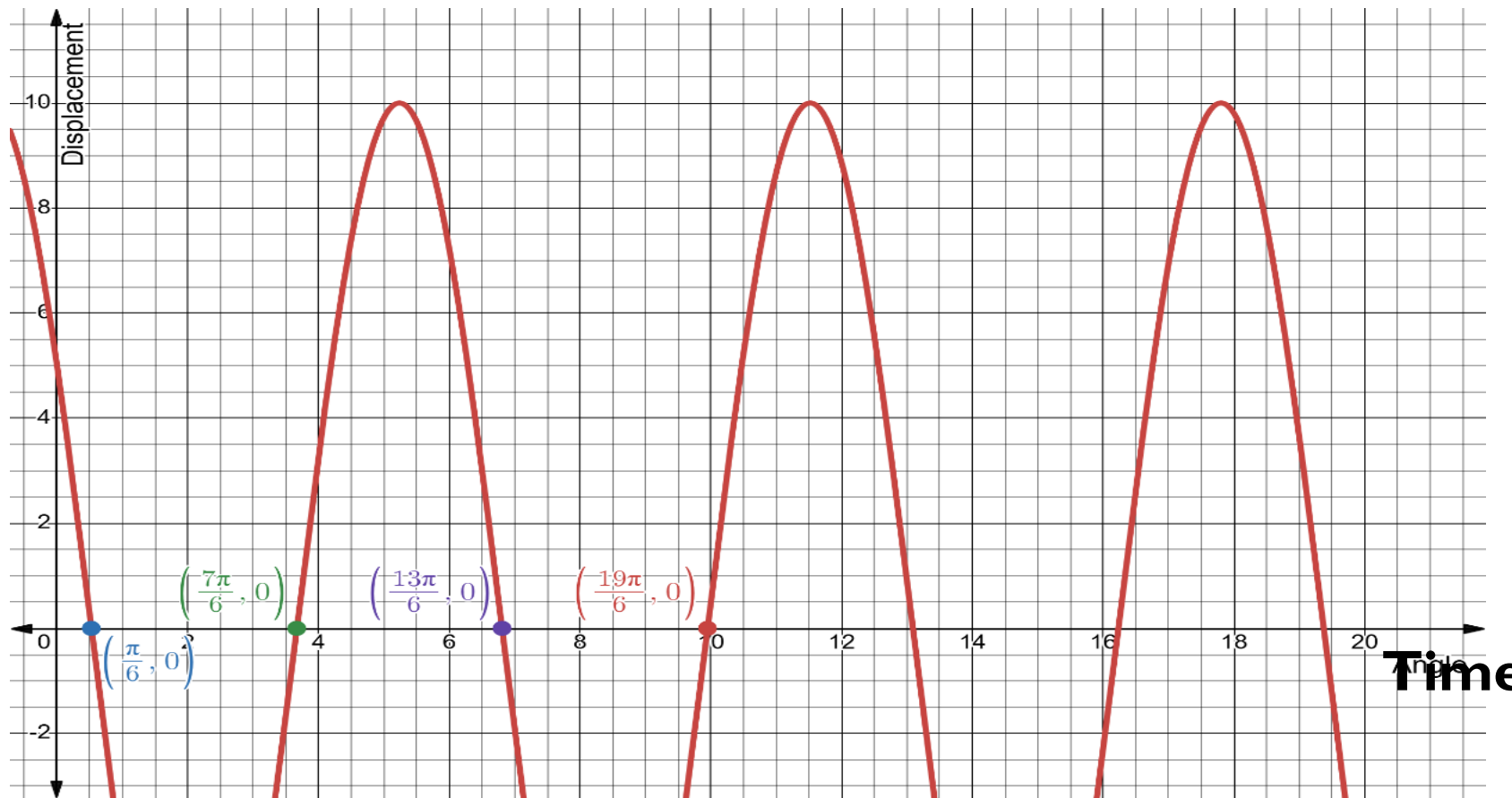


Phase, Phase shift, and Phase difference

Example-0.5 :

Construct the equation of displacement of two objects oscillating in simple harmonic oscillations and find out their phase difference according to Graph.

Example-0.6 :



Phase, Phase shift, and Phase difference

Example-0.6 :

Graph illustrates the displacement vs. time graph for a mass spring system oscillating in Simple Harmonic Oscillation. The mass of the mass-spring system is 500 g. (i) Find out the equation of displacement vs. time, (ii) Calculate the velocity at $t = 3$ sec, (iii) Calculate the acceleration at $t = 2$ sec, (iv) Determine the kinetic energy at $t = 0.5$ sec, (v) Determine the potential energy at $t = 0.8$ sec, and (vi) Determine the total energy.

Example-0.7 :

A block of mass $m = 680$ g is fastened to a spring whose spring constant k is 65 N/m. The block is pulled a distance of $x = 11$ cm from its equilibrium position at $x = 0$ on a frictionless surface and released from rest at $t = 0$.

- (a) What are the angular frequency, the frequency and the period of the resulting motion?
- (b) What is the amplitude of the motion?
- (c) What is the maximum speed of the block and the where the block when it has this speed?
- (d) What is the magnitude of a_{max} of the oscillation?
- (e) What is the phase constant?
- (f) What is the displacement function $x(t)$ for the spring-block system at $t = 1.5$ s?

Solution:

(a) Angular frequency

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{65 \text{ N/m}}{0.68 \text{ kg}}} = 9.78 \text{ rad/s} \approx 9.8 \text{ rad/s}$$

The frequency

$$\nu = \frac{\omega}{2\pi} = \frac{9.78 \text{ rad/s}}{2\pi \text{ rad}} = 1.56 \text{ Hz}$$

The time period

$$T = \frac{1}{\nu} = \frac{1}{1.56 \text{ Hz}} = 0.64 \text{ s} = 640 \text{ ms}$$

(b) The amplitude is the maximum displacement. So,

$$A = 11 \text{ cm} = 0.11 \text{ m}$$

(c) The maximum speed is

$$v_m = \omega A = (9.78 \text{ rad/s})(0.11 \text{ m}) = 1.1 \text{ m/s}$$

The maximum speed occurs whwn the block crosses the equilibrium position i.e. $x = 0$.

(d) The maximum acceleration

$$a_m = \omega^2 A = (9.78 \text{ rad/s})(0.11 \text{ m}) = 11 \text{ m/s}^2$$

(e) The phase constant ϕ can be evaluted in the following way

$$x(t) = A \cos(\omega t - \phi)$$

$$\text{At } t = 0, x = A, \quad \text{So}$$

$$1 = \cos \phi \quad \therefore \quad \phi = \cos^{-1} 1 = 0 \text{ rad}$$

(f)

$$x(t) = A \cos(\omega t - \phi) = (0.11 \text{ m})(\cos(9.8 \times 1.5 - 0)) = -0.059 \text{ m}$$

Example-0.8 :

An object undergoing simple harmonic motion takes 0.25 s to travel from one point of zero velocity to the next such point. The distance between those points is 36 cm. Calculate the (i) period, (ii) frequency, and (iii) amplitude of the motion.

Example-0.9:

What is the maximum acceleration of a platform that oscillates at amplitudes 2.20 cm and frequency 6.60 Hz.

Example-0.10:

Given $x(t) = 6 \cos\left(2.5t - \frac{\pi}{3}\right)$. Find the velocity of the body and acceleration at $t = 1.4\text{s}$.

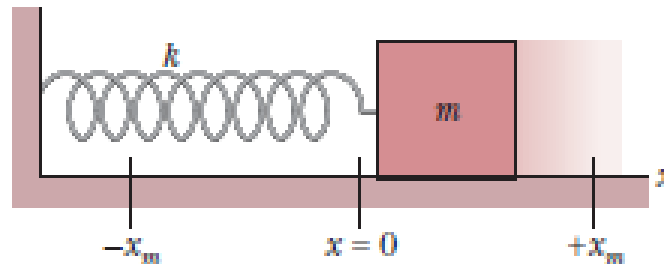
Example-0.11:

A body executes SHM such that its velocity at mean position is 1 m/s and acceleration at one extremity is 1.57 m/s^2 . Calculate time period of oscillation.

Example-0.12:

At $t = 0$, the displacement $x(0)$ of the block in a linear oscillator is -8.50 cm (Read $x(0)$ as “ x at time zero.”). The block’s velocity $v(0)$ then is 0.920 m/s, and its acceleration $a(0)$ is 47.0 m/s².

(a) What is the angular frequency ω of this system? (b) What are the phase constant ϕ and amplitude x_m ?



Example-0.13:

Suppose a spring block-system moves between top and bottom point of a tall buildings as a mass dampers. The block has mass $m=2.72 \times 10^5$ kg and designed to oscillate at a frequency $f=10$ Hz with amplitude $x_m=20$ cm. Calculate (i) the mechanical energy, (ii) what is the block speed as it passes through the equilibrium point?

Example-0.14:

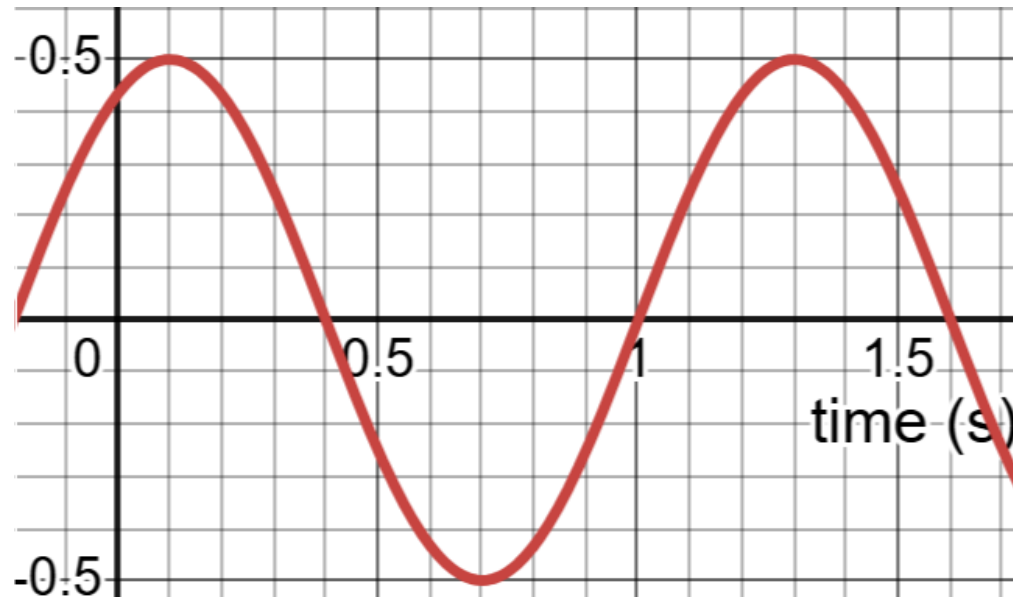
An oscillator consists of a block attached to a spring ($k = 460$ N/m). At some time t , the position (measured from the system's equilibrium location), velocity, and acceleration of the block are $x=0.150$ m, $v = -15$ m/s, and $a = -125$ m/s². Calculate (i) the frequency of oscillation, (ii) the mass of the block, and (iii) the amplitude of the motion.

Example-0.15 :

A particle executes SHM of amplitude 8 m when the particle is 4 m from its mean position, its acceleration is found to be 80 m/s^2 . Find (i) velocity (ii) time period and (iii) maximum velocity.

Example-0.16 :

Construct the equation of displacement from the below Graph which illustrates the displacement vs. time graph for a mass spring system oscillating in simple harmonic motion. If the mass of the system is 500g, calculate the (i) potential energy of the system at $t = 0.25 \text{ sec}$, (ii) velocity at $t = 0.5 \text{ sec}$, (iii) acceleration at $t = 0.9 \text{ sec}$, and (iv) displacement at $t = 2 \text{ sec}$.



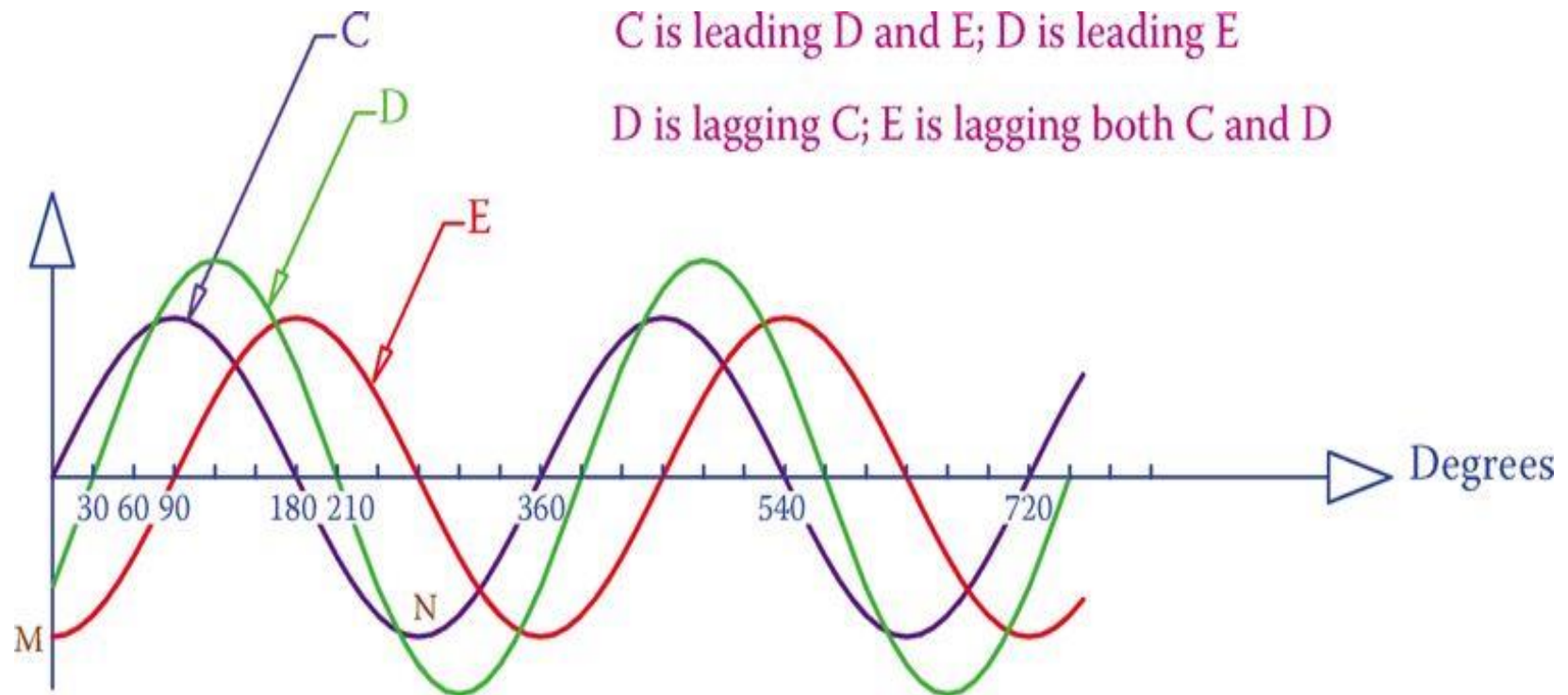
Leading and Lagging of a wave:

Phase leading and Phase lagging

The wave that cuts the origin first i.e., of the horizontal axis, the wave is called leading wave.

On the other hand, lagging is just the opposite of leading.

In this phenomena, the former one is called leading wave and the later one is called lagging wave.



- **“Amplitude A calculation” related to displacement, velocity, and frequency from wave equation** or ‘Amplitude A calculation’ from wave equation

Simple harmonic oscillator

At $t = 0$, write $x = x_0$ and $v = v_0$.

Then at $t = 0$:

$$x_0 = A \cos(\phi)$$

$$v_0 = -\omega_0 A \sin(\phi)$$

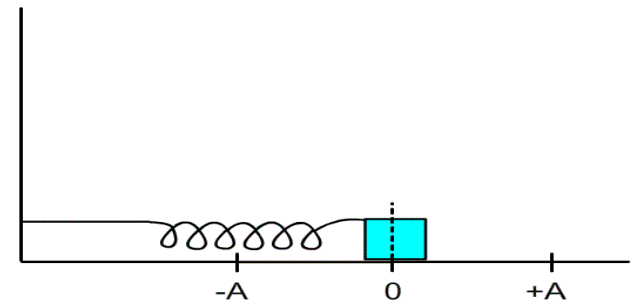
$$\left. \begin{array}{l} x_0 = A \cos(\phi) \\ v_0 = -\omega_0 A \sin(\phi) \end{array} \right\} \tan \phi = -\frac{v_0}{\omega_0 x_0}$$

$$\dots \text{ and } x_0^2 + \left(\frac{v_0}{\omega_0} \right)^2 = A^2 \cos^2(\phi) + A^2 \sin^2(\phi) = A^2$$

$$\therefore A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0} \right)^2}$$

Example-1: A block of mass 680gm is fastened to a spring of spring constant 65N/m . The block is pulled a distance 11 cm from its equilibrium on a frictionless table and released.

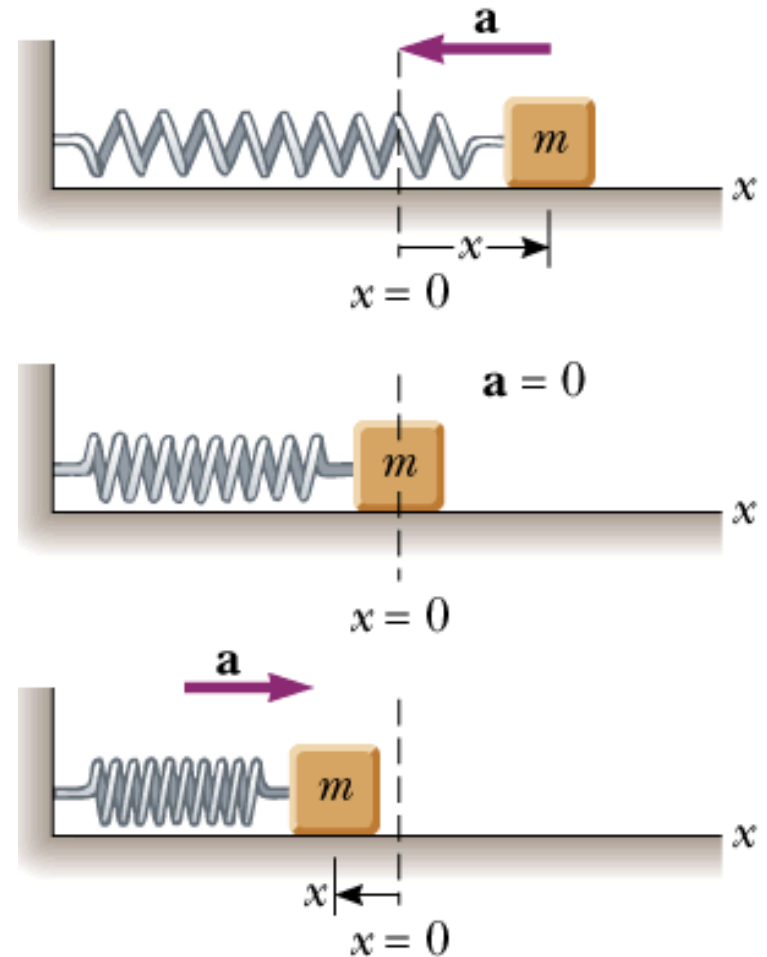
- (a) What are the angular frequency, the frequency, and the period of the motion?
- (b) What is amplitude of the motion?
- (c) What is the maximum speed of the block?



Example-2 :

A spring stretches by 3.90 cm when a 10.0 g mass is hung from it. A 25.0 g mass attached to this spring oscillates in simple harmonic motion.

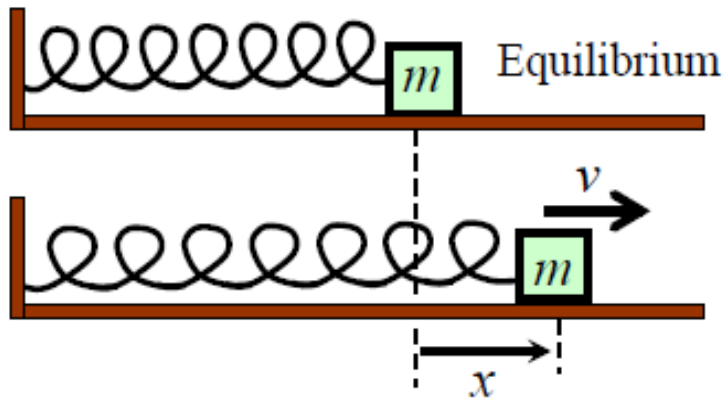
- (a) Calculate the period of the motion.
- (b) Calculate frequency and the angular velocity of the motion.



□ *Derive the differential equation of SHM and find its solution (Mechanical system, Energy Approach) : Alternative Method-3*

SHM with Mechanical System (Energy Approach):

Mass-spring oscillator: an energy approach



Suppose that the mass has a speed v when it has displacement x

Kinetic energy of mass = $\frac{1}{2}mv^2$

Potential energy of spring = $\int_0^x F dx' = \int_0^x kx' dx' = \frac{1}{2}kx^2$

There are no dissipative mechanisms in our model (no friction).
... the total energy of the mass-spring system is conserved.

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{constant}$$

Alternative Method-3

Mass-spring oscillator: an energy approach ...2

SHM with Mechanical System (Energy Approach):

For our mass-spring system: $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{constant}$

$$\therefore \frac{d}{dt} \left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2 \right) = 0$$

$$\therefore mv \frac{dv}{dt} + kx \frac{dx}{dt} = 0$$

$$\therefore mv \frac{dv}{dt} + kxv = 0$$

$$\therefore m \frac{dv}{dt} + kx = 0$$

$$\therefore \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

... as before

□ Total Energy of SHM calculation and its Graphical Representation: Mass-spring oscillator: an energy approach ...3

For the mass-spring system: $x = A \cos(\omega_0 t + \phi)$

$$\text{Potential energy} = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega_0 t + \phi)$$

$$\text{k.e.} = \frac{1}{2} m v^2 = \frac{1}{2} m [-A \omega_0 \sin(\omega_0 t + \phi)]^2 = \frac{1}{2} m A^2 \omega_0^2 \sin^2(\omega_0 t + \phi)$$

Total energy = p.e. + k.e

$$= \frac{1}{2} k A^2 \cos^2(\omega_0 t + \phi) + \frac{1}{2} m A^2 \omega_0^2 \sin^2(\omega_0 t + \phi)$$

$$= \frac{1}{2} k A^2 \quad (= \frac{1}{2} m \omega_0^2 A^2) \quad (\because E \propto A^2)$$

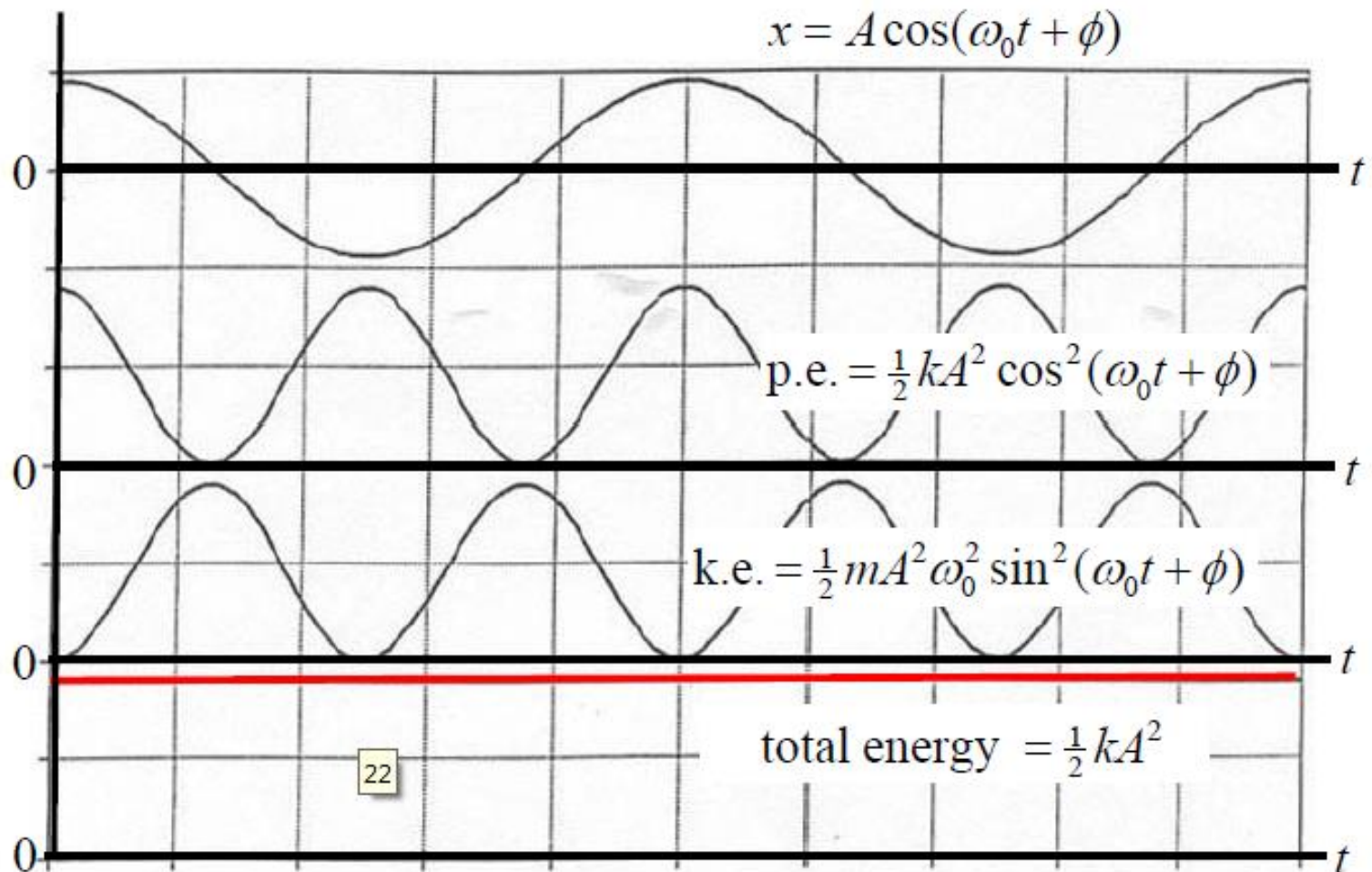
We can now write: $\frac{1}{2} k x^2 + \frac{1}{2} m v^2 = \frac{1}{2} k A^2$

$$\therefore v = \pm \sqrt{\frac{k}{m} (A^2 - x^2)} \quad \text{or} \quad v(x) = \pm \omega_0 \sqrt{A^2 - x^2}$$

(useful)

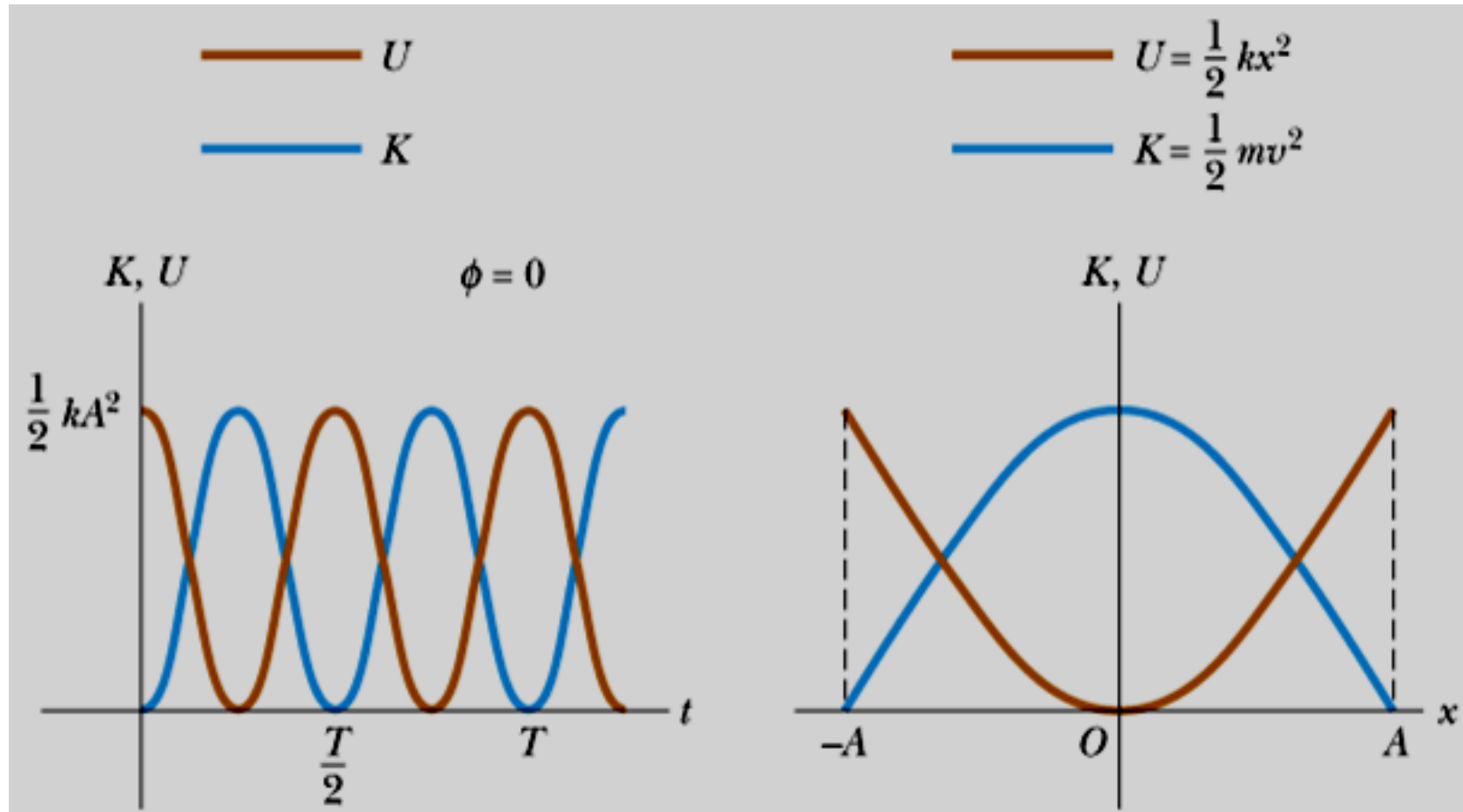
Total Energy or Mechanical energy of SHM calculation and its Graphical Representation:

Energy of the mass-spring simple harmonic oscillator



Total Energy of SHM calculation and its Graphical Representation:

K.E and P.E of SHM



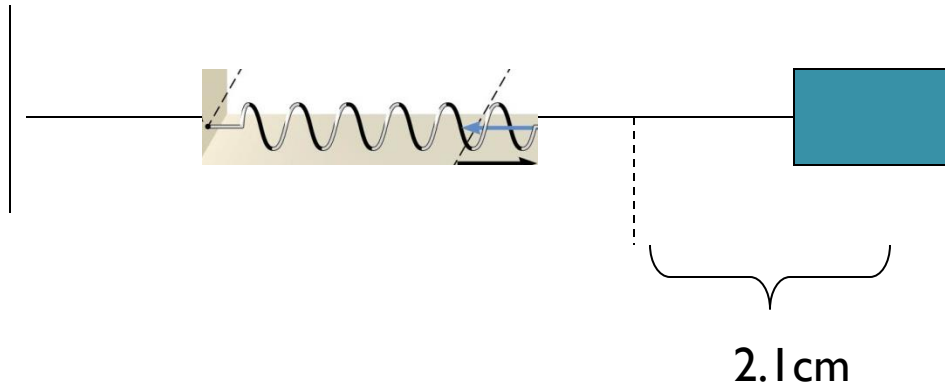
E vs. t graph

E vs. x graph

Check Your Understanding

Example-3: A 0.42-kg block is attached to the end of a horizontal ideal spring and rests on a frictionless surface. The block is pulled so that the spring stretches by 2.1 cm relative to its unstrained length. When the block is released, it moves with an acceleration of 9.0 m/s^2 . What is the spring constant of the spring?

180 N/m



$$kx = ma$$

$$k \times \frac{2.1}{100} = 0.42 \times 9.0 m / s^2$$

$$k = \frac{0.42 \times 9.0}{2.1} \times 100 = 180 N / m$$

Energy in the SHO

Energy calculations.

Example-4: For the simple harmonic oscillation where $k = 19.6 \text{ N/m}$, $A = 0.100 \text{ m}$, $x = -(0.100 \text{ m}) \cos 8.08t$, and $v = (0.808 \text{ m/s}) \sin 8.08t$, determine (a) the total energy, (b) the kinetic and potential energies as a function of time, (c) the velocity when the mass is 0.050 m from equilibrium, (d) the kinetic and potential energies at half amplitude ($x = \pm A/2$).

Solution:

a. $E = \frac{1}{2}kA^2 = \frac{1}{2} \cdot 19.6\text{N/m} \cdot (0.100\text{m})^2 = 9.80 \times 10^{-2} \text{ J}.$

b. $U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2 \omega t = (9.80 \times 10^{-2} \text{ J}) \cos^2 8.08t,$

$$K = E - U = (9.80 \times 10^{-2} \text{ J}) \sin^2 8.08t.$$

Solution:

$$\text{c. } K = E - U, \quad \frac{1}{2}mv^2 = \frac{1}{2}kA^2 - \frac{1}{2}kx^2,$$

$$v = \sqrt{\frac{k}{m}(A^2 - x^2)} = \omega\sqrt{A^2 - x^2}$$

$$= 8.08\text{Hz} \cdot \sqrt{(0.100\text{m})^2 - (0.050\text{m})^2} = 0.70\text{m/s}.$$

$$\text{d. } U = \frac{1}{2}kx^2 = \frac{1}{2}k\left(\pm \frac{A}{2}\right)^2 = \frac{1}{4}E = 2.5 \times 10^{-2} \text{ J},$$

$$K = E - U = 7.3 \times 10^{-2} \text{ J}.$$

Using Conservation of Energy

Example-5: A 500 g block on a spring is pulled a distance of 20 cm and released. The subsequent oscillations are measured to have a period of 0.80 s. At what position (or positions) is the speed of the block 1.0 m/s?

$$T = 0.80 \text{ s} \quad \text{so} \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{(0.80 \text{ s})} = 7.85 \text{ rad/s}$$

$$v = \sqrt{\frac{k}{m}(A^2 - x^2)} = \omega\sqrt{A^2 - x^2}$$

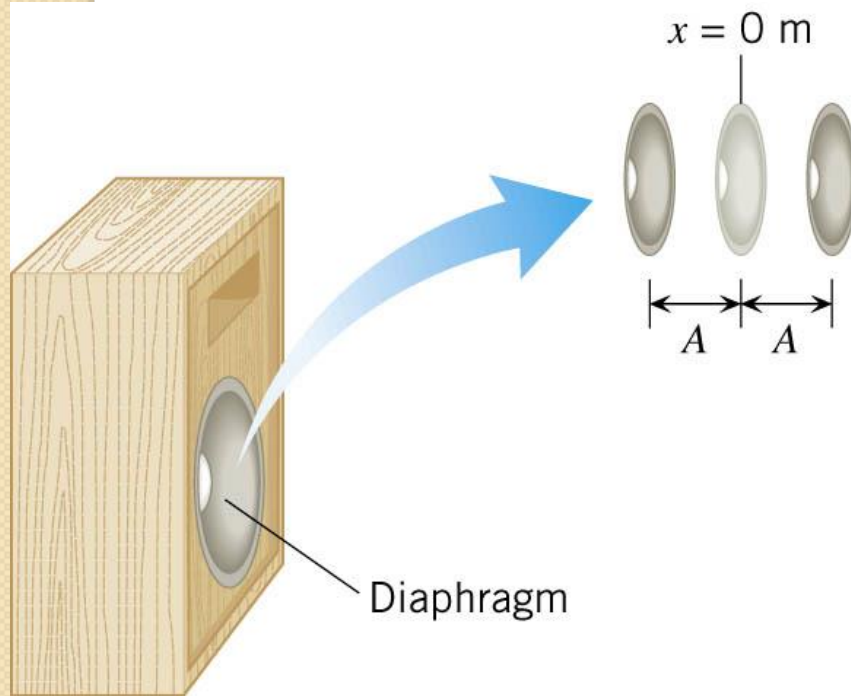
$$x = \pm\sqrt{A^2 - \left(\frac{v}{\omega}\right)^2} = \pm\sqrt{(0.20 \text{ m})^2 - \left(\frac{(1.0 \text{ m/s})}{(7.85 \text{ rad/s})}\right)^2} = \pm 0.154 \text{ m} = \pm 15.4 \text{ cm}$$

Using Conservation of Energy

Example-0.17: Suppose a spring block-system starts moving from the equilibrium as we apply force on it. The block has mass $m = 6.4 \text{ kg}$ and is designed to oscillate with an angular frequency $\omega = 56 \text{ rads}^{-1}$ with amplitude 15 cm. Calculate (i) the kinetic energy at $x = 14 \text{ cm}$ from the equilibrium point and (ii) mathematically calculate the position where the kinetic energy is 0.

Example: The Maximum Speed of a Loudspeaker Diaphragm

Example-6: The diaphragm of a loudspeaker moves back and forth in simple harmonic motion to create sound. The frequency of the motion is $f = 1.0$ kHz and the amplitude is $A = 0.20$ mm.



- (a) What is the maximum speed of the diaphragm?
- (b) Where in the motion does this maximum speed occur?

(a)

$$v_{\max} = A\omega = A(2\pi f) = (0.20 \times 10^{-3} \text{ m})(2\pi)(1.0 \times 10^3 \text{ Hz}) = \boxed{1.3 \text{ m/s}}$$

(b) The speed of the diaphragm is zero when the diaphragm momentarily comes to rest at either end of its motion: $x = +A$ and $x = -A$. Its maximum speed occurs midway between these two positions, or at $x = 0 \text{ m}$.

Example: Radio Station Frequency and Period

Example-7: What is the oscillation period of an FM radio station that broadcasts at 100 MHz?

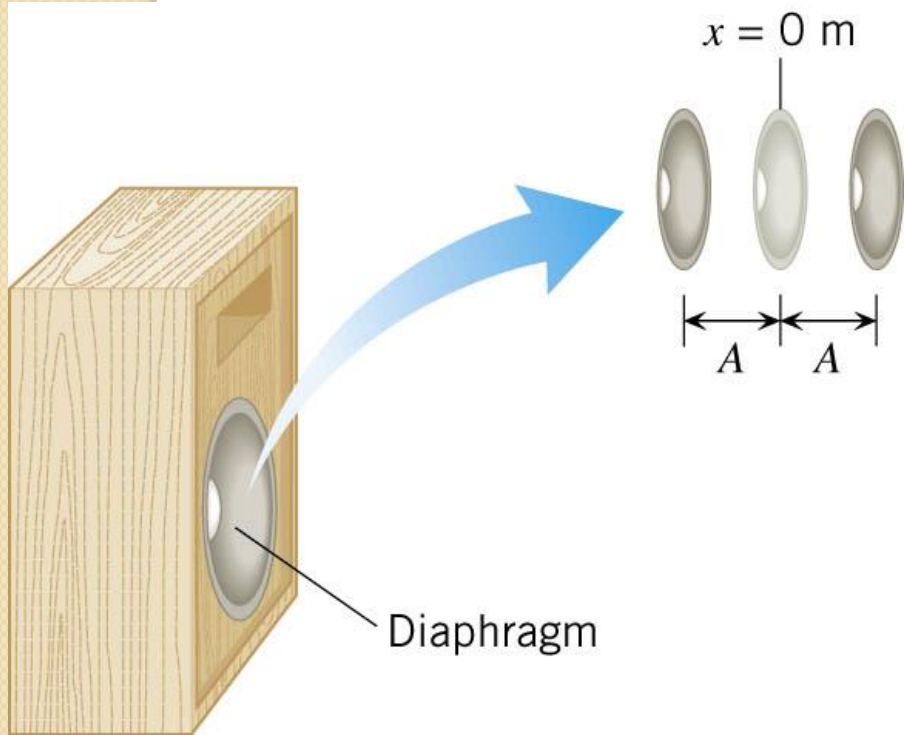
$$f = 100 \text{ MHz} = 1.0 \times 10^8 \text{ Hz}$$

$$T = 1/f = \frac{1}{1.0 \times 10^8 \text{ Hz}} = 1.0 \times 10^{-8} \text{ s} = 10 \text{ ns}$$

Note that $1/\text{Hz} = \text{s}$

Example-8 :

The Loudspeaker Revisited—The Maximum Acceleration



A loudspeaker diaphragm is vibrating at a frequency of $f = 1.0 \text{ kHz}$, and the amplitude of the motion is $A = 0.20 \text{ mm}$.

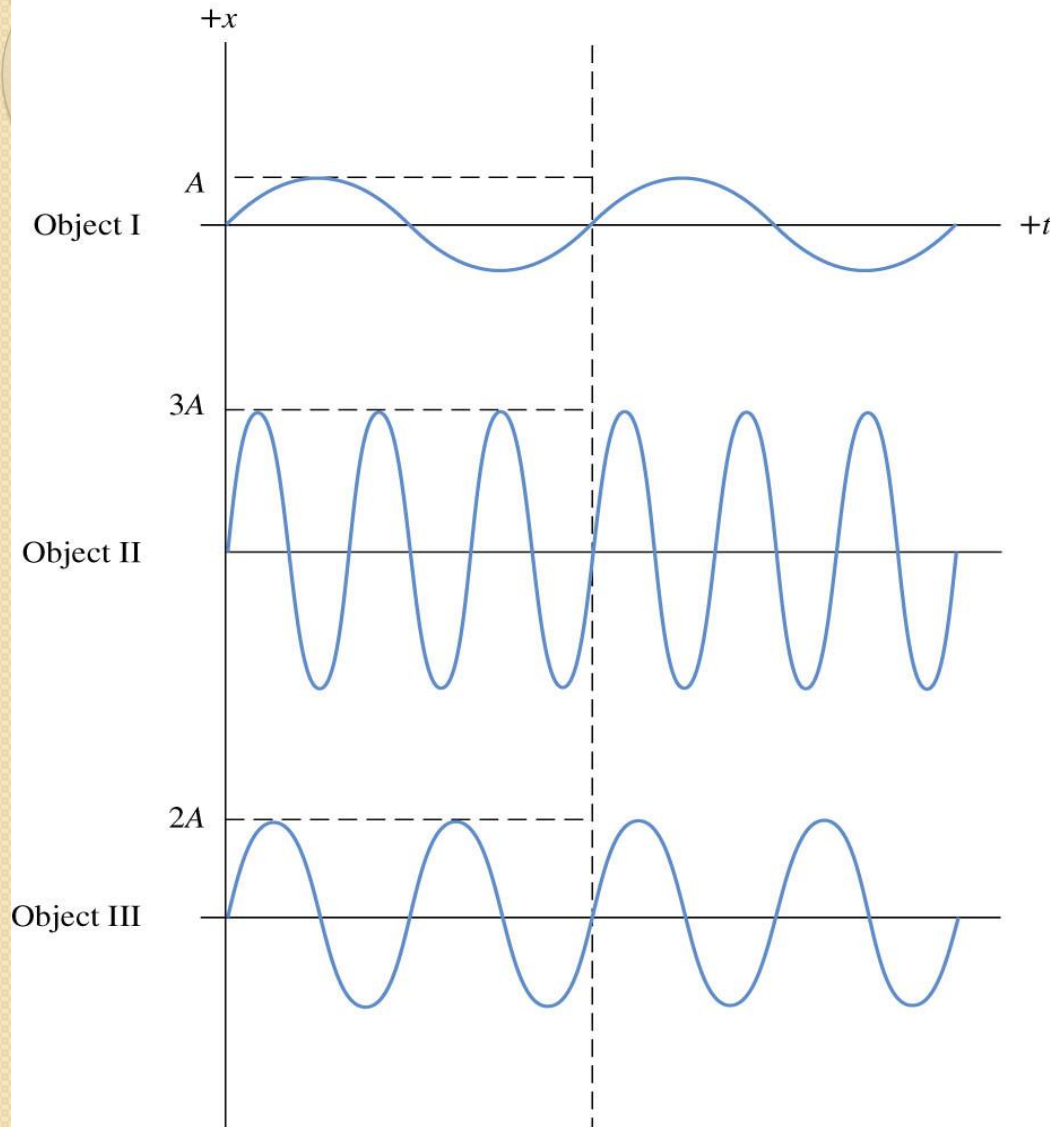
- (a) What is the maximum acceleration of the diaphragm, and
- (b) Where does this maximum acceleration occur?

(a)

$$\begin{aligned} a_{\max} &= A\omega^2 = A(2\pi f)^2 = (0.20 \times 10^{-3} \text{ m}) [2\pi(1.0 \times 10^3 \text{ Hz})]^2 \\ &= \boxed{7.9 \times 10^3 \text{ m/s}^2} \end{aligned}$$

(b) The maximum acceleration occurs at $x = +A$ and $x = -A$

Check Your Understanding



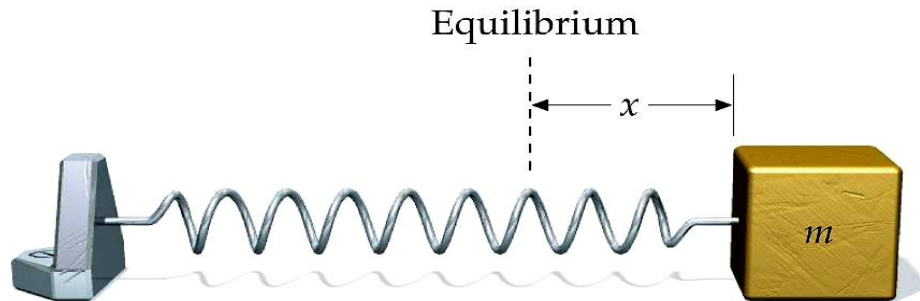
The drawing shows plots of the displacement x versus the time t for three objects undergoing simple harmonic motion. Which object, I, II, or III, has the greatest maximum velocity?

II

Example-9: A Block on a Spring

A 2.00 kg block is attached to a spring as shown. The force constant of the spring is $k = 196 \text{ N/m}$. The block is held a distance of 5.00 cm from equilibrium and released at $t = 0$.

- (a) Find the angular frequency ω , the frequency f , and the period T .
- (b) Write an equation for x vs. time.



$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{(196 \text{ N/m})}{(2.00 \text{ kg})}} = 9.90 \text{ rad/s}$$

$$f = \frac{\omega}{2\pi} = \frac{(9.90 \text{ rad/s})}{2\pi} = 1.58 \text{ Hz}$$

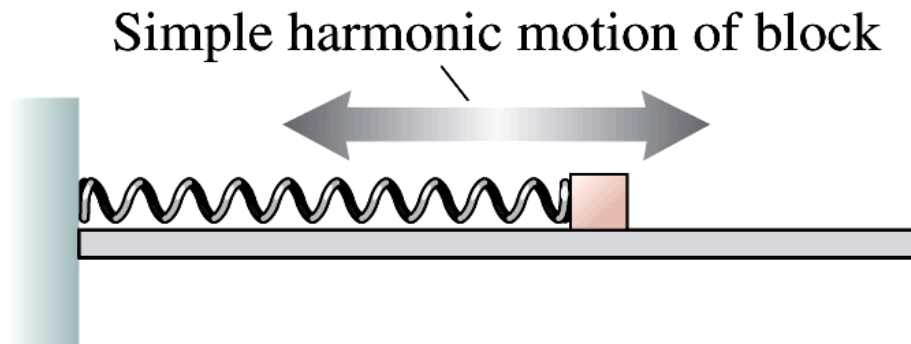
$$T = 1/f = 0.635 \text{ s} \quad A = 5.00 \text{ cm} \text{ and } \delta = 0$$

$$x = (5.00 \text{ cm}) \cos[(9.90 \text{ rad/s})t]$$

Example-10: A System in SHM

An air-track glider is attached to a spring, pulled 20 cm to the right, and released at $t = 0$. It makes 15 complete oscillations in 10 s.

- What is the period of oscillation?
- What is the object's maximum speed?
- What is its position and velocity at $t = 0.80$ s?



$$f = \frac{15 \text{ oscillations}}{10 \text{ s}}$$

$$= 1.5 \text{ oscillations/s} = 1.5 \text{ Hz}$$

$$T = 1/f = 0.667 \text{ s}$$

$$v_{\text{max}} = \frac{2\pi A}{T} = \frac{2\pi(0.20 \text{ m})}{(0.667 \text{ s})} = 1.88 \text{ m/s}$$

$$x = A \cos \frac{2\pi t}{T} = (0.20 \text{ m}) \cos \frac{2\pi(0.80 \text{ s})}{(0.667 \text{ s})} = 0.062 \text{ m} = 6.2 \text{ cm}$$

$$v = -v_{\text{max}} \sin \frac{2\pi t}{T} = -(1.88 \text{ m/s}) \sin \frac{2\pi(0.80 \text{ s})}{(0.667 \text{ s})} = -1.79 \text{ m/s}$$

Example-11: Finding the Time

A mass oscillating in simple harmonic motion, starts at $x = A$ and has period T .

At what time, as a fraction of T , does the mass first pass through $x = \frac{1}{2} A$?

$$x = \frac{1}{2} A = A \cos \frac{2\pi t}{T}$$

$$t = \frac{T}{2\pi} \cos^{-1} \left(\frac{1}{2} \right) = \frac{T}{2\pi} \frac{\pi}{3} = \frac{1}{6} T$$

Example-12: A particle execute s simple harmonic motion given by the equation

$$y = 12 \sin\left(\frac{2\pi t}{10} + \frac{\pi}{4}\right)$$

Calculate (i) amplitude, (ii) frequency, (iii) displacement at $t = 1.25\text{s}$, (iv) velocity at $t = 2.5\text{s}$ (v) acceleration at $t = 5\text{s}$.

Example-13: A particle execute s simple harmonic motion given by the equation

$$y = 10 \sin(10t - \frac{\pi}{6})$$

Calculate (i) frequency, (ii) time period (iii) the maximum displacement (iv) the maximum velocity (v) the maximum acceleration.

Example-0.18 : A particle of mass 500 gm executes simple harmonic motion given by the equation

$$x = 5 \cos\left(\frac{3\pi t}{10} + \frac{\pi}{4}\right) m$$

Calculate (i) amplitude, (ii) angular frequency, (iii) frequency, (iv) time period, (v) displacement at $t = 1$ s, (vi) velocity at $t = 2$ s, (vii) acceleration at $t = 3$ s, (viii) potential energy at $t = 4$ s, (ix) kinetic energy at $t = 5$ s, (x) mechanical energy, (xi) total energy at $t = 6$ s, (xii) draw the displacement graph, (xiii) draw the velocity graph, (xiv) draw the acceleration graph, (xv) draw the potential energy graph, (xvi) draw the kinetic energy graph, and (xvii) draw the total energy graph.

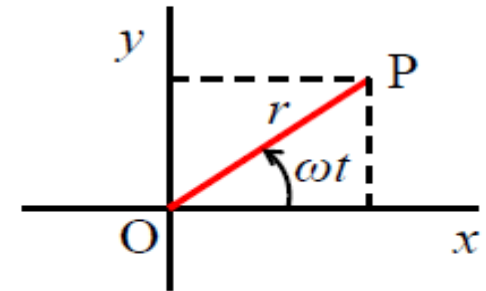
□ *Derive the differential equation of SHM and find its solution (Mechanical system, Complex Variable Approach) :*

Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):

Complex numbers

Consider a vector $\overline{OP}$ of length r which rotates with angular velocity ω



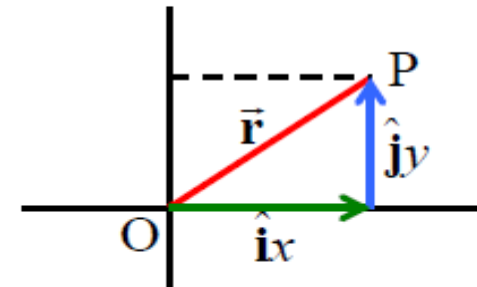
The point P has coordinates

$$x = r \cos \omega t \quad y = r \sin \omega t$$

We see that the x coordinate of P, or the projection of $\overline{OP}$ onto the x -axis, executes SHM

Can also introduce the unit vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$

and write $\vec{\mathbf{r}} = \hat{\mathbf{i}}x + \hat{\mathbf{j}}y$



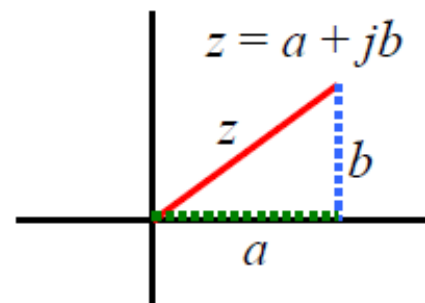
Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):

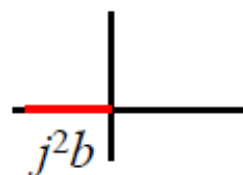
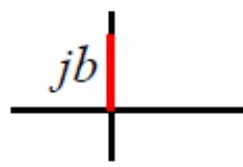
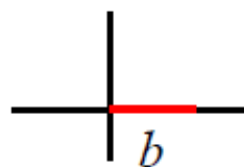
Complex numbers ...2

Modify our notation to $z = x + jy$

... where x means a displacement in the x -direction and jy means a displacement in the y -direction

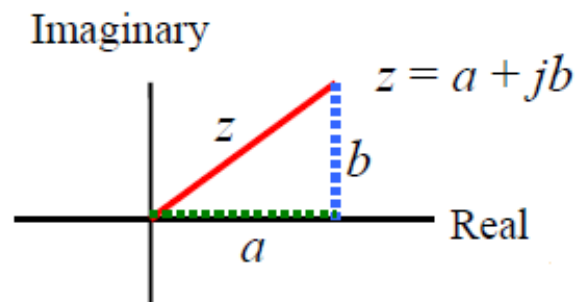


We can also think of j as a rotation through $\pi/2$ anticlockwise



Hence $j^2 = -1$

... really talking about vectors in the complex number plane:



Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):

Complex numbers ...3

From Taylor's theorem: $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$$\text{therefore } e^{j\theta} = 1 + j\theta - \frac{\theta^2}{2!} - \frac{j\theta^3}{3!} + \frac{\theta^4}{4!} + \dots$$

$$\text{and } \cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots \text{ and } j \sin \theta = j\theta - \frac{j\theta^3}{3!} + \dots$$

Hence

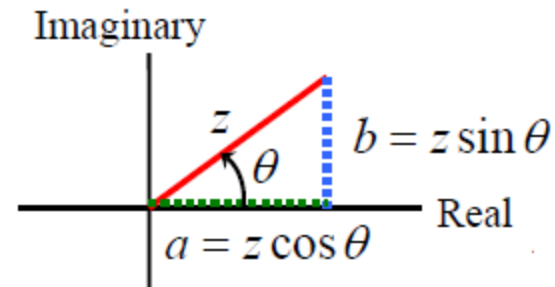
$$e^{j\theta} = \cos \theta + j \sin \theta$$

Euler relation

$$\text{Then } z = a + jb = |z| e^{j\theta}$$

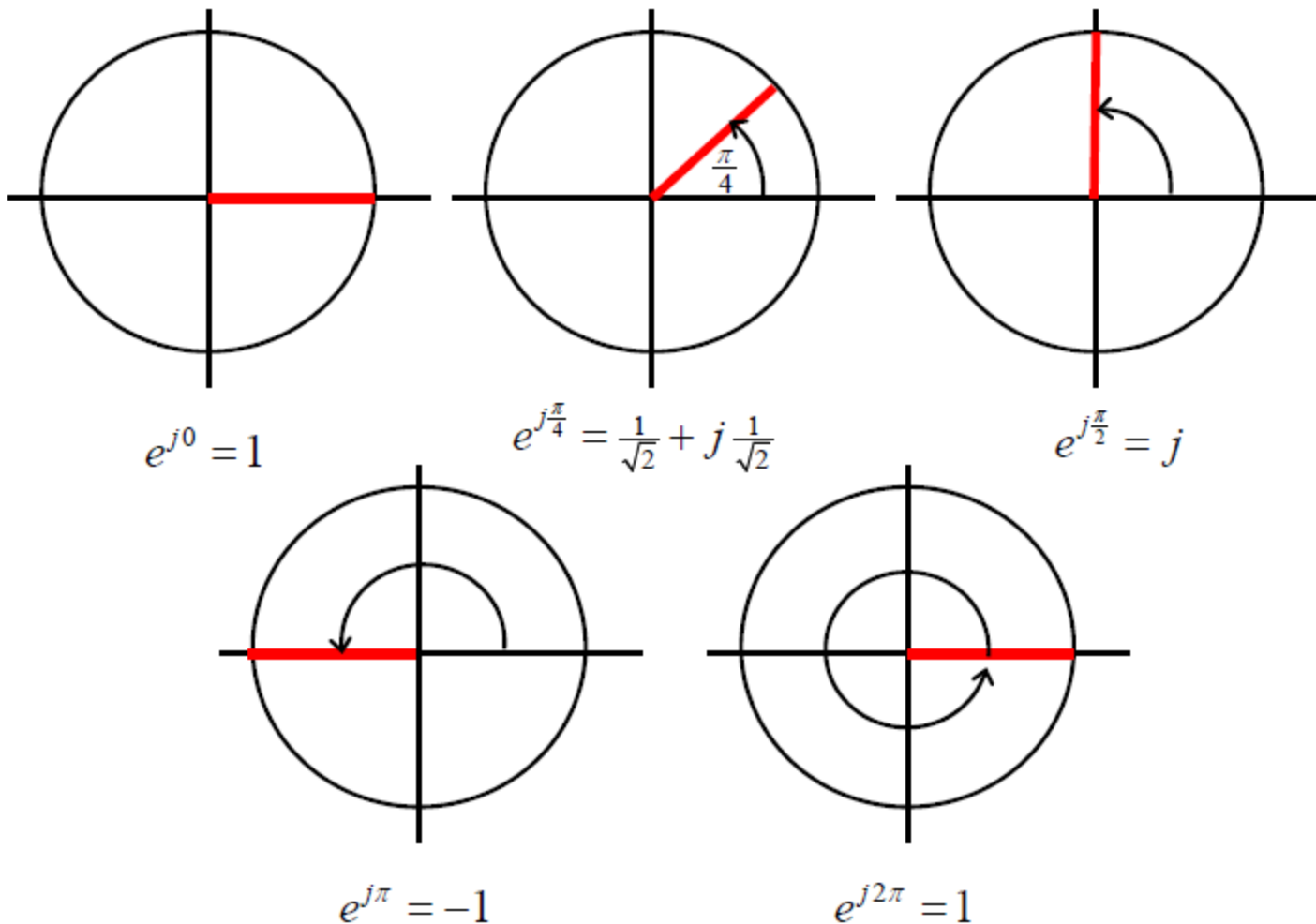
$$\text{where } |z| = \sqrt{a^2 + b^2}$$

$$\tan \theta = \frac{b}{a}$$



Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):



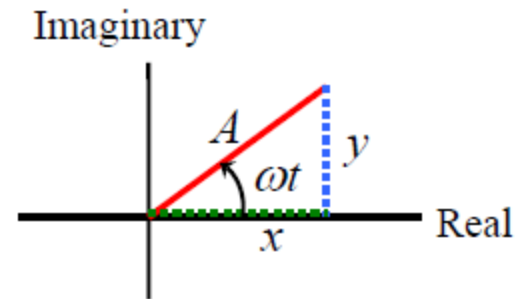
Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):

Complex numbers ...4

For our rotating vectors:

$$\begin{aligned}z &= x + jy \\&= A \cos \omega t + jA \sin \omega t \\&= A(\cos \omega t + j \sin \omega t) \\&= Ae^{j\omega t}\end{aligned}$$



Now write: $Ae^{j(\omega t + \phi)} = A \cos(\omega t + \phi) + jA \sin(\omega t + \phi)$

... and remember that the physical quantity x (e.g. a displacement) is the real part of z :

$$\text{i.e. } x = \text{Re}[z]$$

Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):

SHM using complex numbers

$$\frac{d^2x}{dt^2} + \omega_0^2 x = 0$$

Using $z = x + jy$ becomes $\frac{d^2z}{dt^2} + \omega_0^2 z = 0$

Try $z = Ae^{j(\omega t + \phi)}$

$$\therefore A(j\omega)^2 e^{j(\omega t + \phi)} + \omega^2 A e^{j(\omega t + \phi)} = 0$$

Therefore $z = Ae^{j(\omega t + \phi)}$ is the most general solution
 A and ϕ are determined from the initial conditions.

Take real part of z :

$$x = \text{Re}[z] = A \cos(\omega_0 t + \phi)$$

Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):

SHM using complex numbers

$$x = A \cos(\omega_0 t + \phi)$$

$$z = A e^{j(\omega t + \phi)}$$

$$x = \text{Re}[z]$$

$$\frac{dx}{dt} = -A\omega_0 \sin(\omega_0 t + \phi)$$

$$\frac{dz}{dt} = j\omega A e^{j(\omega t + \phi)} = j\omega z$$

$$\frac{dx}{dt} = \text{Re} \left[\frac{dz}{dt} \right]$$

$$\frac{d^2 x}{dt^2} = -A\omega_0^2 \cos(\omega_0 t + \phi)$$

$$\frac{d^2 z}{dt^2} = (j\omega)^2 A e^{j(\omega t + \phi)} = -\omega^2 z$$

$$\frac{d^2 x}{dt^2} = \text{Re} \left[\frac{d^2 z}{dt^2} \right]$$

Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):

Solving $\frac{d^2x(t)}{dt^2} + \omega_0^2 x(t) = 0$

Let $x = Be^{pt}$

Then $\frac{dx}{dt} = Bpe^{pt}$ and $\frac{d^2x}{dt^2} = Bp^2e^{pt}$

Substituting into DE: $Bp^2e^{pt} + B\omega_0^2e^{pt} = 0$

This holds true for all t if and only if $p^2 = -\omega_0^2$ or $p = \pm j\omega_0$

$$\therefore x = B_1e^{j\omega_0t} + B_2e^{-j\omega_0t}$$

How to get B_1 and B_2 ? ... need to know the initial conditions

... but consider $v = \frac{dx}{dt} = j\omega_0B_1e^{j\omega_0t} - j\omega_0B_2e^{-j\omega_0t}$

Alternative Method-4

SHM with Mechanical System (Complex Variable Approach):

Solving $\frac{d^2x(t)}{dt^2} + \omega_0^2 x(t) = 0$ **continued**

$$v = \frac{dx}{dt} = j\omega_0 B_1 e^{j\omega_0 t} - j\omega_0 B_2 e^{-j\omega_0 t}$$

$$\text{At } t = 0, \quad v(0) = j\omega_0 B_1 - j\omega_0 B_2$$

$$\text{Choose } v(0) = 0$$

Since v must be real, then $B_1 = B_2 = B$

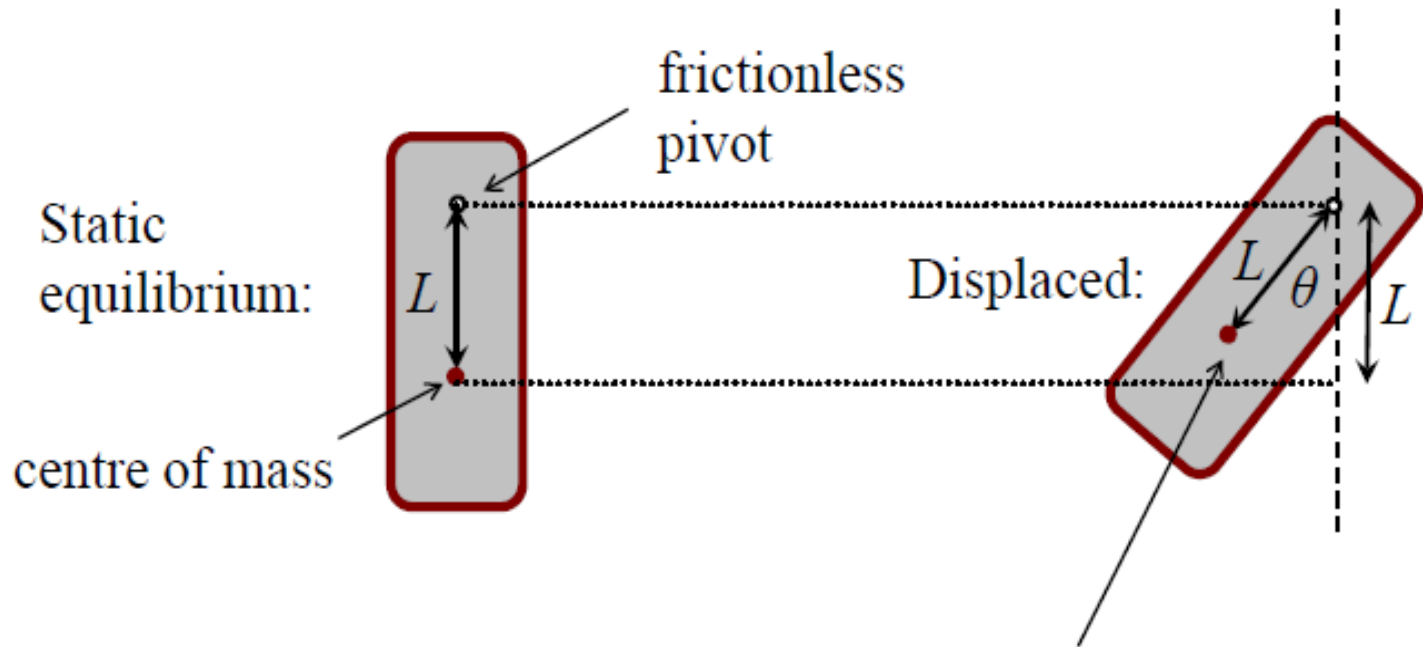
$$\begin{aligned} \text{i.e. } x &= B e^{j\omega_0 t} + B e^{-j\omega_0 t} \\ &= 2B \cos \omega_0 t \end{aligned}$$

$$\therefore x = A \cos \omega_0 t$$

[... with a little more effort we could have got the more general solution $x = A \cos(\omega_0 t + \phi)$]

□ *Derive the differential equation of Simple Pendulum and find its solution (Mechanical system, General Case):* **Alternative Method**

The pendulum: general case



In displaced position, centre of mass is $L - L \cos \theta$ above the equilibrium position.

Recall $\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots$

For small angles, $\cos \theta \approx 1 - \frac{\theta^2}{2}$

Gravitational potential energy = $mgL(1 - \cos \theta) = mgL \frac{\theta^2}{2}$

Alternative Method

Simple Pendulum with Mechanical System (Alternative General Approach):

The pendulum: general case ...2

$$\text{Gravitational potential energy} = \frac{1}{2}mgL\theta^2$$

$$\text{Kinetic energy} = \frac{1}{2}I\left(\frac{d\theta}{dt}\right)^2 \quad I = \text{Moment of inertia}$$

$$\text{Total energy} = \frac{1}{2}I\left(\frac{d\theta}{dt}\right)^2 + \frac{1}{2}mgL\theta^2 = \text{constant}$$

$$\therefore I \frac{d\theta}{dt} \frac{d^2\theta}{dt^2} + mgL\theta \frac{d\theta}{dt} = 0 \quad \dots \text{true for all } \frac{d\theta}{dt}$$

$$\therefore \frac{d^2\theta}{dt^2} = -\frac{mgL}{I}\theta = -\omega_0^2\theta \quad \text{where } \omega_0 = \sqrt{\frac{mgL}{I}}$$

Equation of SHM

Alternative Method

Simple Pendulum with Mechanical System (Alternative General Approach):

The moment of inertia of the pendulum about an axis passing through the point of suspension is

$$= mK^2 + mL^2$$

Therefore, $\omega_0 = \sqrt{\frac{gL}{K^2 + L^2}}$

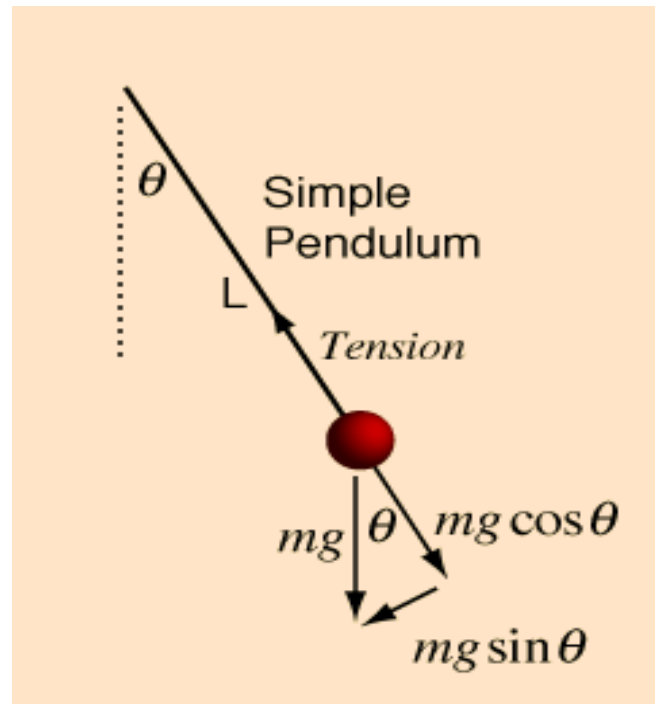
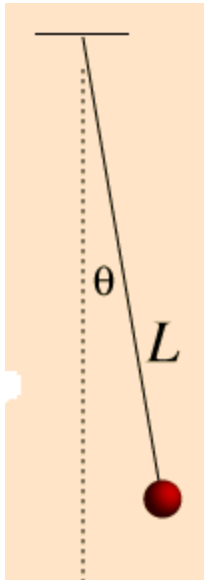
Time Period

$$T = 2\pi \sqrt{\frac{K^2 + L^2}{Lg}}$$

□ *Derive the differential equation of Simple Pendulum and find its solution (Mechanical system):*

Simple Pendulum

A *simple pendulum* consists of a particle of mass m , attached to a frictionless point by a cable of length L and negligible mass.



Simple Pendulum with Mechanical System:

From the above figure restoring force

$$F = -mg \sin \theta$$

If the angle θ is very small $\sin \theta$ is very nearly equal to θ . The displacement along the arc is

$$x = L \theta$$

Therefore,
$$F = -mg \theta$$

Simple Pendulum with Mechanical System:

Acceleration $\frac{d^2 x}{dt^2} = L \frac{d^2 \theta}{dt^2}$

$$\text{Force} = mL \frac{d^2 \theta}{dt^2}$$

$$mL \frac{d^2 \theta}{dt^2} = -mg \theta$$

$$\frac{d^2 \theta}{dt^2} + \frac{g}{L} \theta = 0$$

$$\therefore \theta(t) = \theta_o \cos(\omega t + \delta)$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

Where

$$\omega^2 = \frac{g}{L}$$

$$\therefore \omega = \sqrt{\frac{g}{L}}$$

And

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$f = \frac{1}{2\pi} \sqrt{g/L}$$

Physical Pendulum with Mechanical System (General Approach):

For a physical pendulum the restoring torque $\tau = -L F_g \sin \theta$ and $\tau = I \alpha = I \frac{d^2 \theta}{dt^2}$, where I is the moment of inertia. Therefore

$$\frac{d^2 \theta}{dt^2} + \frac{Lmg}{I} \theta = 0 \quad [\sin \theta \approx \theta]$$

The angular frequency

$$\omega = \sqrt{\frac{Lmg}{I}}$$

and the time period

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{Lmg}}$$

Example 14: Keeping Time

Determine the length of a simple pendulum that will swing back and forth in simple harmonic motion with a period of 1.00 s.

$$f = \frac{1}{2\pi} \sqrt{g / L}$$

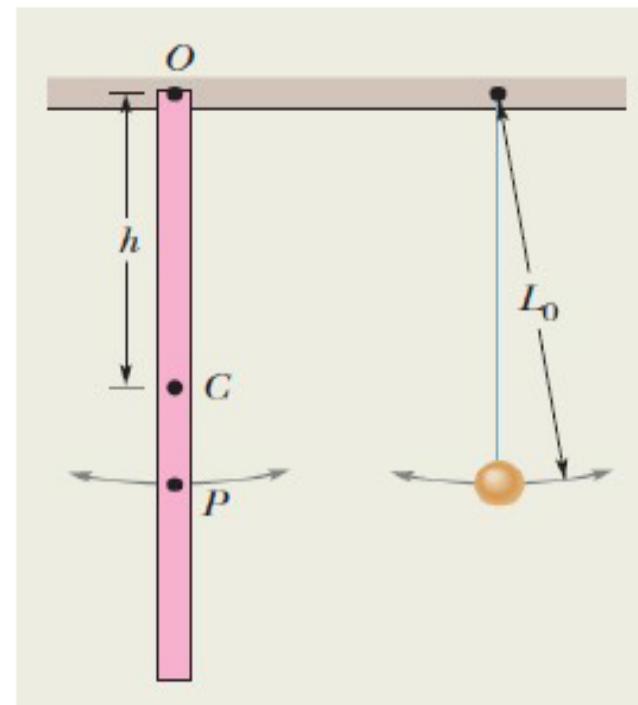
$$L = \frac{T^2 g}{4\pi^2} = \frac{(1.00 \text{ s})^2 (9.80 \text{ m/s}^2)}{4\pi^2} = \boxed{0.248 \text{ m}}$$

Example-0.19:

A meter stick swings about a pivot point at one end, at distance h from the stick's center of mass. What is the period of oscillation?

Solution:

$$\begin{aligned} T &= 2\pi \sqrt{\frac{I}{hmg}} = 2\pi \sqrt{\frac{\frac{1}{3}mL^2}{\frac{L}{2}mg}} \\ &= 2\pi \sqrt{\frac{2L}{3g}} \\ &= 2\pi \sqrt{\frac{2 \times 1.0m}{3 \times 9.8ms^{-2}}} = 1.64 s \end{aligned}$$



□ *Derive the differential equation of Torsional Pendulum and find its solution (Mechanical system):*

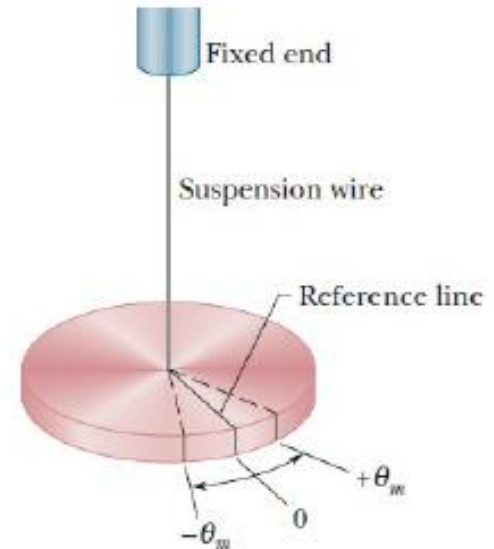
Torsional Pendulum:

According to the Hook's law

$$\tau \propto \theta$$

$$\text{or, } \tau = -\kappa \theta \quad (13)$$

$$\text{Again, } \tau = I \alpha = I \frac{d^2 \theta}{dt^2} \quad (14)$$



where I denotes the moment of inertia. From equation (4) and (5) we can write

$$I \frac{d^2 \theta}{dt^2} = -\kappa \theta$$

$$\text{or, } \frac{d^2 \theta}{dt^2} + \frac{k}{I} \theta = 0$$

Torsional Pendulum with Mechanical System (General Approach):

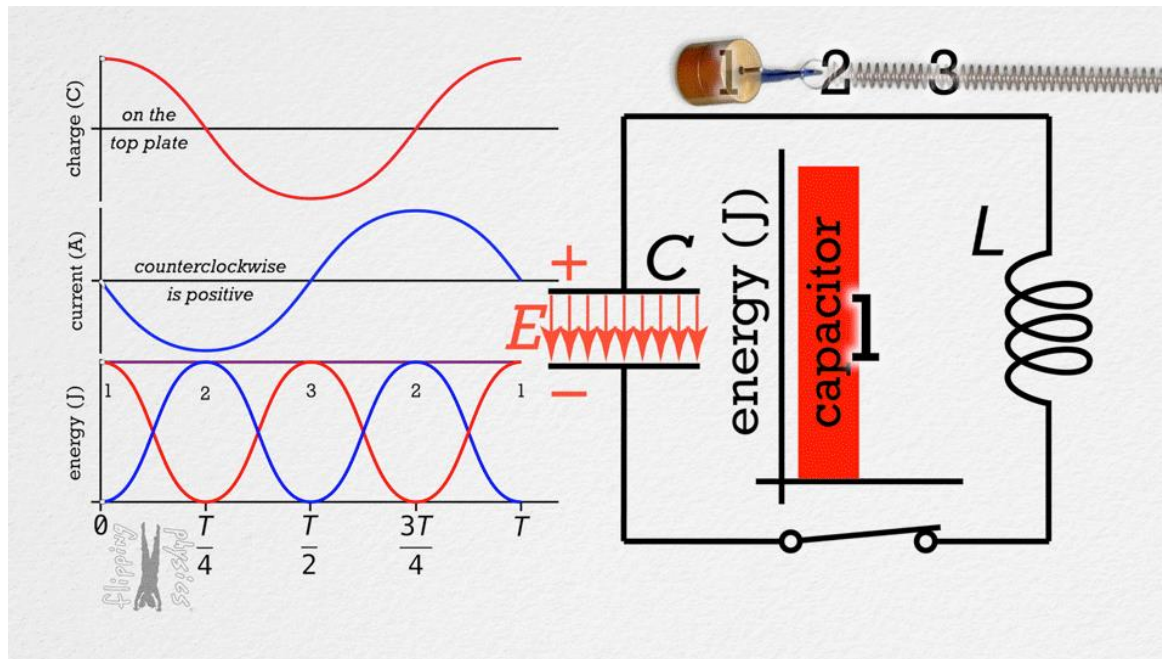
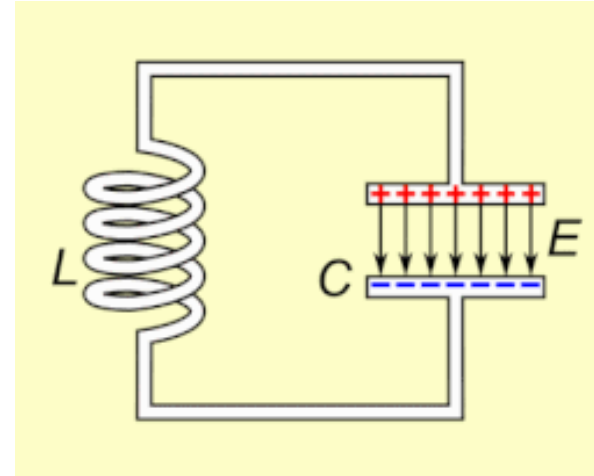
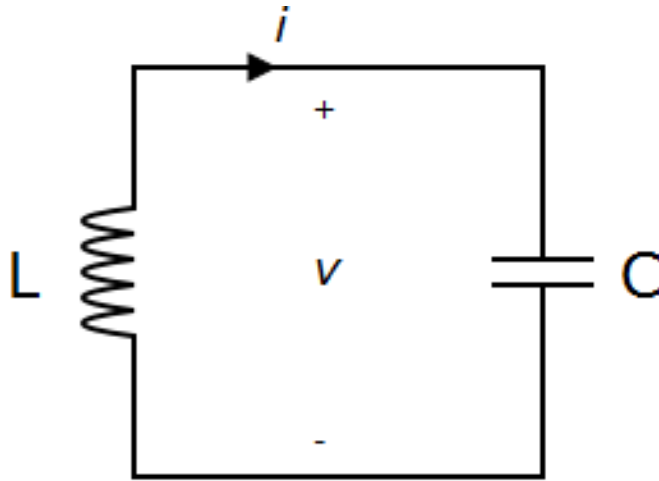
The solution of the above equation is

$$\theta = \theta_m \cos(\omega t - \phi) \quad [\because \omega = \sqrt{\frac{\kappa}{I}}]$$

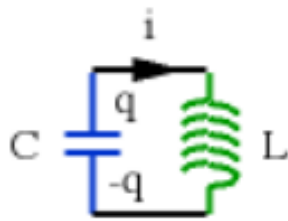
and the period of oscillation is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{\kappa}}$$

LC Circuit (SHM in Electric Circuit):

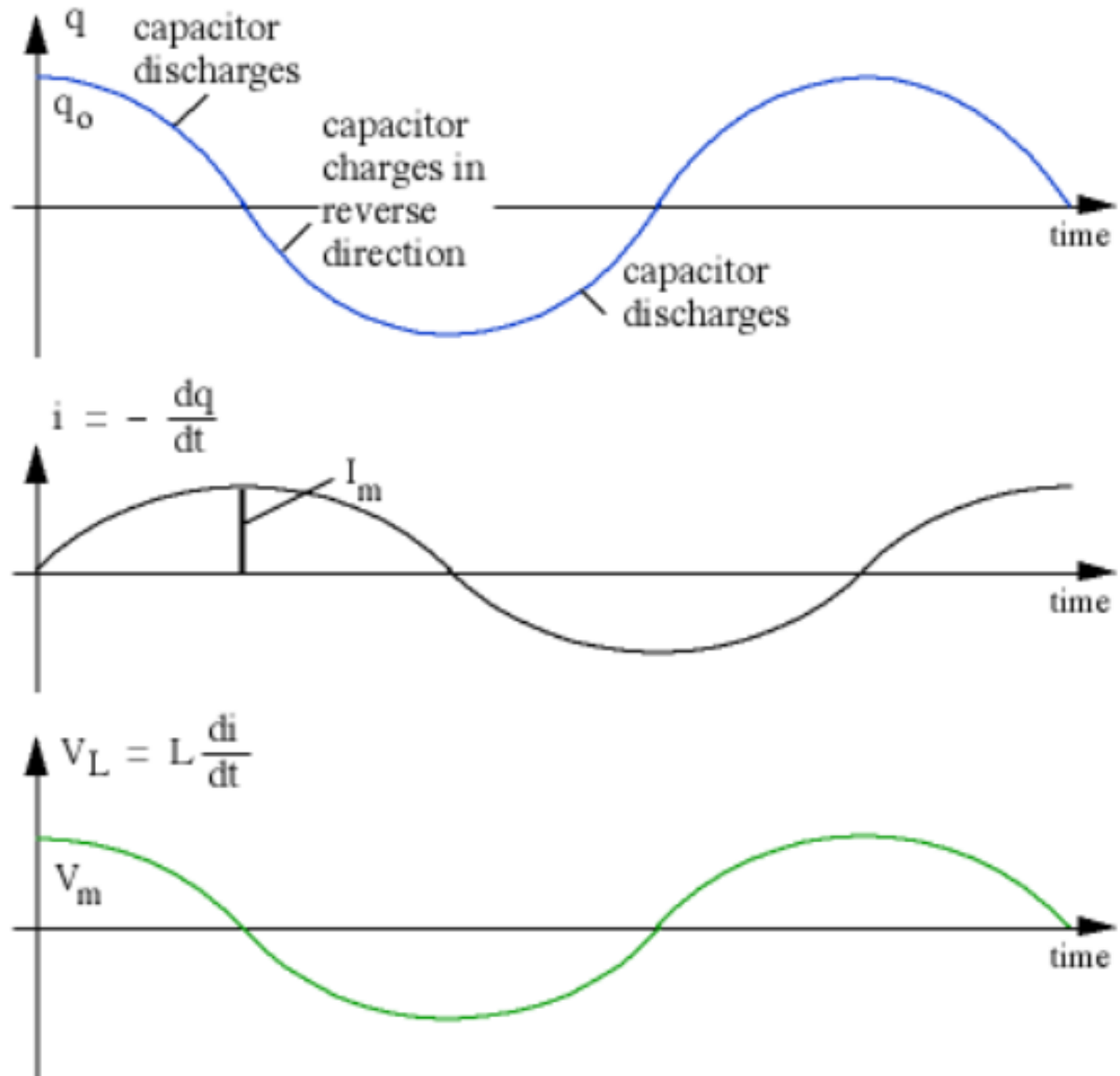


LC Circuit (SHM in Electric Circuit):

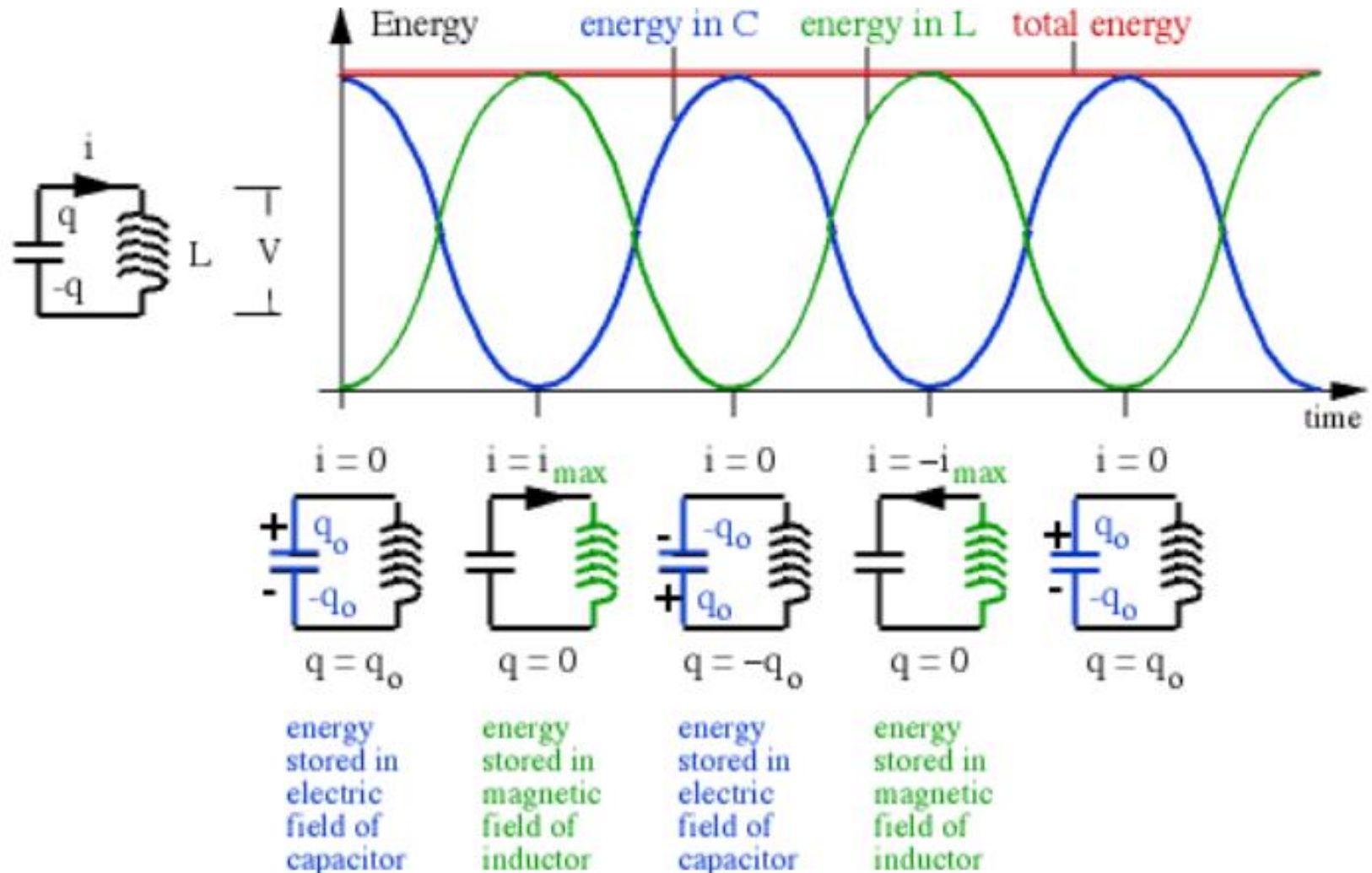


In the direction of the arrow, the current is the rate at which the capacitor is discharging

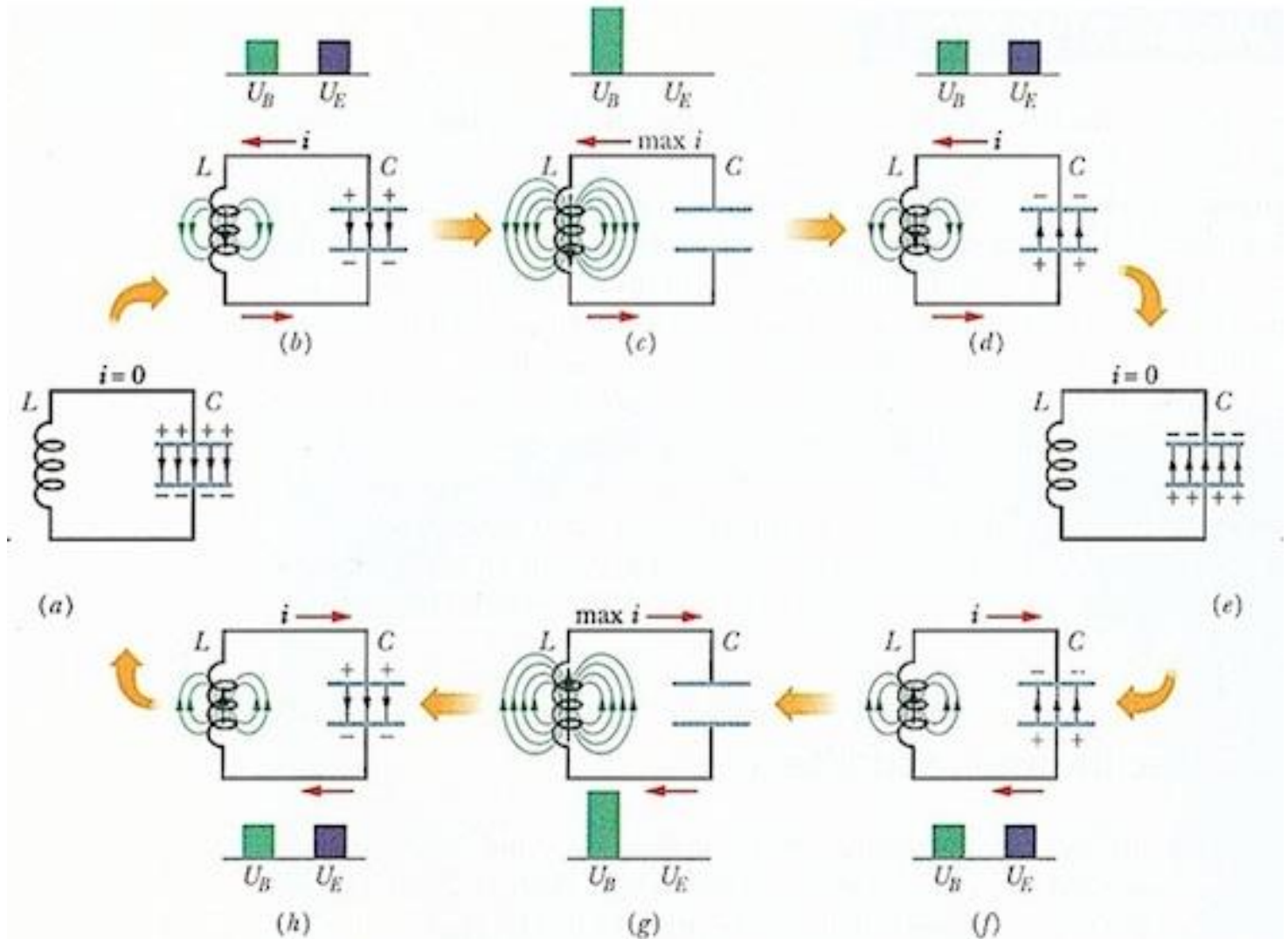
The voltage drop across the inductor (-1 times its emf) depends upon the rate at which the current changes and the inductance L .



LC Circuit (SHM in Electric Circuit):

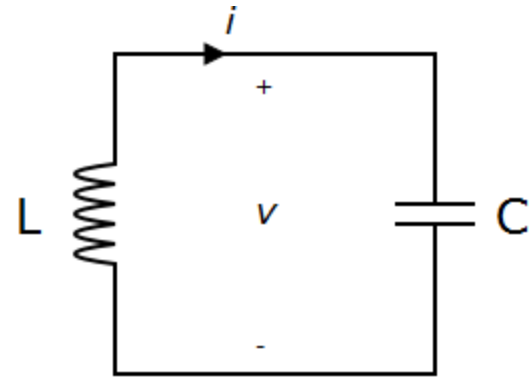
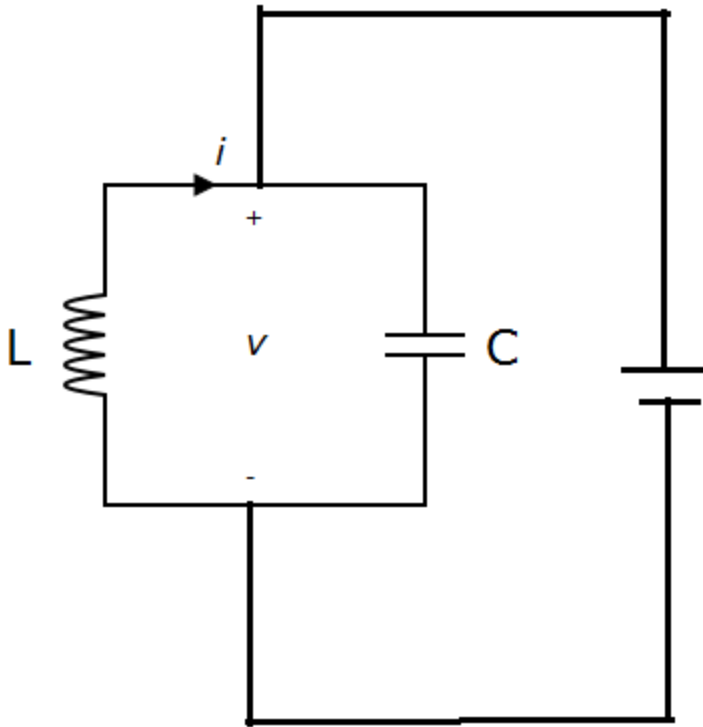


Charging discharging of an LC Circuit



□ *Derive the differential equation of SHM and find its solution (Electrical system, LC Circuit):*

LC Circuit



SHM with Electrical System (LC Circuit):

An **LC circuit**, also called a **resonant circuit**, **tank circuit**, or **tuned circuit**, consists of an inductor, represented by the letter L, and a capacitor, represented by the letter C. When connected together, they can act as an electrical resonator.

Voltage across capacitor at any instant

$$V_C = \frac{Q}{C}$$

Q is the charge on the capacitor and C is capacitance of capacitor.

Voltage across inductor at the same instant

$$V_L = L \frac{di}{dt}$$

$$\frac{Q}{C} + L \frac{di}{dt} = 0 \quad \text{Kirchhoff's voltage law}$$

$$\frac{d^2 i}{dt^2} + \frac{1}{LC} i = 0$$

Similar to differential equation of SHM

$$\frac{d^2 x}{dt^2} + \omega_0^2 x = 0, \quad \omega_0^2 = \frac{1}{LC}$$

SHM with Electrical System (LC Circuit):

Time Period

$$T = 2\pi\sqrt{LC}$$

Frequency

$$f = \frac{1}{2\pi\sqrt{LC}}$$

SHM with Electrical System (LC Circuit):

Solution of the differential equation is

$$Q(t) = Q_0 \cos(\omega_0 t + \phi)$$

Current in the circuit

$$i(t) = -i_0 \sin(\omega_0 t + \phi)$$

SHM with Mechanical and Electrical System Comparison

Mechanical

displacement x

velocity v

mass m

spring constant k

$$\omega_0 = \sqrt{\frac{k}{m}}$$

potential energy: $\frac{1}{2}kx^2$

kinetic energy: $\frac{1}{2}mv^2$

Electrical

charge Q

current I

inductance L

$\frac{1}{\text{capacitance}}$ $\frac{1}{C}$

$$\omega_0 = \sqrt{\frac{1}{LC}}$$

Electric energy
stored in capacitor: $\frac{1}{2}\frac{Q^2}{C}$

Magnetic energy
stored in inductor: $\frac{1}{2}LI^2$

Damped Harmonic Motion (DHM) :

Damped oscillations

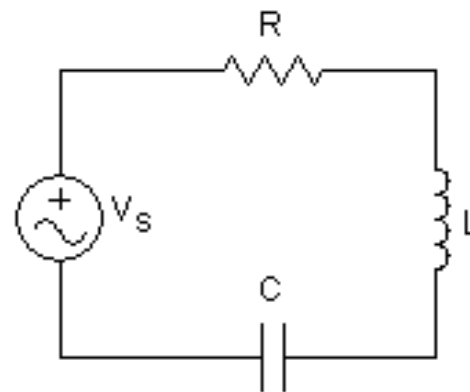
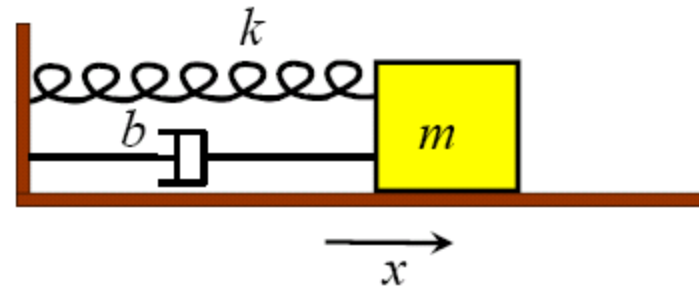
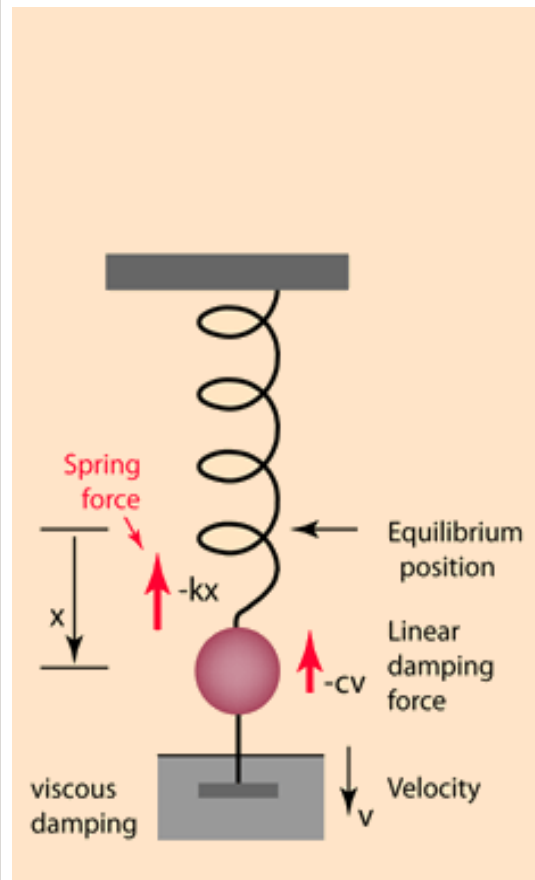
We have thus far neglected all dissipative mechanisms
... our oscillations can continue oscillating with the
same amplitude forever ...

Various physical damping mechanisms will contribute towards
the damping...

- friction between mass and table
- “air resistance”
- internal friction in spring
-

... model these by introducing a damping force which is
proportional to the velocity of the oscillator ...

Examples some damped oscillating systems





Damped Harmonic motion: When oscillating bodies do not move back and forth between precisely fixed limits because frictional force dissipates the energy and amplitude of oscillation decreases with time and finally die out. Such harmonic motion is called Damped Harmonic Motion.

□ *Derive the differential equation of DHM and find its solution (Mechanical system):*

DHM with Mechanical System (Spring mass oscillator):

Damped mass-spring system

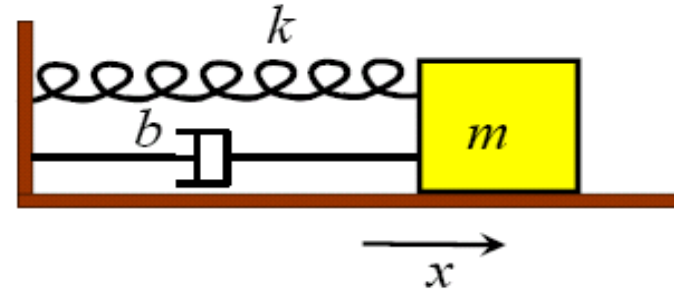
In these systems the damping force

$$F' = -bv$$

For horizontal forces on the mass: $ma = -kx - bv$

$$\text{or } m \frac{d^2x}{dt^2} = -kx - b \frac{dx}{dt}$$

$$\text{or } \frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0 \quad \text{where } \begin{cases} \omega_0 = \sqrt{\frac{k}{m}} \\ \gamma = \frac{b}{m} \end{cases}$$



γ : "damping constant" unit: s^{-1} • "life time" = $\frac{1}{\gamma/2}$

DHM with Mechanical System (Spring mass oscillator):

Damped oscillator equation

$$\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

Let $x = Be^{pt}$

Then $\frac{dx}{dt} = Bpe^{pt}$ and $\frac{d^2x}{dt^2} = Bp^2e^{pt}$

Substituting into DE: $Bp^2e^{pt} + \gamma Bpe^{pt} + \omega_0^2 Be^{pt} = 0$

Thus $p^2 + \gamma p + \omega_0^2 = 0$

$$\therefore p = \frac{1}{2} \left\{ -\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2} \right\}$$

or $p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

DHM with Mechanical System (Spring mass oscillator):

Damped oscillator equation ...2

$$p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

We can distinguish three cases:

(i) $\omega_0^2 > \frac{\gamma^2}{4}$ **Oscillatory behaviour**

(ii) $\omega_0^2 = \frac{\gamma^2}{4}$ **Critical damping**

(iii) $\omega_0^2 < \frac{\gamma^2}{4}$ **Overdamping**

$$\text{Case (i):} \quad \omega_0^2 > \frac{\gamma^2}{4}$$

$$\therefore \sqrt{\gamma^2/4 - \omega_0^2} = \sqrt{-(\omega_0^2 - \gamma^2/4)}$$

$$\text{Put} \quad \omega_1^2 = \omega_0^2 - \gamma^2/4$$

$$\therefore p = -\frac{\gamma}{2} \pm \sqrt{-\omega_1^2} = -\frac{\gamma}{2} \pm j\omega_1$$

The solution will be:

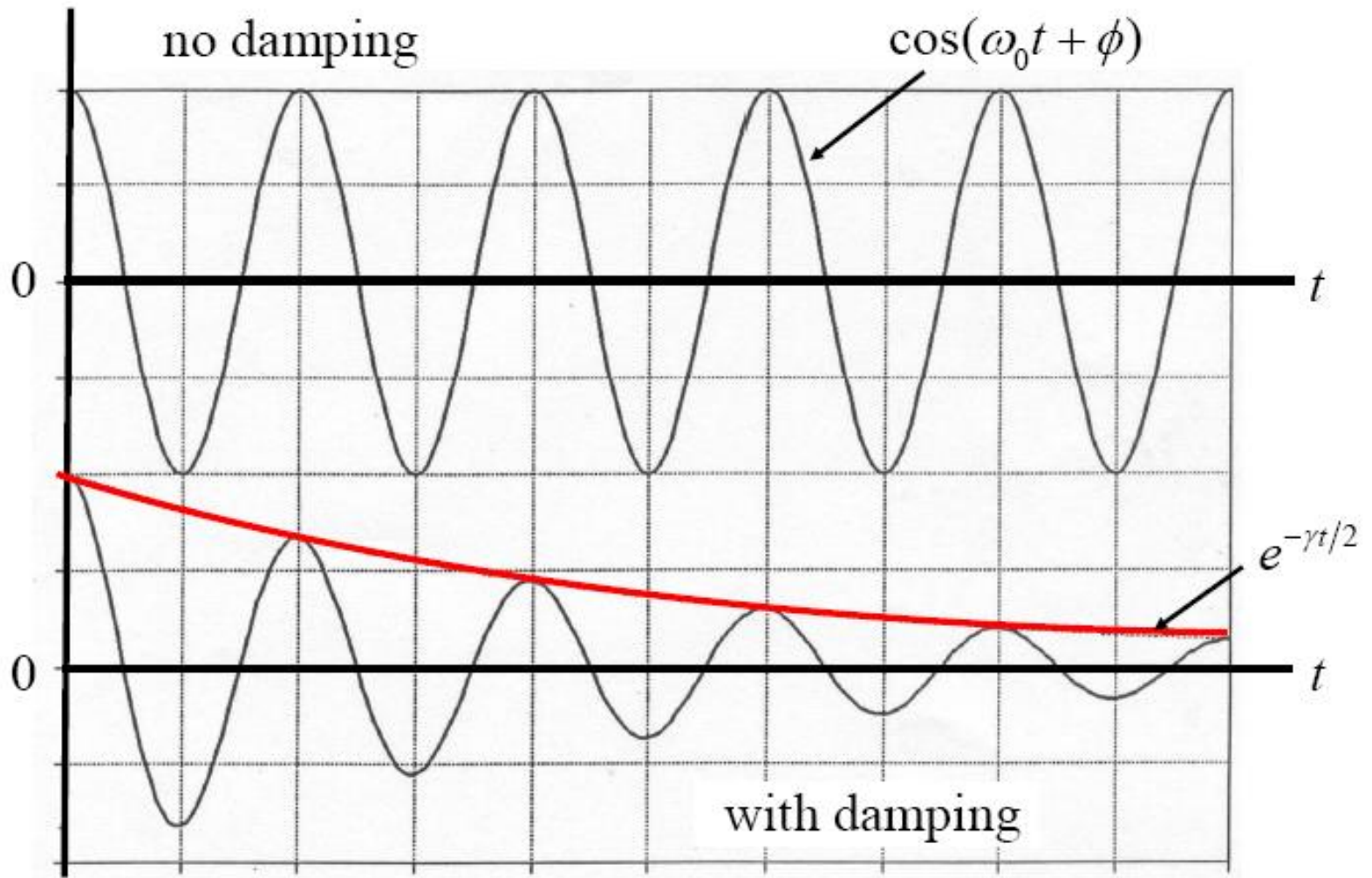
$$x = B_1 e^{(-\frac{\gamma}{2} + j\omega_1)t} + B_2 e^{(-\frac{\gamma}{2} - j\omega_1)t} = e^{-\frac{\gamma}{2}t} \{ B_1 e^{j\omega_1 t} + B_2 e^{-j\omega_1 t} \}$$

$$\dots \text{ leading to } x(t) = A e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \phi)$$

This is an **oscillatory solution** $A \cos(\omega_1 t + \phi)$ multiplied by a damping factor $e^{-\gamma t/2}$.

As $\gamma \rightarrow 0$ we approach our undamped oscillator.

DHM with Mechanical System (Spring mass oscillator):



DHM with Mechanical System (Spring mass oscillator):

$$\text{Case (ii): } \omega_0^2 = \frac{\gamma^2}{4}$$

The two roots coincide: $p = -\frac{\gamma}{2}$

The solution will be $x(t) = (A + Bt)e^{-\frac{\gamma}{2}t}$

The condition $\omega_0^2 = \gamma^2/4$ is referred to as the “**critical damping**” condition.

If $\omega_0^2 < \gamma^2/4$ a system released from rest will oscillate.

As γ is increased the oscillations decay more rapidly, until at $\omega_0^2 = \gamma^2/4$ oscillation no longer occurs.

[... many practical applications ...]

DHM with Mechanical System (Spring mass oscillator):

$$\text{Case (iii): } \omega_0^2 < \frac{\gamma^2}{4}$$

$$\begin{aligned} p &= -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2} \\ &= -\frac{\gamma}{2} \pm \lambda \quad \text{say} \end{aligned}$$

The solution will be $x(t) = B_1 e^{(-\frac{\gamma}{2} + \lambda)t} + B_2 e^{(-\frac{\gamma}{2} - \lambda)t}$

The condition $\omega_0^2 < \frac{\gamma^2}{4}$ is referred to as **overdamping**

... a slower approach to the rest position is observed.

Oscillatory

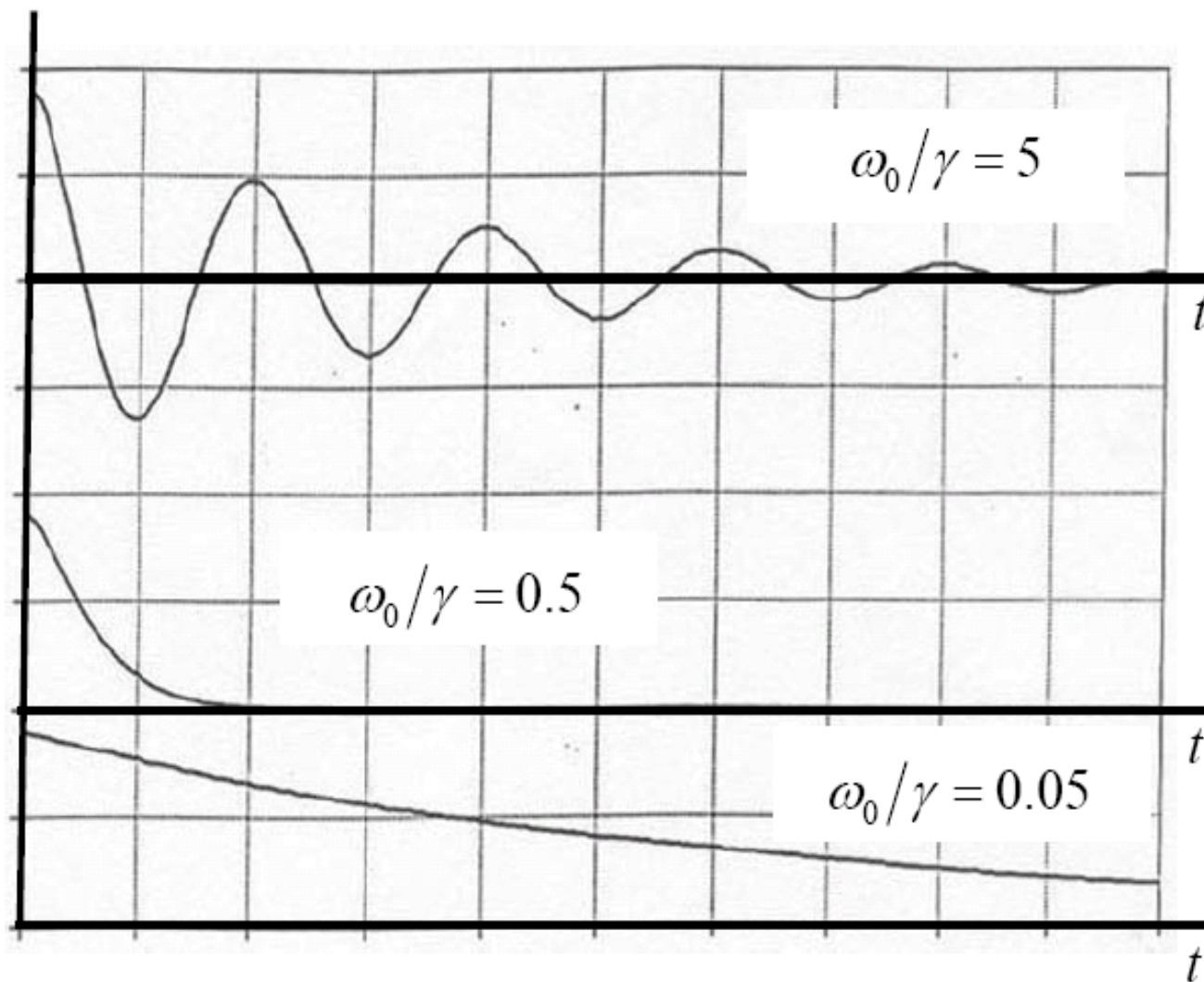
$$\omega_0/\gamma = 5$$

**Critical
damping**

$$\omega_0/\gamma = 0.5$$

Overdamping

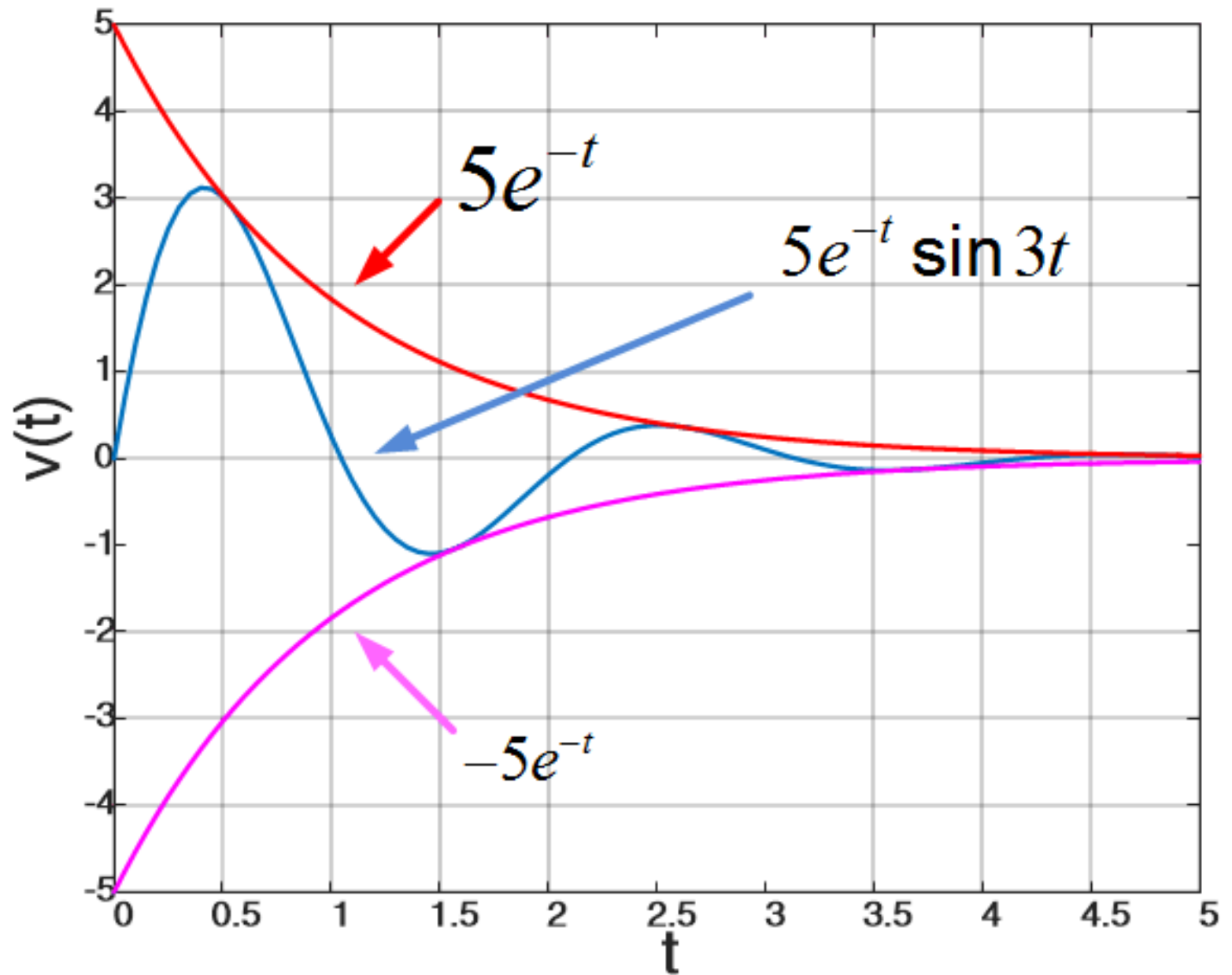
$$\omega_0/\gamma = 0.05$$



Distinction between the SHM and the DHM:

SHM	DHM
(1) Life time infinite.	(1) Life time is finite.
(2) Angular frequency $\omega = \sqrt{k/m}$	(2) Angular frequency $\omega_0 < \omega$
(3) Time period is smaller.	(3) Time period is larger.
(4) Energy remains constant, i.e. $E = \frac{1}{2}kA^2$	(4) E is gradually dissipated and falls to zero in time.

Typical DHM



Example-0.20 : For a damped oscillator $m = 200 \text{ gm}$, $k = 80 \text{ N/m}$ and $b = 65 \text{ gm/s}$. (i) What is the period of the motion? (ii) How long does it take for the amplitude of the damped oscillations to drop to half its initial value? (iii) How many oscillations does it complete in life time? (iv) How long does it take for the mechanical energy to drop to one-half its initial value? (v) What is its life time? (vi) The maximum displacement of undamped oscillator is 35 cm . If the damping is stopped after 20 cycles, What is the damping energy? (vii) What is the ratio of the oscillation amplitude to the initial oscillation amplitude at this cycle? (viii) What is damping frequency? and (ix) What is the damping amplitude?

Example-0.21:

A damped harmonic oscillation of a body of mass 0.3 kg is represented by the following equation

$$x(t) = Ae^{0.5t} \cos(3\pi t - \frac{\pi}{6})$$

Find (i) damping coefficient, (ii) natural frequency, (iii) force constant, and (iv) damping factor at $t = 2$ ms.

□ *Derive the differential equation of DHM and find its solution (Mechanical system):*

Alternative Method (DHM in Spring mass oscillator):

Damped Harmonic Motion

Net force on the mass F

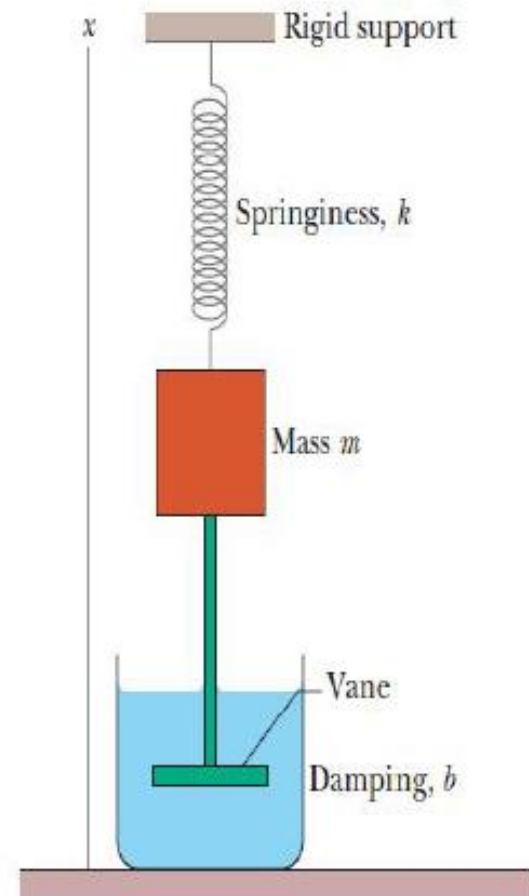
$$F = F_{restor} + F_{fric} \quad (1)$$

$$F_{restor} = -kx$$

$$F_{fric} = -bv = -b \frac{dx}{dt} \quad (2)$$

where b is called the damping coefficient. Therefore

$$\begin{aligned} F &= -kx - b \frac{dx}{dt} \\ \text{or, } m \frac{d^2x}{dt^2} &= -kx - b \frac{dx}{dt} \\ \frac{d^2x}{dt^2} &= -\frac{k}{m}x - \frac{b}{m} \frac{dx}{dt} \end{aligned}$$



Alternative Method (DHM in Spring mass oscillator):

DHM with Mechanical System (Spring mass oscillator) Alternative Method

Putting $k/m = \omega^2$ and $b/m = 2\gamma$ we can write

$$\frac{d^2x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega^2 x = 0 \quad (3)$$

This is the differential equation for a damped simple harmonic motion

Solution

Let us consider a trial solution

$$x = A e^{\lambda t} \quad (4)$$

$$\text{or, } \frac{dx}{dt} = \lambda A e^{\lambda t} \quad (5)$$

$$\text{and, } \frac{d^2x}{dt^2} = \lambda^2 A e^{\lambda t} \quad (6)$$

Alternative Method (DHM in Spring mass oscillator):

Substituting equation (4), (5) and (6) in to (3) we have

$$\begin{aligned}(\lambda^2 + 2\gamma\lambda + \omega^2) A e^{\lambda t} &= 0 \\ \text{So, } \lambda^2 + 2\gamma\lambda + \omega^2 &= 0 \quad [\because A e^{\lambda t} \neq 0] \\ \therefore \lambda &= -\gamma \pm \sqrt{\gamma^2 - \omega^2}\end{aligned}\tag{7}$$

Two independent solutions are

$$\begin{aligned}x_1 &= e^{-\gamma t + \sqrt{\gamma^2 - \omega^2} t} \\ x_2 &= e^{-\gamma t - \sqrt{\gamma^2 - \omega^2} t}\end{aligned}$$

The general solution is

$$\begin{aligned}x(t) &= e^{-\gamma t} \left(C e^{\sqrt{\gamma^2 - \omega^2} t} + D e^{-\sqrt{\gamma^2 - \omega^2} t} \right) \\ &= e^{-\frac{b}{2m} t} \left(C e^{\sqrt{(\frac{b}{2m})^2 - \omega^2} t} + D e^{-\sqrt{(\frac{b}{2m})^2 - \omega^2} t} \right)\end{aligned}\tag{8}$$

where C and D are constants of integration. Eq (8) is the solution for the DHM. $e^{-\gamma t}$ is known as the damping factor.

Alternative Method (DHM in Spring mass oscillator):

Different cases:

Case-I: Over damped motion:

If $\gamma > \omega$, that is the damping force is greater than the restoring force, $\sqrt{\gamma^2 - \omega^2}$ is positive. The solution will be of the form

$$\begin{aligned}x(t) &= e^{-\gamma t} \left(C e^{\sqrt{\gamma^2 - \omega^2} t} + D e^{-\sqrt{\gamma^2 - \omega^2} t} \right) \\&= e^{-\gamma t} (C e^{\alpha t} + D e^{-\alpha t})\end{aligned}$$

where $\alpha = \sqrt{\gamma^2 - \omega^2}$. The motion is overdamped. That is the motion will come to stop without oscillation after some time.

Case-II: Critically damped motion:

If $\gamma = \omega$, $\alpha = 0$. Therefore

$$\begin{aligned}x(t) &= e^{-\gamma t} (C + D) \\&= B e^{-\gamma t} \quad [B = C + D]\end{aligned}$$

This motion is critically damped. That is the body comes to its equilibrium position in the shortest possible time.

Alternative Method (DHM in Spring mass oscillator):

Case-III: Underdamped oscillation:

If $\gamma < \omega$, that is the damping force is weaker than the restoring force,

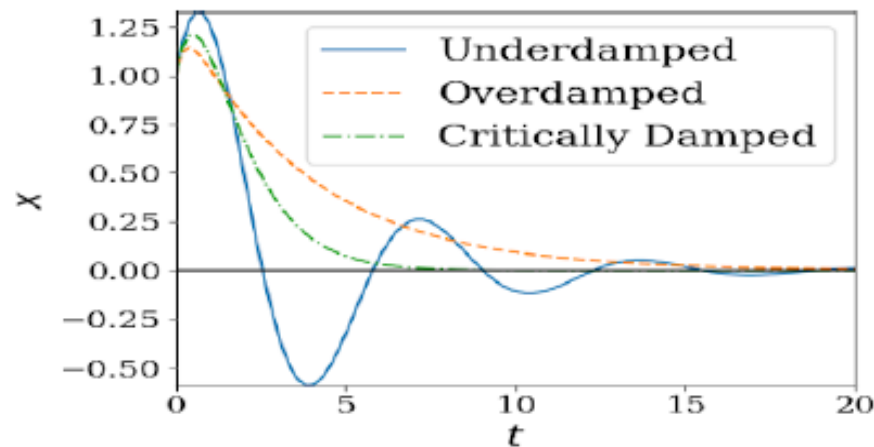
$$\sqrt{\gamma^2 - \omega^2} = \sqrt{-(\omega^2 - \gamma^2)} = \pm i\sqrt{\omega^2 - \gamma^2} = \pm i\omega_0 \text{ (say)}$$

$$\text{So, } x(t) = e^{-\gamma t} [C e^{i\omega_0 t} + D e^{-i\omega_0 t}]$$

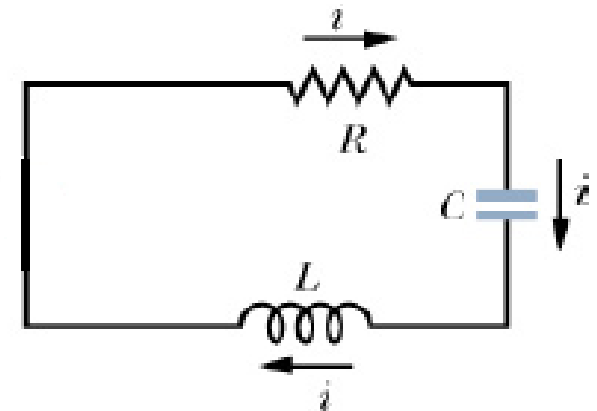
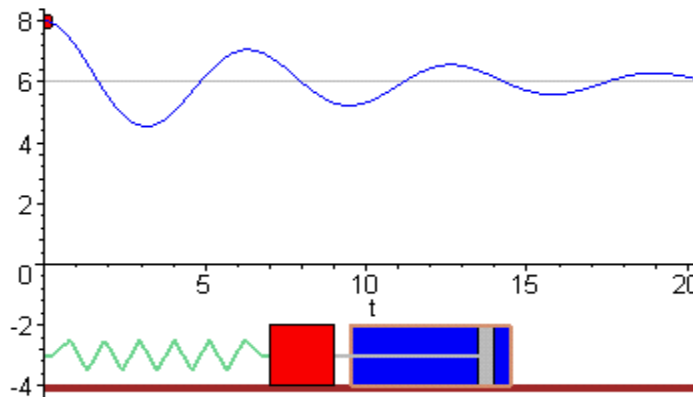
$$\therefore x(t) = A e^{-\gamma t} \cos(\omega_0 t - \phi)$$

where A and ϕ are two constants. The time period of oscillation is

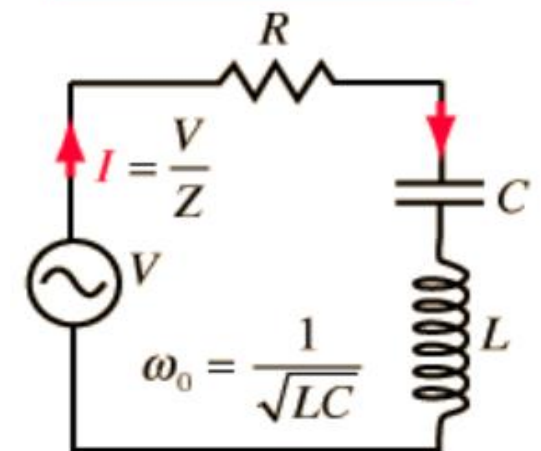
$$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{\sqrt{\omega^2 - \gamma^2}}$$



RLC circuit (DHM in Electric Circuit):



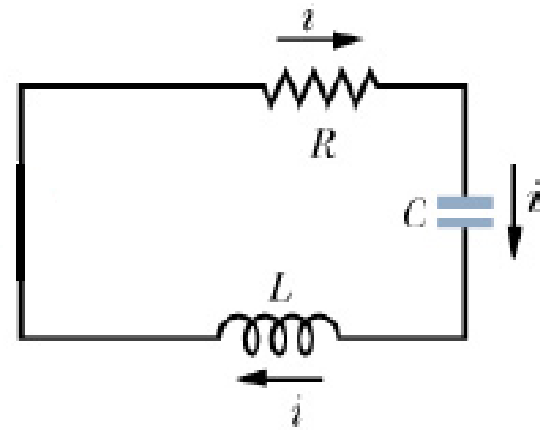
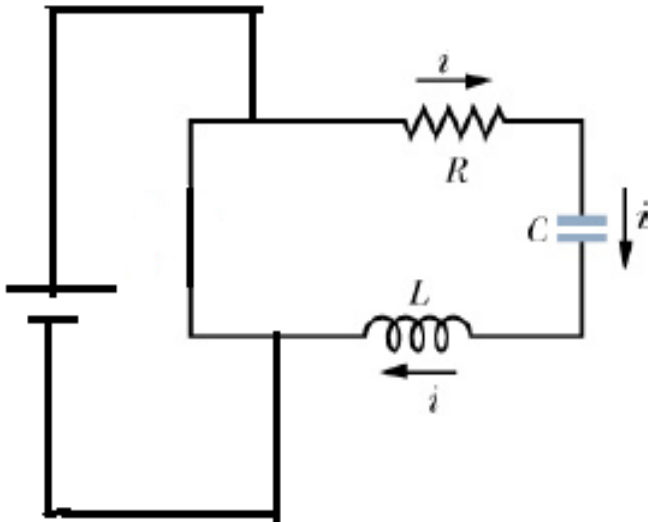
Minimum impedance
at resonant frequency



Series Resonance

❑ *Derive the differential equation of DHM and find its solution (Electrical system, RLC Circuit):*

RLC circuit (DHM in Electric Circuit):



RLC circuit (DHM in Electric Circuit):

- Voltage across resistor R $V_R = iR$
- Voltage across capacitor C $V_C = \frac{Q}{C}$
- Voltage across inductor L $V_L = L \frac{di}{dt}$
- According to Kirchhoff's voltage law

$$iR + \frac{Q}{C} + L \frac{di}{dt} = 0$$

DHM with Electrical System (RLC Circuit):

Rewrite the equation

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$$

Comparing with the equation

$$\frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

Where

$$\gamma = \frac{R}{L} \quad \omega_0 = \sqrt{\frac{1}{LC}}$$

DHM with Electrical System (RLC Circuit):

Three distinguish cases are

i) $\frac{1}{LC} > \frac{R^2}{4L^2}$ Oscillatory behavior

ii) $\frac{1}{LC} = \frac{R^2}{4L^2}$ Critical damping

iii) $\frac{1}{LC} < \frac{R^2}{4L^2}$ Over damping

DHM with Electrical System (RLC Circuit):

Case i) $\frac{1}{LC} > \frac{R^2}{4L^2}$

Solution of the differential equation

$$Q(t) = Ae^{-\frac{R}{2L}t} \cos(\omega_1 t + \phi)$$

Where $\omega_1 = \sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)}$

Frequency of oscillation

$$f = \frac{1}{2\pi} \sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)}$$

❑ **Derive the differential equation of DHM and find its solution (Electrical system, RLC Circuit): Alternative Method**

A RLC Circuit:

Let us consider an LCR oscillating circuit. Using Kirchhoff loop rule we can write

$$L \frac{di}{dt} + iR + \frac{q}{C} = 0$$

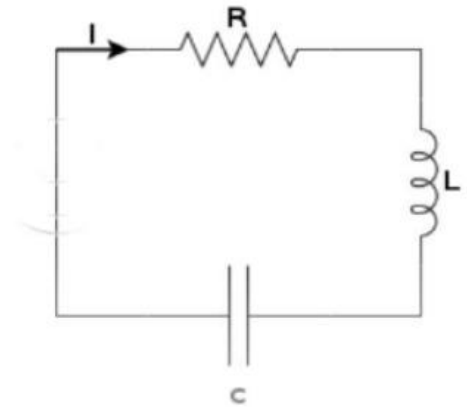
$$\text{or, } \frac{d^2q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{LC} = 0 \quad (9)$$

In order to solve this second order ordinary differential equation let consider a trial solution

$$q = e^{\alpha t} \quad (10)$$

$$\text{So, } \frac{dq}{dt} = \alpha e^{\alpha t} \quad (11)$$

$$\frac{d^2q}{dt^2} = \alpha^2 e^{\alpha t} \quad (12)$$



Inserting equations (9), (10) and (11) in (8) and setting damping constant $R/L = 2\gamma$, and $1/LC = \omega^2$ we have

Alternative Method (DHM in Electrical System: RLC Circuit)

$$(\alpha^2 + 2\gamma\alpha + \omega^2) e^{\alpha t} = 0$$

But $e^{\alpha t} \neq 0$. Therefore,

$$\begin{aligned}\alpha^2 + 2\gamma\alpha + \omega^2 &= 0 \\ \therefore \alpha &= -\gamma \pm \sqrt{\gamma^2 - \omega^2}\end{aligned}$$

The general solution is

$$\begin{aligned}q &= A e^{-\gamma t + \sqrt{\gamma^2 - \omega^2} t} + B e^{-\gamma t - \sqrt{\gamma^2 - \omega^2} t} \\ &= e^{-\gamma t} (A e^{\sqrt{\gamma^2 - \omega^2} t} + B e^{-\sqrt{\gamma^2 - \omega^2} t})\end{aligned}$$

Alternative Method (DHM in Electrical System: RLC Circuit)

Case-I: If $\gamma > \omega$, $\sqrt{\gamma^2 - \omega^2}$ is positive. This is the case of overdamped motion.

Case-II: If $\gamma = \omega$

$$q = e^{-\gamma t}(A + B)$$

This is the situation of critically damped motion.

Case-III: If $\gamma < \omega$, $\sqrt{\gamma^2 - \omega^2} = i\sqrt{\omega^2 - \gamma^2}$. So,

$$\begin{aligned} q(t) &= e^{-\gamma t}(Ae^{i\sqrt{\omega^2 - \gamma^2}t} + Be^{-i\sqrt{\omega^2 - \gamma^2}t}) \\ &= q_m e^{-\gamma t} \cos(\omega_0 t - \delta) \end{aligned}$$

where $\omega_0 = \sqrt{\omega^2 - \gamma^2}$. This is the case of damped oscillation.

EX-15: A capacitor $1.0\mu\text{F}$, an inductor 0.2h and a resistance 800Ω are joined in series. Is the circuit oscillatory?
Find the frequency of oscillation.

Ex-16: Find whether the discharge of capacitor through the following inductive circuit is oscillatory.
 $C = 0.1\mu\text{F}$, $L = 10\text{mh}$, $R = 200\Omega$
If Oscillatory, find the frequency of oscillation.

Example-0.22:

Given $L = 10 \text{ mh}$, $C = 1.0 \text{ } \mu\text{F}$ and $R = 0.1 \text{ } \Omega$. After how long a time will the charge oscillations decay to half amplitude?

Solution:

Suppose that after time t_0 the charge oscillation will decay to half amplitude. So

$$\begin{aligned} q_m e^{-\gamma(t+t_0)} &= \frac{1}{2} q_m e^{-\gamma t} \\ \text{or, } e^{-\gamma t_0} &= \frac{1}{2} \end{aligned}$$

Taking logarithm on both sides we have

$$\begin{aligned} -\gamma t_0 &= \ln\left(\frac{1}{2}\right) \\ \therefore t_0 &= \frac{1}{\gamma} \ln 2 = \frac{2L}{R} \ln 2 \\ &= \frac{2(10 \times 10^{-3} \text{ h}) \ln 2}{0.1 \text{ } \Omega} = 0.14 \text{ s} \end{aligned}$$

The angular frequency

$$\begin{aligned}\omega_0 &= \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} \\&= \sqrt{\frac{1}{10 \times 10^{-3} \text{ h} \times 1.0 \times 10^{-6} \text{ F}} - \left(\frac{0.1 \text{ } \Omega}{2 \times 10.0 \times 10^{-3} \text{ h}}\right)^2} \\&= 1.0 \times 10^4 \text{ rad/s}\end{aligned}$$



Lissajous' Figures: When particle is influenced simultaneously by two simple harmonic motion at right angles to each other, the resultant motion of the particle traces a curve. These curves are called Lissajous' figures. The shape of the curves depends on the time period, phase difference and amplitude of the constituent vibrations.

Lissajou's Figure Design

Combination of two vibrations at right angles

$$x = A_1 \cos(\omega_1 t + \phi_1)$$

$$y = A_2 \cos(\omega_2 t + \phi_2)$$

Consider case where frequencies are equal and let initial phase difference be ϕ

$$\text{Write } x = A_1 \cos(\omega_0 t) \quad \text{and} \quad y = A_2 \cos(\omega_0 t + \phi)$$

$$\text{Case 1 : } \phi = 0 \quad \left. \begin{array}{l} x = A_1 \cos(\omega_0 t) \\ y = A_2 \cos(\omega_0 t) \end{array} \right\} y = \frac{A_2}{A_1} x \quad \text{Rectilinear motion}$$

$$\begin{aligned} \text{Case 2 : } \phi = \pi/2 \quad x &= A_1 \cos(\omega_0 t) \\ y &= A_2 \cos(\omega_0 t + \pi/2) = -A_2 \sin(\omega_0 t) \\ \therefore \frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} &= 1 \end{aligned} \quad \begin{array}{l} \text{Elliptical path in} \\ \text{clockwise direction} \end{array}$$

Lissajou's Figure Design

Combination of two vibrations at right angles ...2

$$\left. \begin{array}{l} \text{Case 3 : } \phi = \pi \quad x = A_1 \cos(\omega_0 t) \\ \quad \quad \quad y = A_2 \cos(\omega_0 t + \pi) = -A_2 \cos(\omega_0 t) \end{array} \right\} y = -\frac{A_2}{A_1} x$$

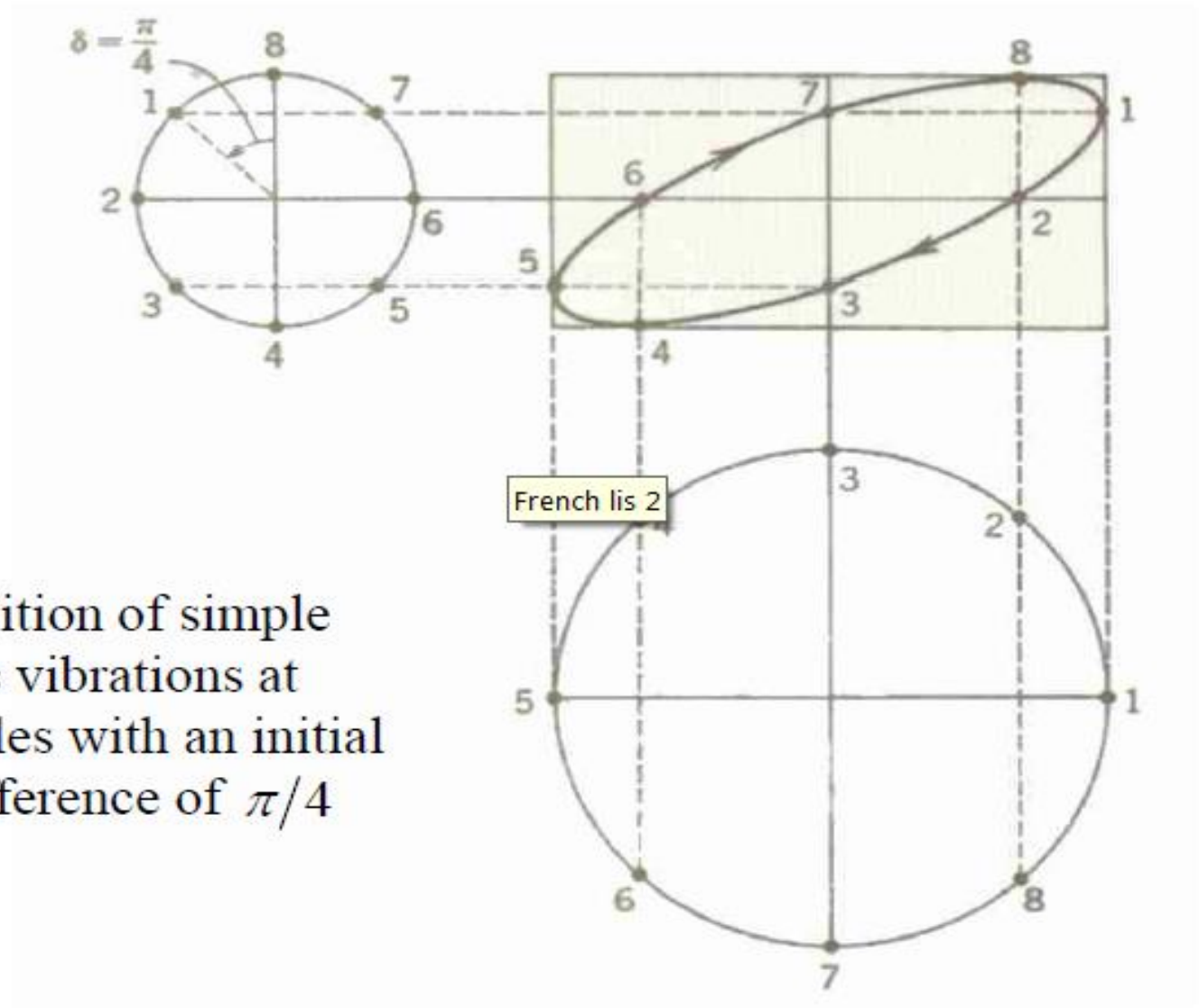
$$\begin{array}{l} \text{Case 4 : } \phi = 3\pi/2 \quad x = A_1 \cos(\omega_0 t) \\ \quad \quad \quad y = A_2 \cos(\omega_0 t + 3\pi/2) = +A_2 \sin(\omega_0 t) \\ \quad \quad \quad \therefore \frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = -1 \quad \text{Elliptical path in} \\ \quad \quad \quad \quad \quad \quad \quad \quad \quad \text{anticlockwise direction} \end{array}$$

$$\begin{array}{l} \text{Case 5 : } \phi = \pi/4 \quad x = A_1 \cos(\omega_0 t) \\ \quad \quad \quad y = A_2 \cos(\omega_0 t + \pi/4) \end{array}$$

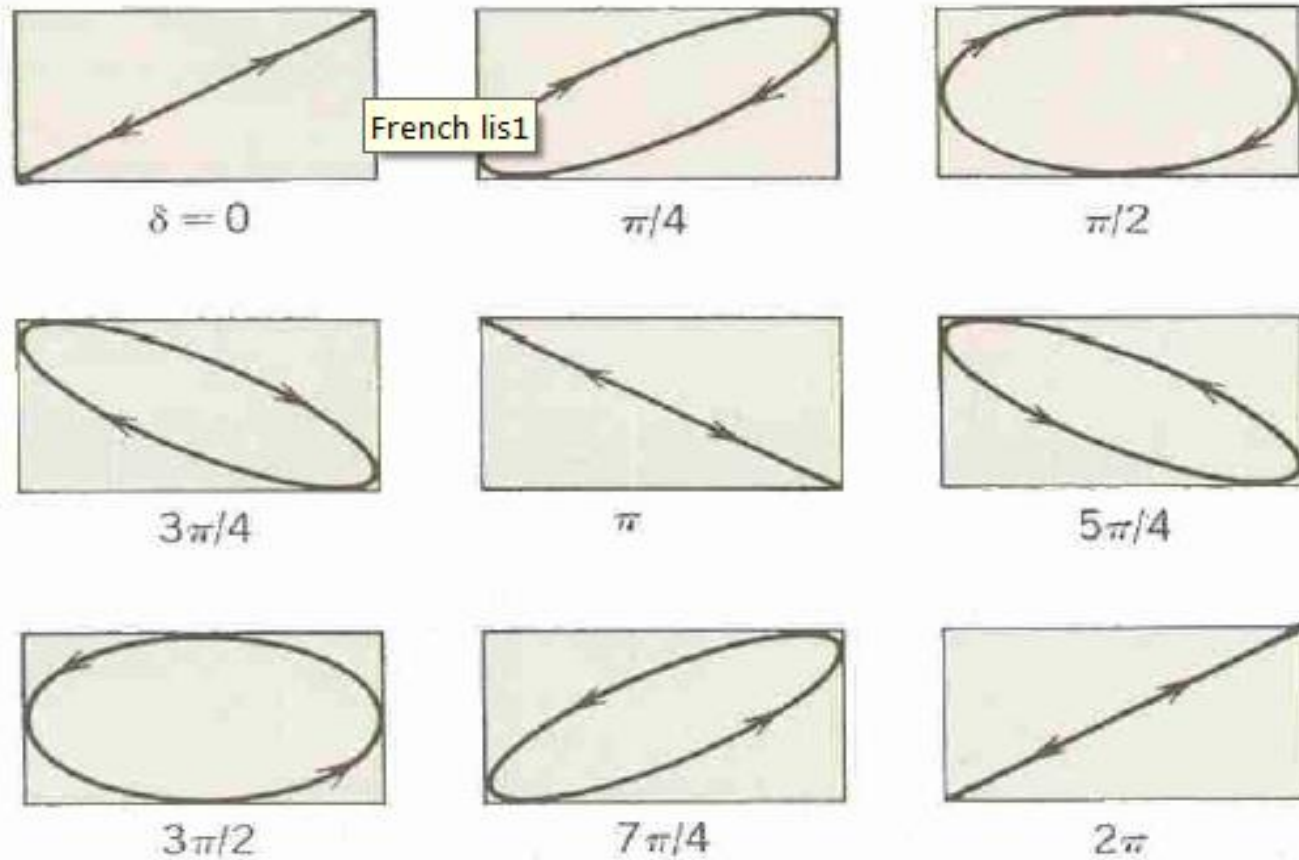
Harder to see ... use a graphical approach ...

Lissajou's Figure Design

Superposition of simple harmonic vibrations at right angles with an initial phase difference of $\pi/4$



Lissajou's Figure Design



Superposition of two perpendicular simple harmonic motions of the same frequency for various initial phase differences.

Wave Motion

- ✦ Differential Equation of Wave motion
- ✦ Progressive and Standing Waves
- ✦ Group and Phase Velocity
- ✦ Power and Intensity of Wave Motion

Review: Simple Harmonic Motion

- The position x of an object moving in simple harmonic motion as a function of time has the following form:

$$x = A \cos (\omega t + \phi)$$

i.e. the object periodically moves back and forth between the amplitudes $x=+A$ and $x=-A$.

The time it takes for the object to make one full cycle is the period $T=2\pi/\omega=1/f$, where f is the frequency of the motion.

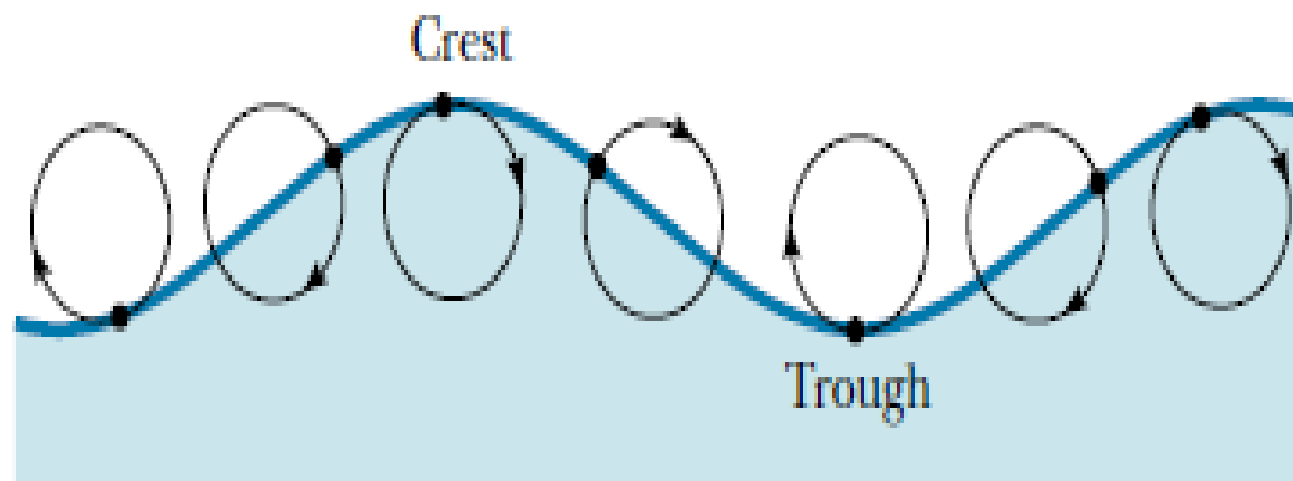
Thus, the angular speed in terms of T and f reads

$$\omega = 2\pi/T \quad \text{and} \quad \omega = 2\pi f$$

What is a wave ?

- Nature of waves:
 - ➔ A wave is a traveling disturbance that transports energy from place to place.
 - ➔ There are two basic types of waves: transverse and longitudinal.
 - ➔ Transverse: the disturbance travels perpendicular to the direction of travel of the wave.
 - ➔ Longitudinal: the disturbance occurs parallel to the line of travel of the wave.

Wave motion 



- Examples:

- ➔ Longitudinal: Sound waves (e.g. air moves back & forth)

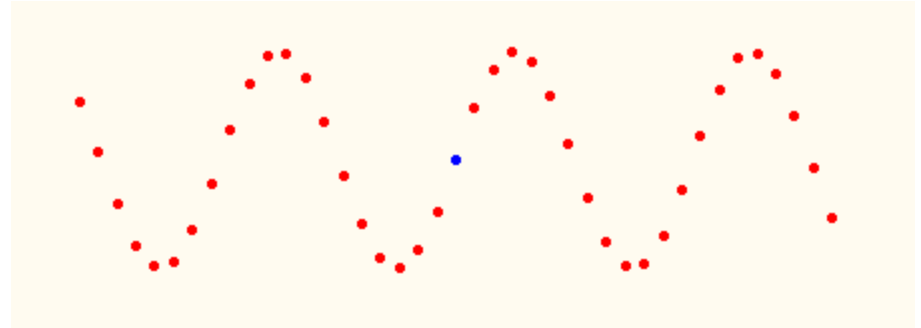
- ➔ Transverse: Light waves (electromagnetic waves, i.e. electric and magnetic disturbances)

The source of the wave, i.e. the disturbance, moves continuously in simple harmonic motion, generating an entire wave, where each part of the wave also performs a simple harmonic motion.

Types of Waves

- **Transverse:** The medium oscillates perpendicular to the direction the wave is moving.

→ Water waves

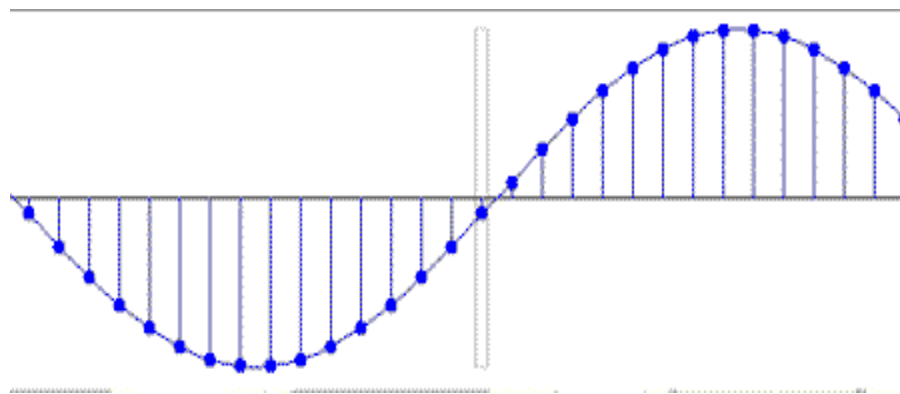
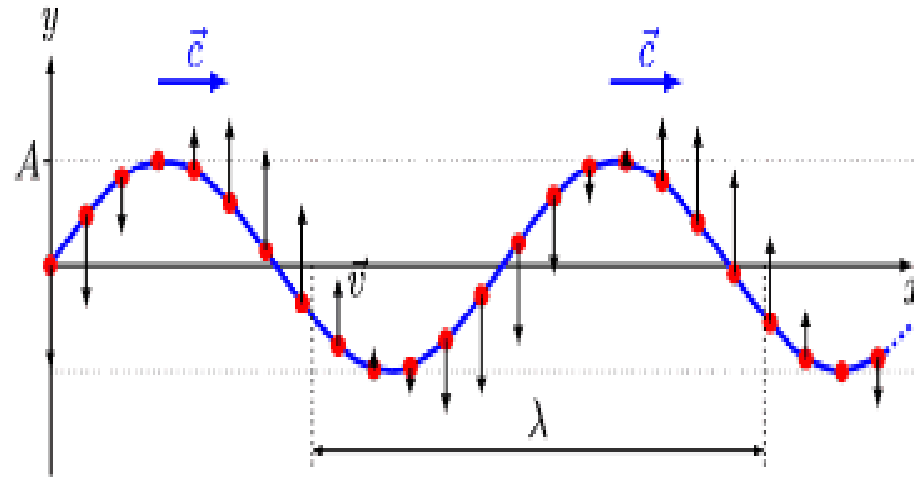


- **Longitudinal:** The medium oscillates in the same direction as the wave is moving

→ Sound

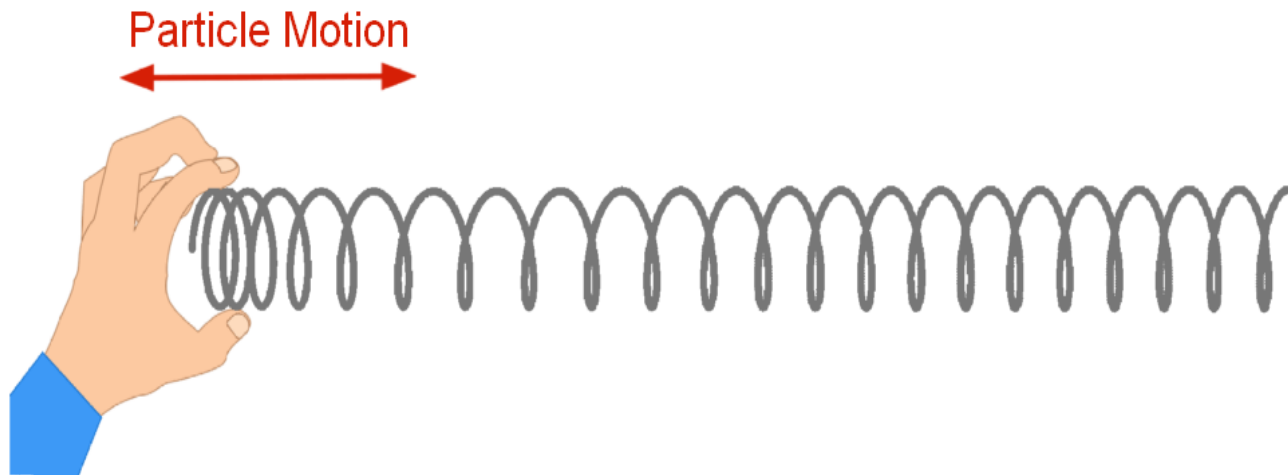
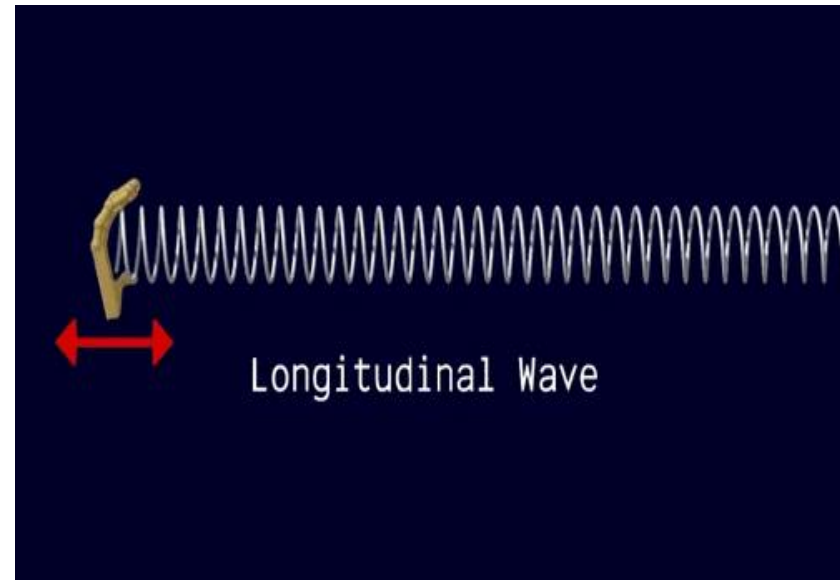
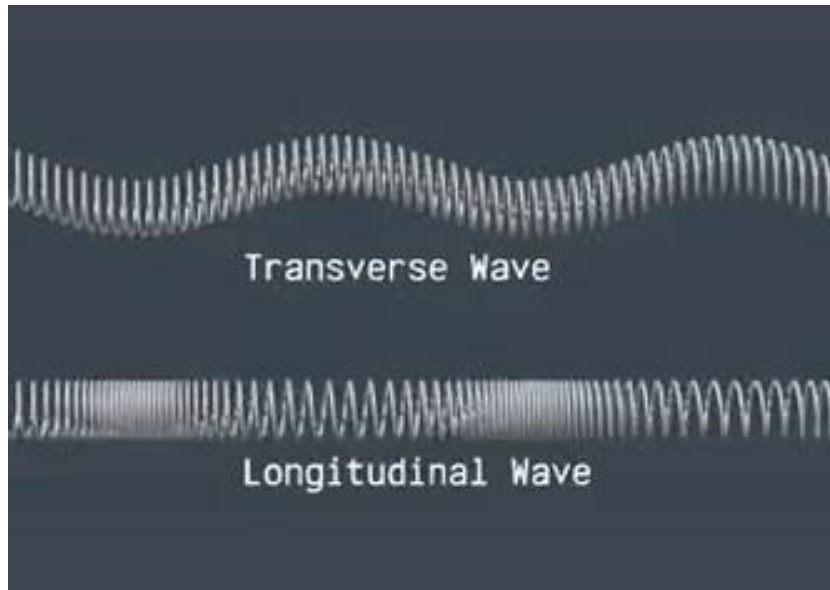


Types of Waves



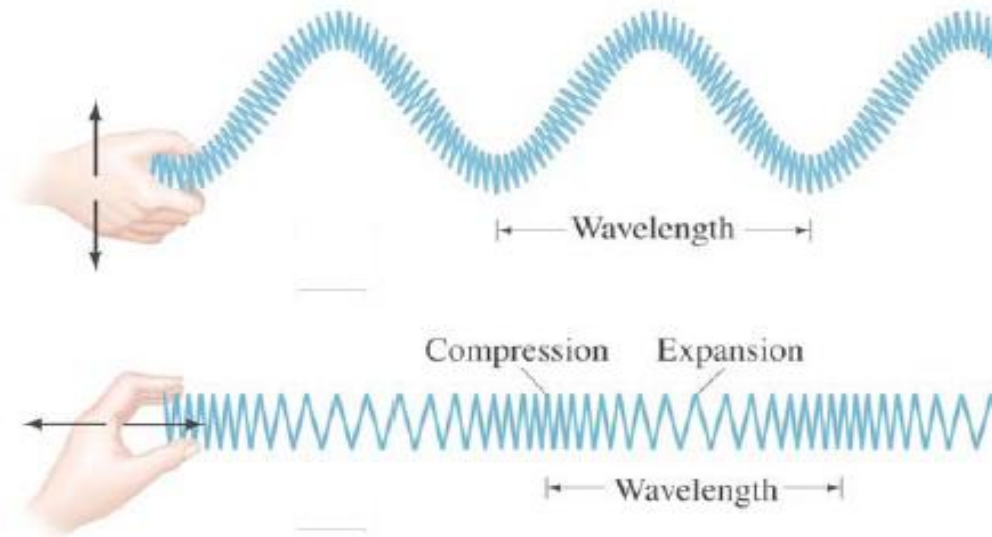
Transverse wave

Types of Waves



Longitudinal wave

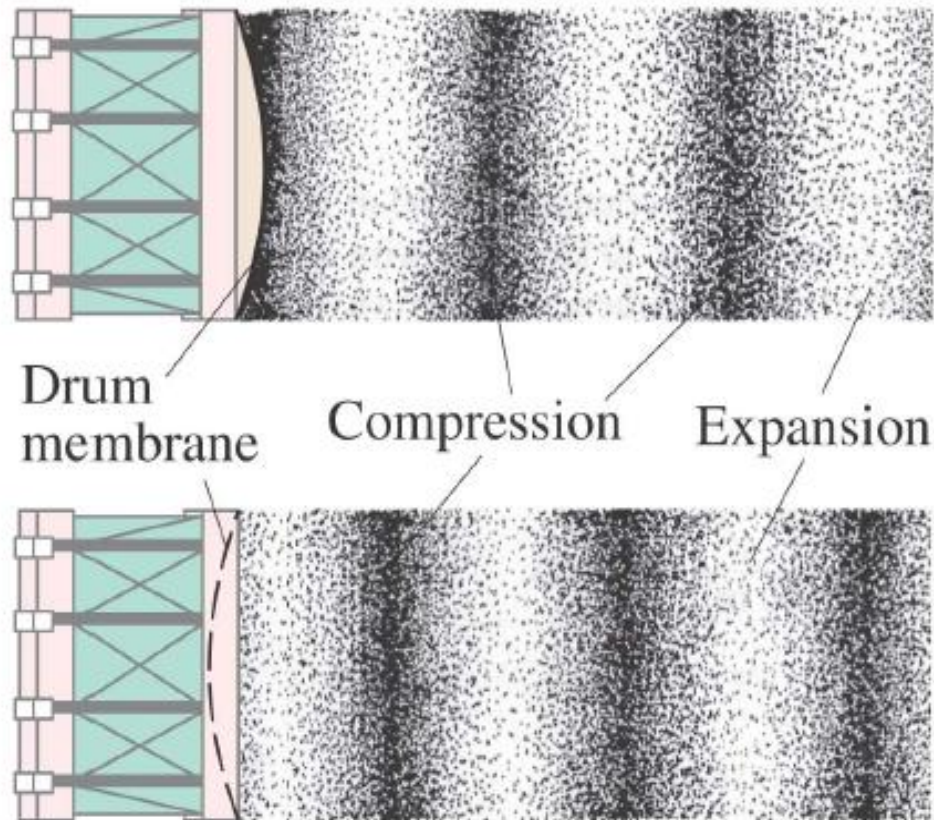
Types of Waves: Transverse and Longitudinal



The motion of particles in a wave can be either perpendicular to the wave direction (transverse) or parallel to it (longitudinal).

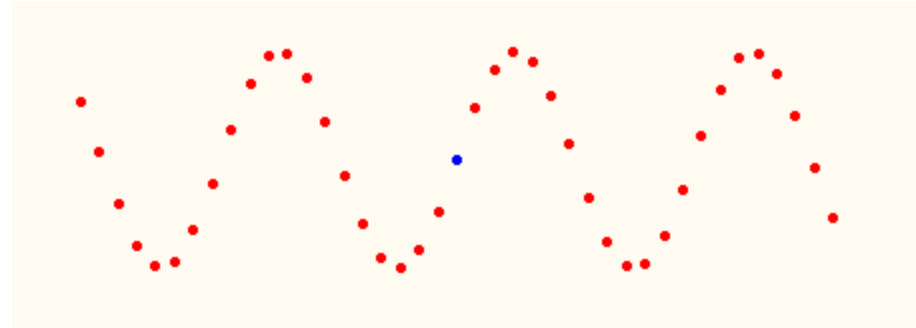
Types of Waves: Transverse and Longitudinal

Sound waves are longitudinal waves:



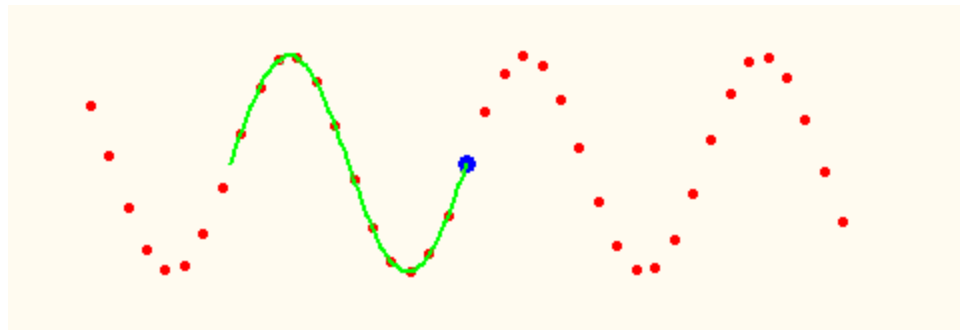
Wave Properties...

- Period: The time T for a point on the wave to undergo one complete oscillation.



- Speed: The wave moves one wavelength λ in one period T so its speed is $v = \lambda / T$.

$$v = \frac{\lambda}{T}$$



Wave Properties...

- The speed of a wave is a constant that depends only on the medium, not on the amplitude, wavelength or period:

λ and T are related !

 $\lambda = v T$ or $\lambda = 2\pi v / \omega$ (since $T = 2\pi / \omega$)

or $\lambda = v / f$ (since $T = 1 / f$)

- Recall $f = \text{cycles/sec}$ or revolutions/sec

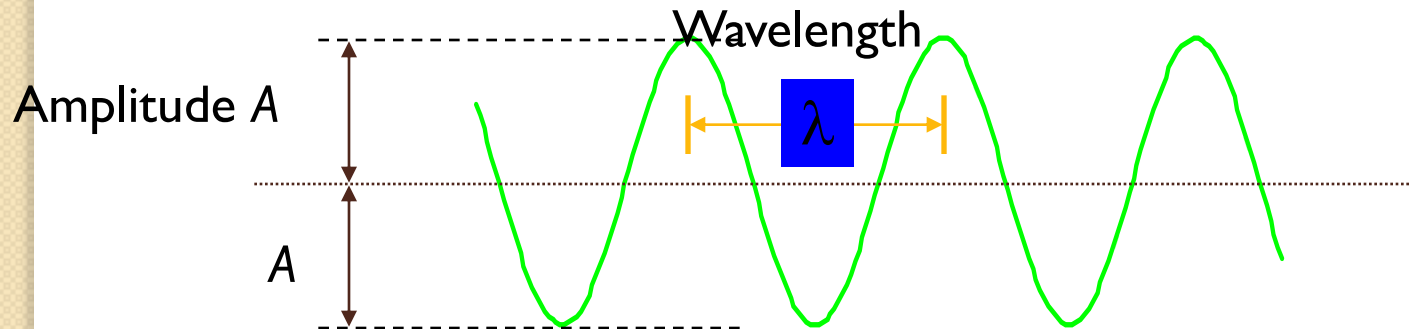
$$\omega = 2\pi f$$

Is the speed of a wave particle the same as the speed of the wave ?

No. Wave particle performs simple harmonic motion: $y = A \sin \omega t$.

Wave Properties

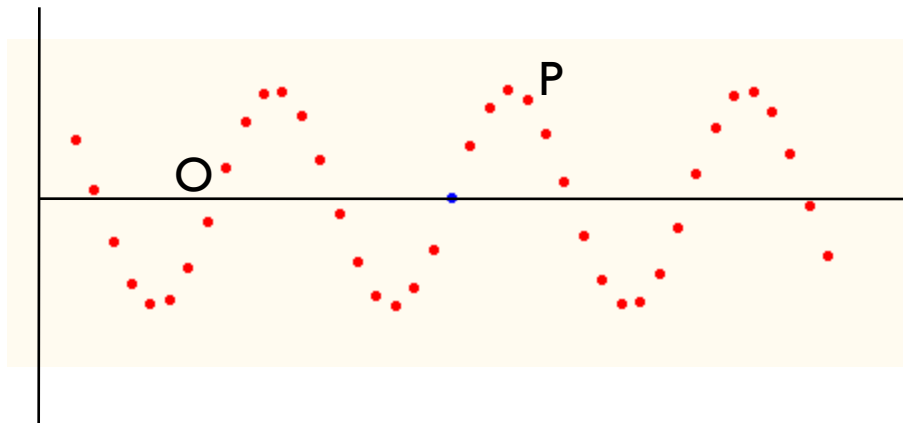
- Amplitude: The maximum displacement A of a point on the wave.
- Wavelength: The distance λ between identical points on the wave.



❑ Derive the linear and differential equation of Travelling wave or Progressive wave

Equation for a Progressive Wave

The simplest type of wave is the one in which the particles of the medium are set into simple harmonic vibrations as the wave passes through it. The wave is then called a simple harmonic wave.



Linear and Differential equation of Travelling wave or Progressive wave

Consider a particle O in the medium.

The displacement at any instant of time is given by

$$y = A \sin \omega t \dots\dots(1)$$

Where A is the amplitude, ω is the angular frequency of the wave. Consider a particle P at a distance x from the particle O on its right.

Let the wave travel with a velocity v from left to right. Since it takes some time for the disturbance to reach P, its displacement can be written as

$$y = A \sin (\omega t - \phi) \dots \dots \dots (2)$$

Where ϕ is the phase difference between the particles O and P.

We know that a path difference of λ corresponds to a phase difference of 2π radians. Hence a path difference of x corresponds to a phase difference of

$$\frac{2\pi}{\lambda} \cdot x$$

Linear and Differential equation of Travelling wave or Progressive wave

$$\phi = \frac{2\pi x}{\lambda}$$

Displacement of particle P is

$$y = A \sin\left(\omega t - \frac{2\pi x}{\lambda}\right) \dots \dots \dots (3)$$

$$\text{But } \omega = \frac{2\pi}{T}$$

$$y = A \sin\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right) \dots \dots \dots (5)$$

$$y = A \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

$$\text{But } v = \frac{\lambda}{T} \text{ or } T = \frac{\lambda}{v}$$

$$y = A \sin 2\pi \left(\frac{vt}{\lambda} - \frac{x}{\lambda} \right)$$

$$y = A \sin \frac{2\pi}{\lambda} (vt - x) \dots \dots \dots (6)$$

Linear and Differential equation of Travelling wave or Progressive wave

Similarly, for a particle at a distance x to the left of 0, the equation for the displacement is given by

$$y = A \sin \frac{2\pi}{\lambda} (vt + x) \dots \dots \dots (7)$$

Differential equation for wave motion

We have wave equation

$$y = A \sin \frac{2\pi}{\lambda} (vt - x) \dots \dots \dots (1)$$

Differentiating equation with respect to x,

We get,

$$\frac{dy}{dx} = -A \frac{2\pi}{\lambda} \cos \frac{2\pi}{\lambda} (vt - x) \dots \dots \dots (2)$$

Differential equation of Travelling wave or Progressive wave

$\frac{dy}{dx}$ represents the strain or the compression. When $\frac{dy}{dx}$ is positive, a rarefaction takes place

and when $\frac{dy}{dx}$ is negative, a compression takes place.

The velocity of the particle whose displacement y is represented by equation, is obtained by differentiating it with respect to t , since velocity is the rate of change of displacement with respect to time.

$$\frac{dy}{dt} = A \frac{2\pi v}{\lambda} \cos \frac{2\pi}{\lambda} (vt - x) \dots \dots \dots (3)$$

Comparing equations (2) and (3) we get,

$$\frac{dy}{dt} = -v \frac{dy}{dx} \dots\dots\dots(4)$$

Particle velocity = wave velocity x slope of the displacement curve or strain.

Differentiating equation (2)

$$\frac{d^2 y}{dx^2} = -A \frac{4\pi^2}{\lambda^2} \sin \frac{2\pi}{\lambda} (vt - x) \dots\dots\dots(5)$$

Differentiating equation (3)

$$\frac{d^2 y}{dt^2} = -A \frac{4\pi^2 v^2}{\lambda^2} \sin \frac{2\pi}{\lambda} (vt - x) \dots \dots \dots (6)$$

Comparing equs. (5) and (6)

$$\frac{d^2 y}{dt^2} = v^2 \frac{d^2 y}{dx^2} \dots \dots \dots (7)$$

Equation (7) represents the differential equation of wave velocity

Group Velocity and Phase velocity

"Freshman" waves go on forever...

(but they are nonetheless instructive)

A wave that propagates forever in one dimension is described by:

$$y = A \cos\left(\frac{2\pi x}{\lambda} - 2\pi f t\right)$$

$$v_p = \lambda f$$

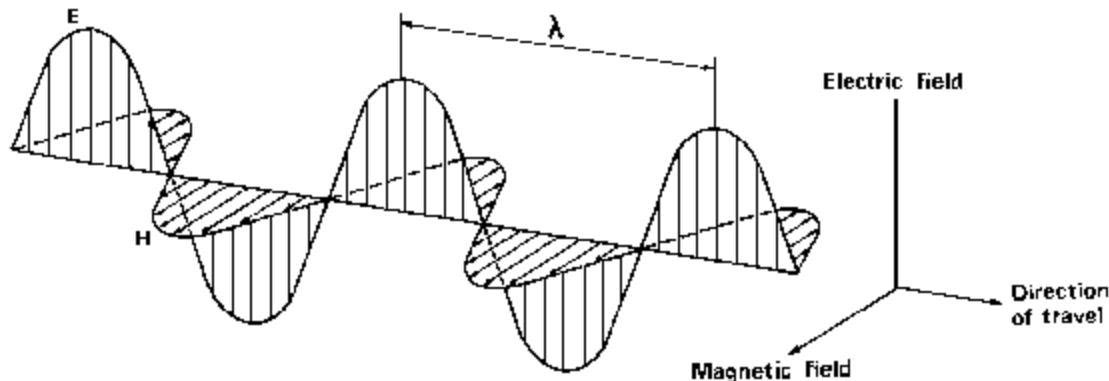
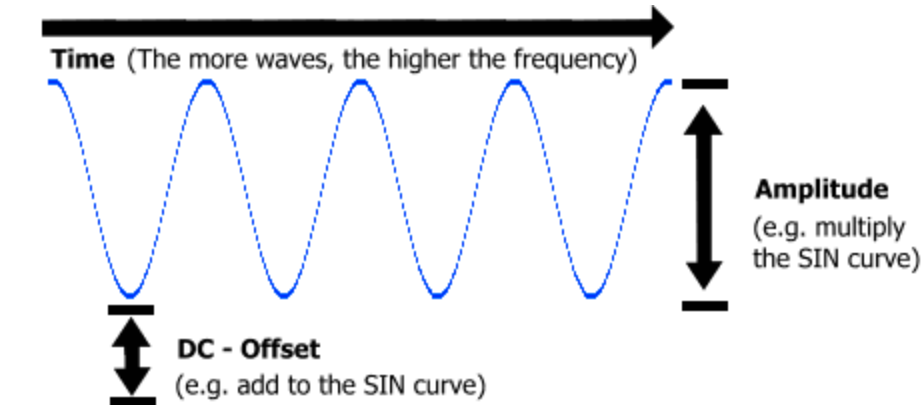
in shorthand:

$$y = A \cos(kx - \omega t)$$

$$\omega = 2\pi f, \quad k = 2\pi / \lambda$$

angular
frequency

wave
number



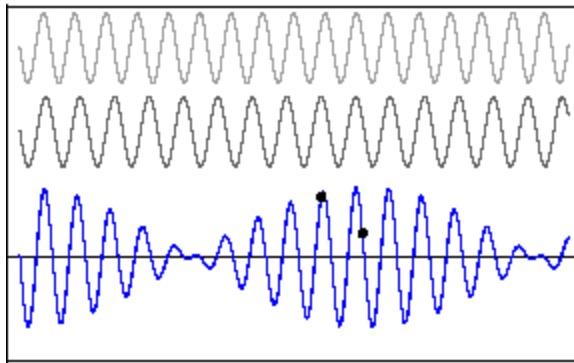
Interfering waves, generally...

Group Velocity and Phase velocity

$$y = y_1 + y_2 = A \cos(k_1 x - \omega_1 t) + A \cos(k_2 x - \omega_2 t)$$

$\Downarrow$

$$y = 2A \cos \frac{1}{2} \{ (k_2 - k_1)x - (\omega_2 - \omega_1)t \} \bullet \cos \frac{1}{2} \{ (k_1 + k_2)x - (\omega_1 + \omega_2)t \}$$



“Beats” occur when you add two waves of slightly different frequency. They will interfere constructively in some areas and destructively in others.

Can be interpreted as a sinusoidal envelope:

$$2A \cos \left(\frac{\Delta k}{2} x - \frac{\Delta \omega}{2} t \right)$$

Modulating a high frequency wave within the envelope: $\cos \left[\frac{1}{2} (k_1 + k_2)x - \frac{1}{2} (\omega_1 + \omega_2)t \right]$

the group
velocity

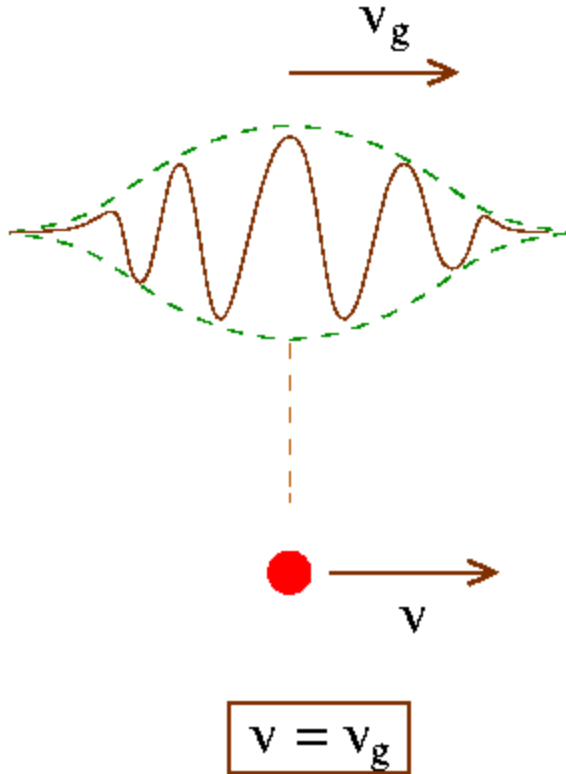
$$v_g = \frac{(\omega_2 - \omega_1)/2}{(k_2 - k_1)/2} = \frac{\Delta \omega}{\Delta k}$$

the phase
velocity

$$v_p = \frac{(\omega_2 + \omega_1)/2}{(k_2 + k_1)/2} \approx \frac{\omega_1}{k_1} = v_1$$

if $v_1 \approx v_2$

Group Velocity and Phase velocity



the group
velocity

$$v_g = \frac{(\omega_2 - \omega_1)/2}{(k_2 - k_1)/2} = \frac{\Delta\omega}{\Delta k}$$

Listen to the beats!



Example-17: The equation of a traveling wave is

$$y = 4.0 \sin \pi(0.10x - 2t)$$

Find (i) wavelength, (ii) speed and
(iii) frequency of oscillating particle of the wave

Example-18: When a simple harmonic wave is propagated through a medium, the displacement of a particle in cm at any instant is

$$y = 10 \sin \frac{2\pi}{100} (36000t - 20)$$

Calculate the amplitude, wave velocity, wavelength, frequency, and period of the oscillating particle.

Example-19: The equation of a traveling wave is

$$y = 4.0 \sin \pi(0.10t + 2x)$$

Find (i) amplitude (ii) wavelength (iii) speed and (iv) frequency of wave

Example-20: When a simple harmonic wave is propagated through a medium, the displacement of the particle at any instant of time is given by

$$y = 5.0 \sin \pi(360t - 0.15x)$$

Calculate

- (i) the amplitude of the vibrating particle,
- (ii) wave velocity,
- (ii) wave length,
- (iv) frequency and
- (v) time period.

Example-21: A simple harmonic wave of amplitude 8units travels a line of particles in the direction of positive X axis. At any instant for a particle at a distance of 10cm from the origin, the displacement is +6units and at a distance a particle from the origin is 25units, the displacement is +4units. Calculate the wavelength.

Standing Wave or Stationary Wave

Introduction to stationary waves

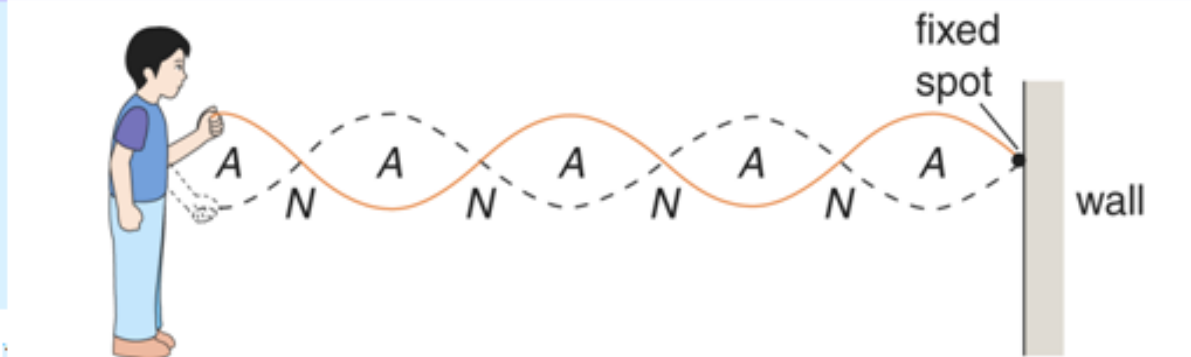


Stationary waves (standing waves) produced in

- string of guitar or piano
- rope (one end tie to wall)

- **Incident wave** is reflected by the wall to form **reflected wave**

- **superposition** of the waves produces stationary wave



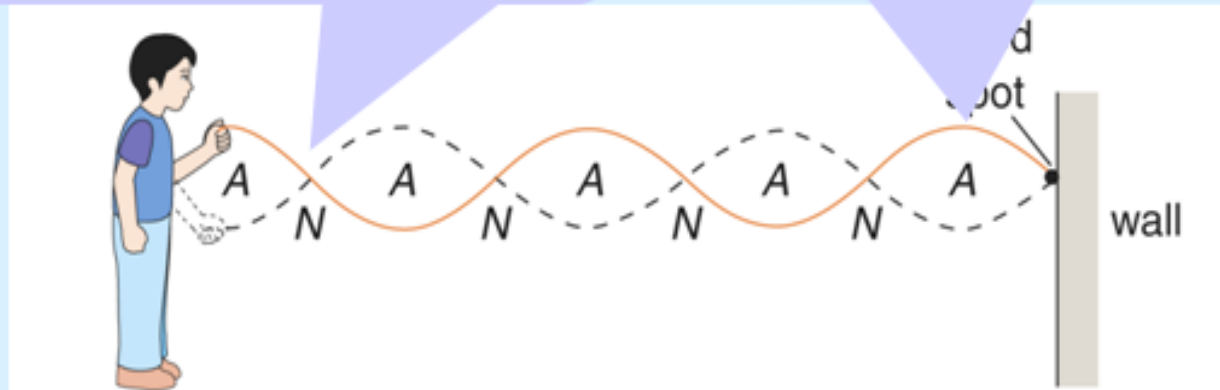
Standing Wave or Stationary Wave

Introduction to stationary waves



N – **nodes** (remain stationary)

A – **antinodes**
(largest amplitude of oscillation)



A stationary wave is produced by the superposition of two **progressive waves** of the same amplitude and frequency but travelling in opposite directions.

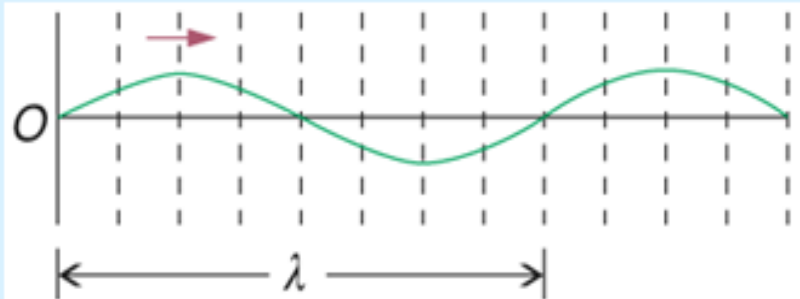
Standing Wave or Stationary Wave

Introduction to stationary waves

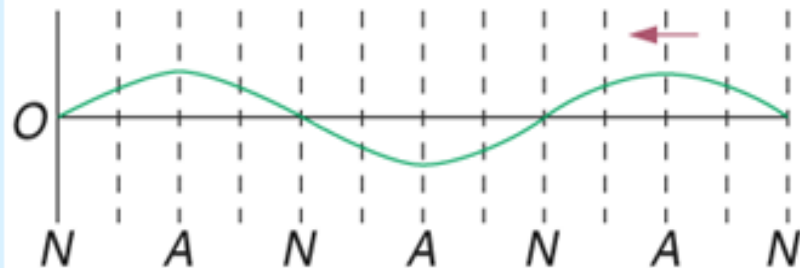


(a) at time t (**constructive interference**)

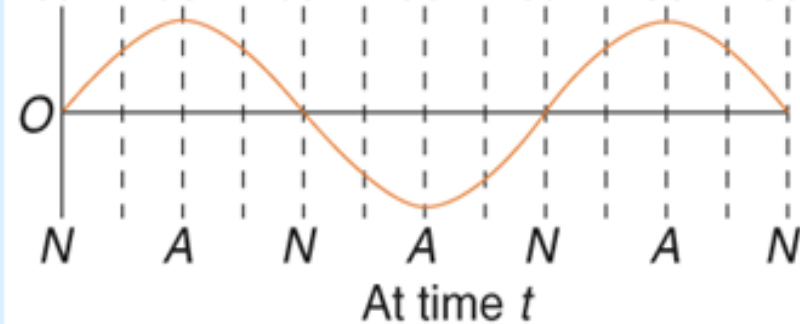
(1) To right



(2) To left



(1) + (2) = (3)



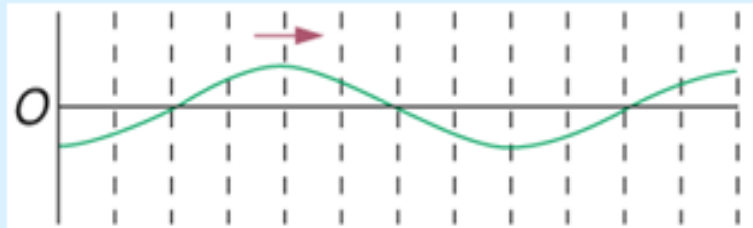
Standing Wave or Stationary Wave

Introduction to stationary waves

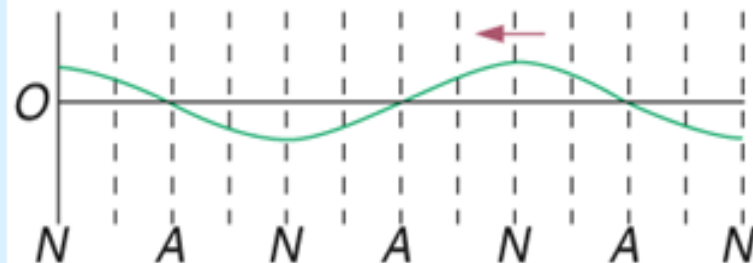


(b) at time $t + T/4$ (destructive interference)

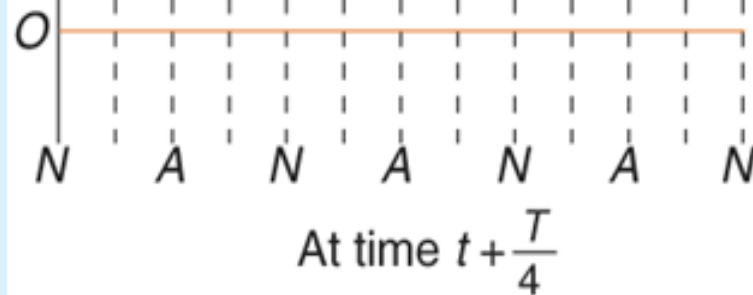
(1)



(2)



(1) + (2) = (3)

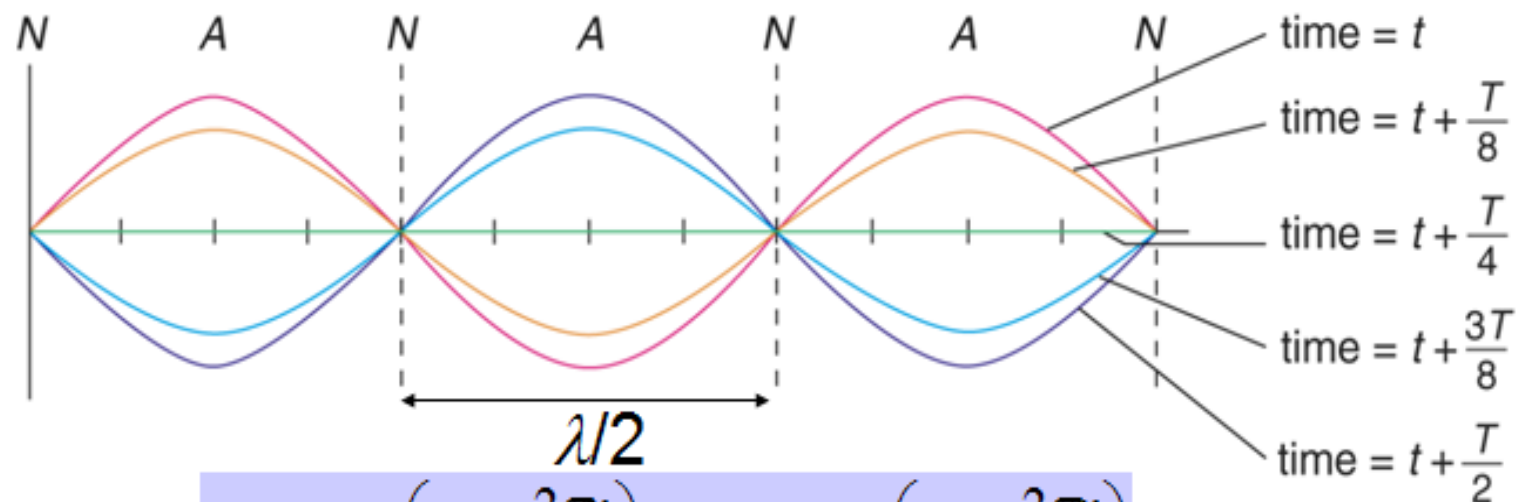


Standing Wave or Stationary Wave

Introduction to stationary waves



Vibration of stationary wave at various instants



$$y_1 = a \sin\left(\omega t - \frac{2\pi x}{\lambda}\right), \quad y_2 = a \sin\left(\omega t + \frac{2\pi x}{\lambda}\right)$$
$$y = y_1 + y_2 = \left(2a \cos \frac{2\pi x}{\lambda}\right) \sin \omega t$$

Amplitude (A) of resultant stationary wave

Standing Wave or Stationary Wave



The positions of antinodes

Progressive wave	Stationary wave
Energy is transferred along the direction of propagation.	No energy is transferred along the direction of propagation.
The wave profile moves in the direction of propagation.	The wave profile does not move in the direction of propagation.
Every point along the direction of propagation is displaced .	There are points known as nodes where no displacement occurs.
Every point has the same amplitude .	Points between two successive nodes have different amplitudes .
Neighbouring points are not in phase .	All points between two successive nodes vibrate in phase with one other.

Forced Harmonic Motion (FHM)

Forced Oscillations, FHM: Resonance

Forced vibrations occur when there is a periodic driving force. This force may or may not have the same period as the natural frequency of the system.

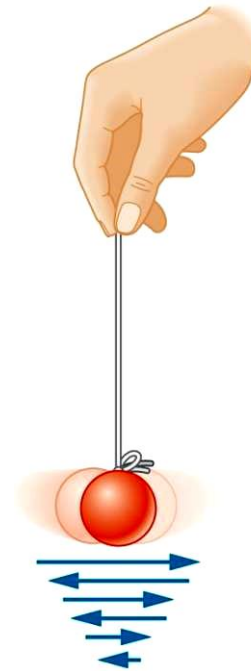
If the frequency is the same as the natural frequency, the amplitude can become quite large. This is called resonance.

Forced Harmonic Motion (FHM)



Figure : Caption: (a) Large-amplitude oscillations of the Tacoma Narrows Bridge, due to gusty winds, led to its collapse (1940). (b) Collapse of a freeway in California, due to the 1989 earthquake.

Forced Harmonic Motion (FHM)



□ *Derive the differential equation of FHM and find its solution (Mechanical system):*

Equation of Driven Harmonic Motion (FHM):

- Consider what happens when we apply a time-dependent force $F(t)$ to a system that normally would carry out SHM with an angular frequency ω_0 .
- Assume the external force $F(t) = mF_0\sin(\omega t)$. The equation of motion can now be written as

$$\frac{d^2x}{dt^2} = -\omega_0^2 x + F_0 \sin(\omega t)$$

- The steady state motion of this system will be harmonic motion with an angular frequency equal to the angular frequency of the driving force.

Forced Harmonic Motion (FHM)

- Consider the general solution

$$x(t) = A \cos(\omega t + \phi)$$

- The parameters in this solution must be chosen such that the equation of motion is satisfied. This requires that

$$-\omega^2 A \cos(\omega t + \phi) + \omega_0^2 A \cos(\omega t + \phi) - F_0 \sin(\omega t) = 0$$

- This equation can be rewritten as

$$\begin{aligned} &(\omega_0^2 - \omega^2) A \cos(\omega t) \cos(\phi) - \\ &(\omega_0^2 - \omega^2) A \sin(\omega t) \sin(\phi) - F_0 \sin(\omega t) = 0 \end{aligned}$$

Forced Harmonic Motion (FHM)

- Our general solution must thus satisfy the following condition:

$$(\omega_0^2 - \omega^2)A \cos(\omega t) \cos(\phi) - \{(\omega_0^2 - \omega^2)A \sin(\phi) - F_0\} \sin(\omega t) = 0$$

- Since this equation must be satisfied at all time, we must require that the coefficients of $\cos(\omega t)$ and $\sin(\omega t)$ are 0. This requires that

$$(\omega_0^2 - \omega^2)A \cos(\phi) = 0$$

and

$$(\omega_0^2 - \omega^2)A \sin(\phi) - F_0 = 0$$

Forced Harmonic Motion (FHM)

- The interesting solutions are solutions where $A \neq 0$ and $\omega \neq \omega_0$. In this case, our general solution can only satisfy the equation of motion if

$$\cos(\phi) = 0$$

and

$$(\omega_0^2 - \omega^2)A \sin(\phi) - F_0 = (\omega_0^2 - \omega^2)A - F_0 = 0$$

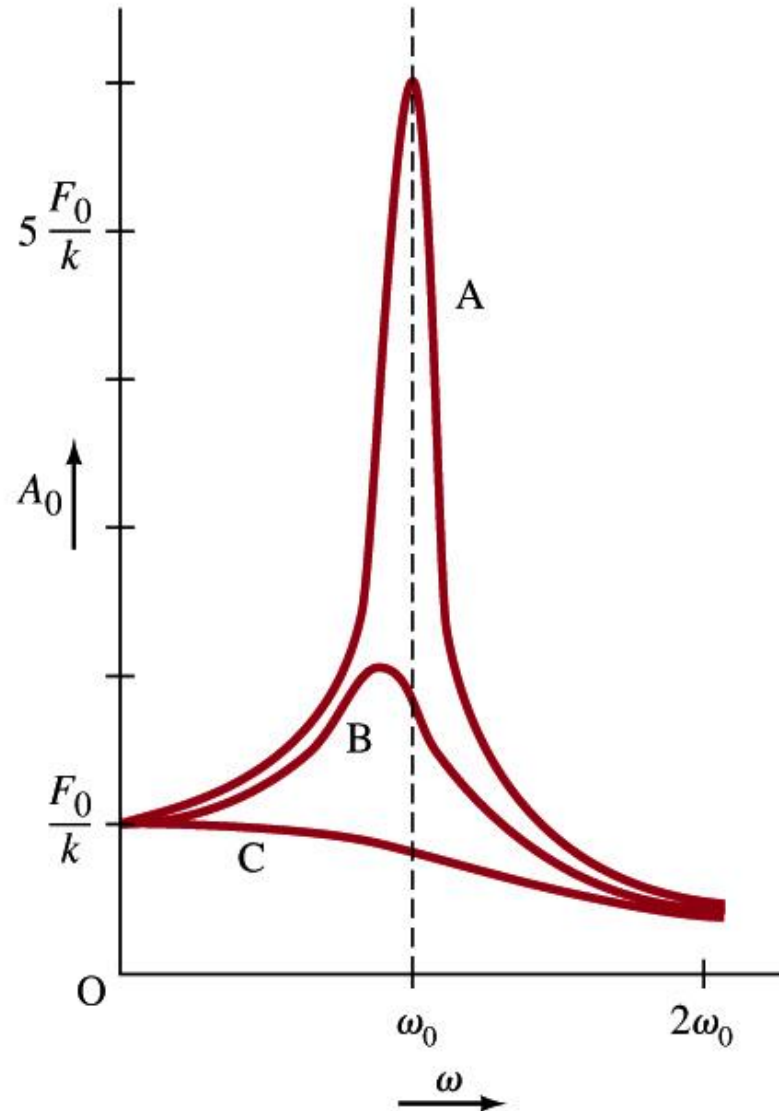
- The amplitude of the motion is thus equal to

$$A = \frac{F_0}{(\omega_0^2 - \omega^2)}$$

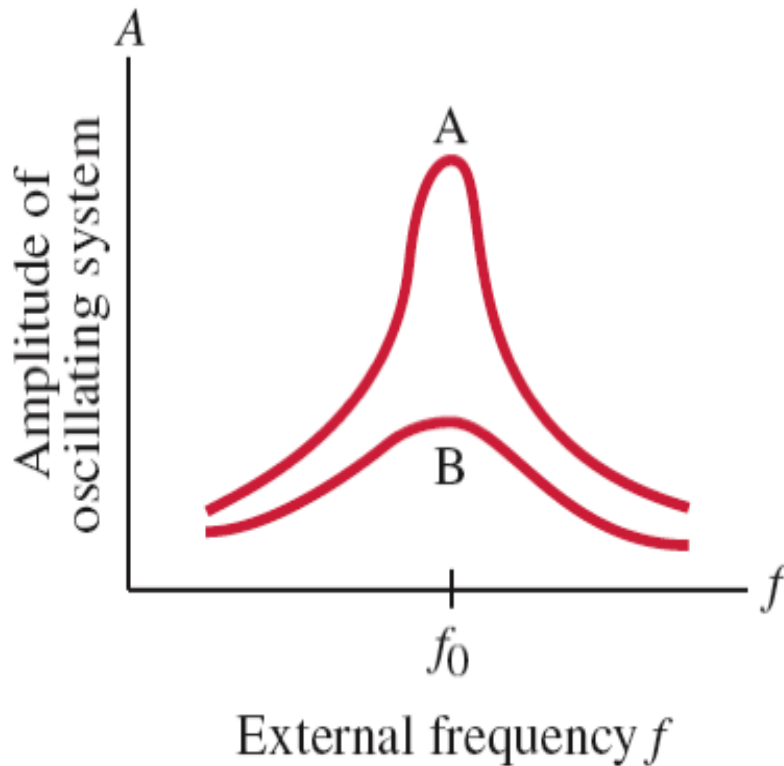
Forced Harmonic Motion (FHM)

- If the driving force has a frequency close to the natural frequency of the system, the resulting amplitudes can be very large even for small driving amplitudes. The system is said to be in resonance.
- In realistic systems, there will also be a damping force. Whether or not resonance behavior will be observed will depend on the strength of the damping term.

Driven Harmonic Motion



Forced Oscillations: Resonance



The sharpness of the resonant peak depends on the damping. If the damping is small (A) it can be quite sharp; if the damping is larger (B) it is less sharp.

Like damping, resonance can be wanted or unwanted. Musical instruments and TV/radio receivers depend on it.

□ *Derive the differential equation of FHM and find its solution (Mechanical system): Summary*

Forced Oscillations: Resonance

The equation of motion for a forced oscillator is:

$$ma = -kx - bv + F_0 \cos \omega t.$$

The solution is: $x = A_0 \sin(\omega t + \phi_0),$

where $A_0 = \frac{F_0}{m \sqrt{(\omega^2 - \omega_0^2)^2 + b^2 \omega^2 / m^2}}$

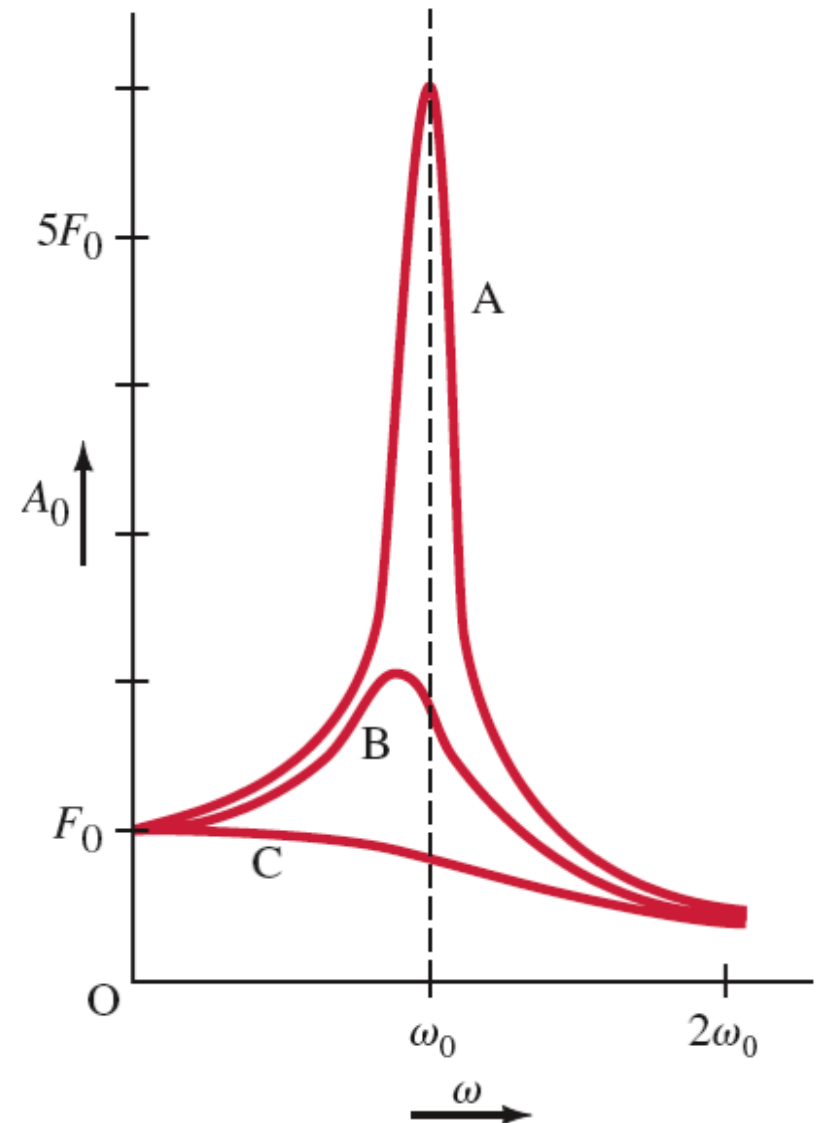
and

$$\phi_0 = \tan^{-1} \frac{\omega_0^2 - \omega^2}{\omega(b/m)}.$$

Forced Oscillations: Resonance

The width of the resonant peak can be characterized by the Q factor:

$$Q = \frac{m\omega_0}{b}.$$



Forced Oscillations: Resonance

Quality Factor

The sharpness of the resonance curve is usually described by a dimensionless parameter known as the quality factor, Q .

$$Q = \omega_0 / \Delta\omega = (\omega_0 L) / R$$

- $\Delta\omega$ is the width of the curve, measured between the two values of ω for which P_{avg} has half its maximum value.
 - These points are called the *half-power points*.

A high- Q circuit responds only to a narrow range of frequencies.

- Narrow peak

A low- Q circuit can detect a much broader range of frequencies.

A radio's receiving circuit is an important application of a resonant circuit.

Forced Oscillations: Resonance

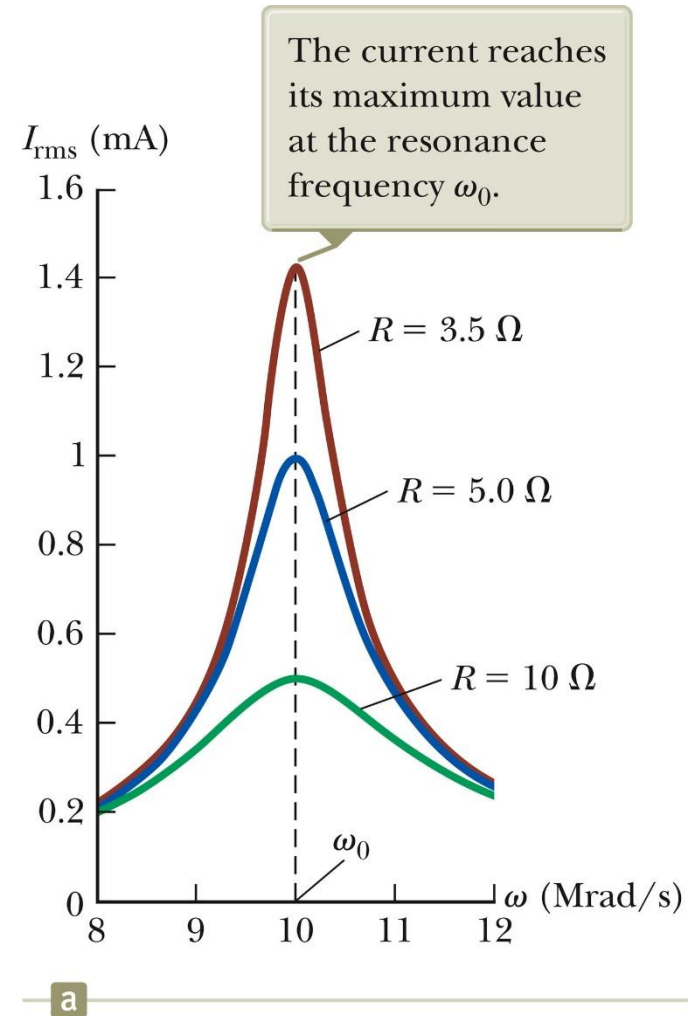
Quality Factor

Resonance occurs at the same frequency regardless of the value of R .

As R decreases, the curve becomes narrower and taller.

Theoretically, if $R = 0$ the current would be infinite at resonance.

- Real circuits always have some resistance.



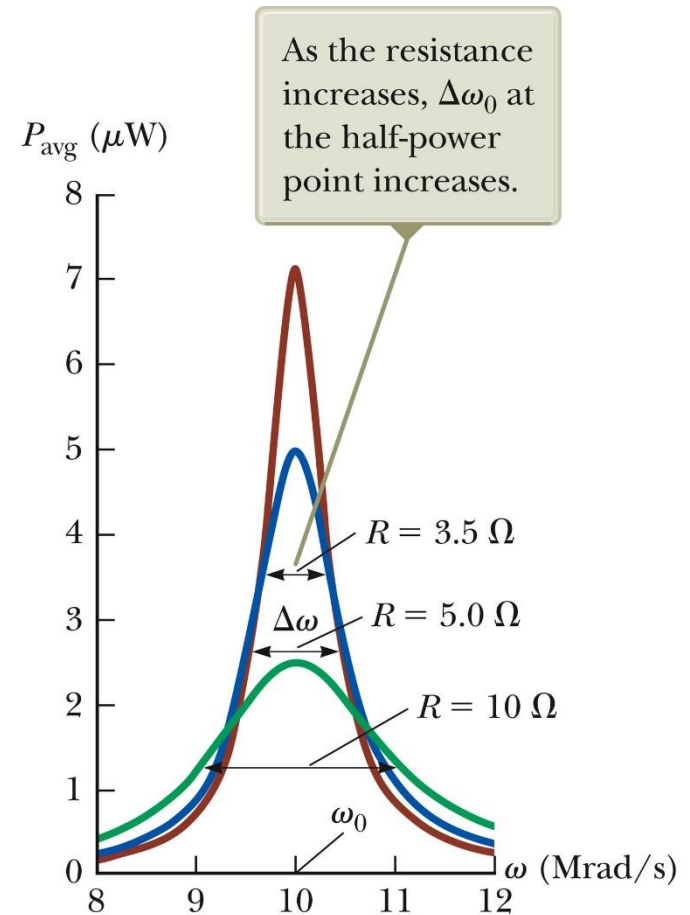
Forced Oscillations, FHM: Resonance

Power as a Function of Frequency

Power can be expressed as a function of frequency in an RLC circuit.

$$P_{av} = \frac{(\Delta V_{rms})^2 R \omega^2}{R^2 \omega^2 + L^2 (\omega^2 - \omega_o^2)^2}$$

This shows that at resonance, the average power is a maximum.



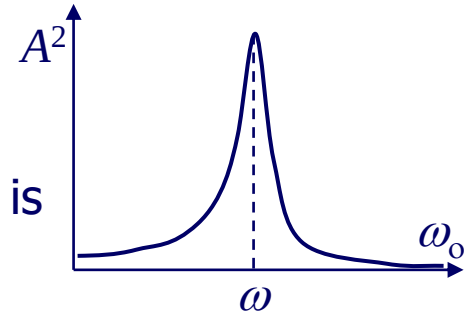
Resonance: FHM

- The long term solution for the driven oscillator is

$$x(t) = A \cos(\omega t - \delta),$$

for $t \gg 1/\beta$, where the amplitude, as we have seen, is

$$A^2 = \frac{f_o^2}{(\omega_o^2 - \omega^2)^2 + 4\beta^2 \omega^2}.$$



- This amplitude expression is interesting, because it says that as you drive a system at frequency ω , its amplitude depends on both how far off you are from the resonant frequency ω_o , and also on how big the damping is.
- If there is no damping at all, and you drive the system at the resonant frequency, then both terms

$$(\omega_o^2 - \omega^2)^2 = 0 \quad \text{and} \quad 4\beta^2 \omega^2 = 0.$$

- In this case, the amplitude goes to infinity. The way to think about this is that the driving force pumps energy into the oscillator (like pushing a child on a swing), and if there is no dissipation, there is no loss of energy and the energy grows to become infinite.
- If you tune the driver frequency (variable ω) for a given oscillator (fixed ω_o), what is the value of ω for which A^2 is maximum?

$$\omega = \omega_2 = \sqrt{\omega_o^2 - 2\beta^2}.$$

Keeping Track of Frequencies

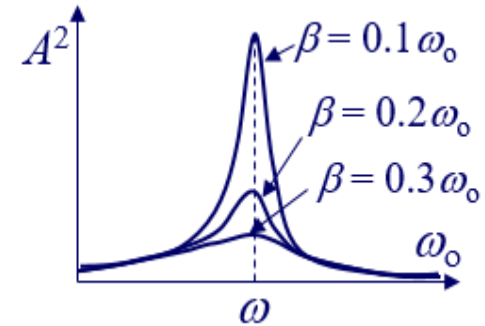
- We have met a number of frequencies in this discussion, so it might be helpful to list them in one place to help keep them straight.

$\omega_o = \sqrt{k / m}$ = natural frequency of undamped oscillator

$\omega_1 = \sqrt{\omega_o^2 - \beta^2}$ = frequency of damped oscillator

ω = frequency of driving force

$\omega_2 = \sqrt{\omega_o^2 - 2\beta^2}$ = value of ω at which response is max



- To find out the maximum amplitude of a particular driven oscillator, just let $\omega \sim \omega_o$ in

$$A^2 = \frac{f_o^2}{(\omega_o^2 - \omega^2)^2 + 4\beta^2 \omega^2}.$$

i.e. $A_{\max} = \frac{f_o}{2\beta \omega_o}.$

- From this you can see that the amplitude goes as β^{-1} . It also turns out that the width of the resonance increases with β as $\text{FWHM} \sim 2\beta$. (Prob. 5.41)

Quality Factor and Phase at Resonance

- There is a quantitative measure of how sharp the resonance is, which you may have heard of before—the **quality factor**, or Q .
- It is just the ratio of the resonance peak frequency ω_0 to the peak width 2β .

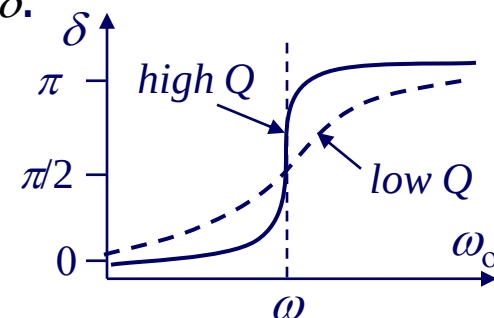
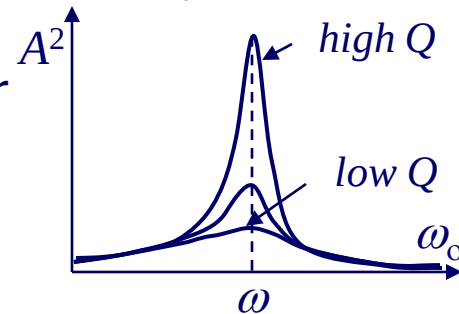
$$Q = \frac{\omega_0}{2\beta}.$$

- In the case of a pendulum, which you might think is a good resonator (keeps good time), the Q might be about 100. For a quartz clock, on the other hand, Q may reach 10000.
- An alternative way of thinking about the Q of a resonator is to consider it as

$$Q = \pi \frac{\text{decay time}}{\text{period}} = \pi \frac{1/\beta}{2\pi/\omega_0} = \frac{\omega_0}{2\beta}.$$

- What about the phase? Remember our expression for δ :

$$\delta = \arctan \frac{2\beta\omega}{\omega_0^2 - \omega^2}.$$



Forced Oscillations & Resonance

If we apply a periodically varying driving force to an oscillator (rather than just leaving it to vibrate on its own) we have a **FORCED OSCILLATION**

Free Oscillation with damping:

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

$$\text{Amplitude} = A_0 e^{-(b/2m)t}$$

$$\text{Frequency } \omega' = \sqrt{\left(\frac{k}{m} - \frac{b^2}{4m^2}\right)} = \sqrt{\left(\omega_0^2 - \frac{b^2}{4m^2}\right)}$$

Forced Oscillation with damping:

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = F_0 \cos \omega_D t$$

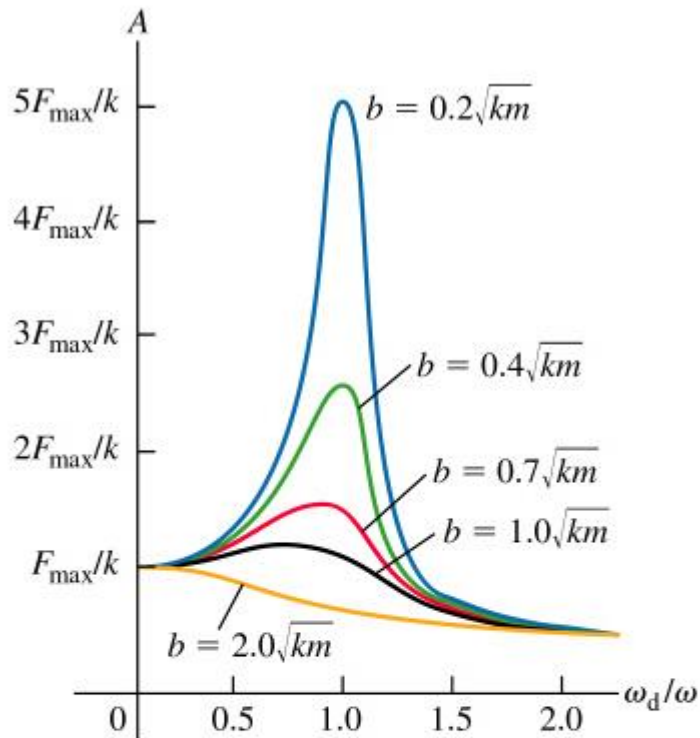
$$\text{Amplitude} = \frac{F_0}{\sqrt{\left((k - m\omega_D^2)^2 + b^2\omega_D^2\right)}}$$

MAXIMUM AMPLITUDE WHEN DENOMINATOR MINIMISED:

$k = m\omega_D^2$ ie when driving frequency = natural frequency of the UNDAMPED oscillator

$$\omega_D = \omega_0 = \sqrt{\frac{k}{m}}$$

Forced Oscillations & Resonance



Greater damping (larger b):

- Peak becomes broader
- Peak becomes less sharp
- Peak shifts toward lower frequencies

If $b > \sqrt{2km}$, peak disappears completely

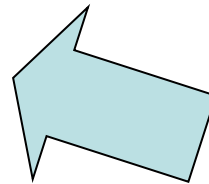
When driving frequency = natural frequency of oscillator, amplitude is maximum.

We say the system is in **RESONANCE**

“Sharpness” of resonance peak described by **quality factor (Q)**

High Q = sharp resonance

Damping reduces Q



Forced Damped Oscillation and Resonance

When a body is subject to an oscillatory external force then a forced oscillation occurs. For example, bridge vibrates under the influence of marching soldiers, tuning fork vibrates when exposed to the periodic force of a sound wave etc. Forced oscillations have the frequency of the external force and not the natural frequency of the body. However, the response of the body depends on the relation between the forced and the natural frequency.

□ *Derive the differential equation of FHM and find its solution (Mechanical system): Alternative Method*

FHM with Mechanical System (General Approach): Alternative Method

In the case of forced damped Oscillation the net force F on the object is

$$\begin{aligned} F &= F_{restor} + F_{damp} + F_{force} \\ m \frac{d^2 x}{dt^2} &= -kx - b \frac{dx}{dt} + F_0 \cos \omega t \\ \frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x &= \frac{F_0}{m} \cos \omega t \\ \therefore \frac{d^2 x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega_0^2 x &= \frac{F_0}{m} \cos \omega t \end{aligned} \quad (13)$$

This is the differential equation for the forced damped harmonic oscillation.

FHM with Mechanical System (General Approach): Alternative Method

Solution:

Let the trial solution of the equation has the form

$$x = A \cos(\omega t - \phi) \quad (14)$$

Substituting equation (14) in to (13) we have

$$\begin{aligned} -A\omega^2 \cos(\omega t - \phi) - 2\gamma A\omega \sin(\omega t - \phi) \\ + \omega_0^2 A \cos(\omega t - \phi) = \frac{F_{ex}}{m} \cos \omega t \end{aligned}$$

$$\begin{aligned} \text{or, } & (-\omega^2 A \cos \phi + 2\gamma\omega A \sin \phi + \omega_0^2 A \cos \phi - \frac{F_{ex}}{m}) \cos \omega t \\ & (-\omega^2 A \sin \phi + 2\gamma\omega A \cos \phi + \omega_0^2 A \sin \phi) \sin \omega t = 0 \end{aligned}$$

When $\omega t = 0$, $\sin \omega t = 0$ and $\cos \omega t = 1$, we then have

$$-\omega^2 A \cos \phi + 2\gamma\omega A \sin \phi + \omega_0^2 A \cos \phi - \frac{F_{ex}}{m} = 0 \quad (15)$$

FHM with Mechanical System (General Approach): Alternative Method

Again when $\omega t = \pi/2$, $\sin \omega t = 1$ and $\cos \omega t = 0$ then we have

$$-\omega^2 A \sin \phi + 2\gamma\omega A \cos \phi + \omega_0^2 A \sin \phi = 0$$

$$\text{or, } A \sin \phi (\omega_0^2 - \omega^2) = 2\gamma\omega A \cos \phi$$

$$\therefore \tan \phi = \frac{\cos \phi}{\sin \phi} = \frac{2\gamma\omega}{(\omega_0^2 - \omega^2)} \quad (16)$$

So,

$$\sin \phi = \frac{2\gamma\omega}{\{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2\}^{1/2}} \quad (17)$$

$$\cos \phi = \frac{(\omega_0^2 - \omega^2)}{\{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2\}^{1/2}} \quad (18)$$

Using eqn (17) and (18) into (15) we can write

$$A = \frac{F_{ex}/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}} \quad (19)$$

FHM with Mechanical System (General Approach): Alternative Method

Therefore the solution has the form

$$x(t) = \frac{F_{ex}}{m\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}} \cos(\omega t - \phi) \quad (20)$$

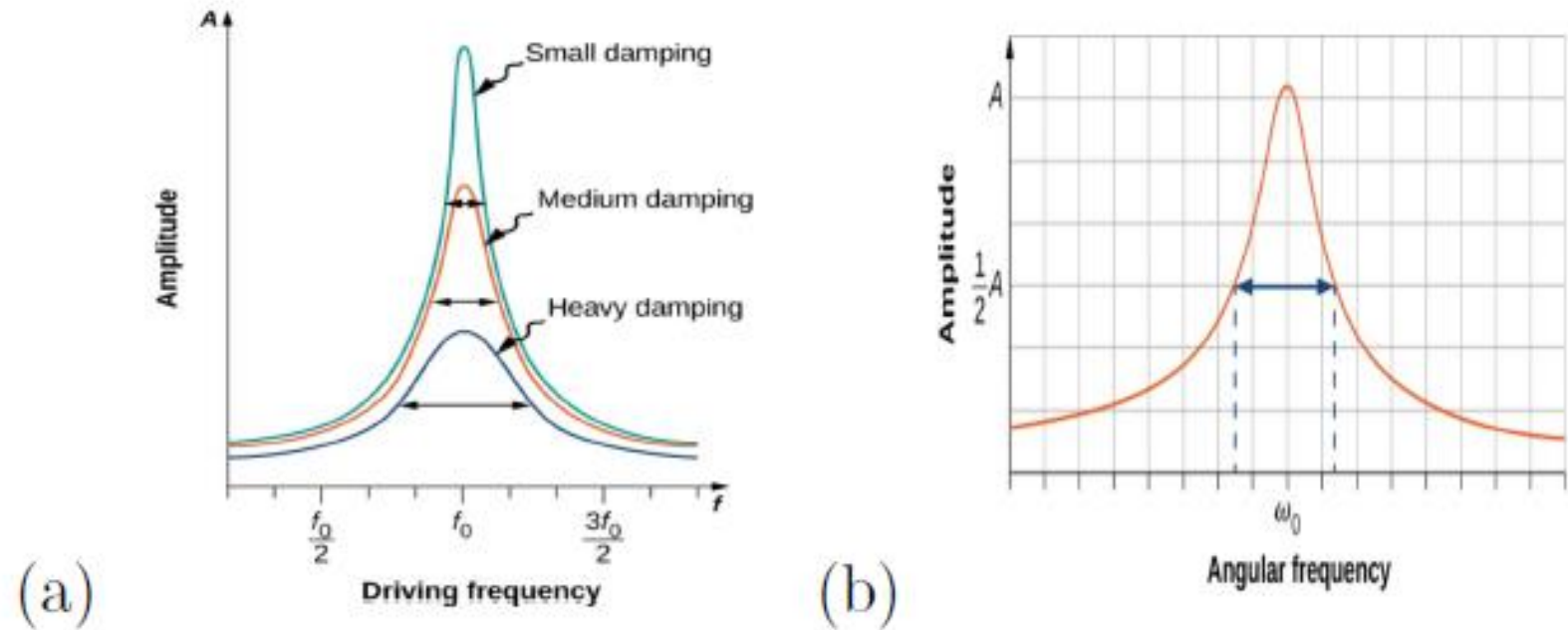
The frequency for which amplitude is maximum is known as the resonant frequency ω_r ,

$$\begin{aligned} \frac{\partial A}{\partial \omega} \bigg|_{\omega=\omega_r} &= 0 \\ \text{or, } 8\gamma^2\omega_r - 4\omega_r(\omega_0^2 - \omega_r^2) &= 0 \\ \therefore \omega_r &= \sqrt{\omega_0^2 - 2\gamma^2} \end{aligned} \quad (21)$$

This is the resonance frequency. So the resonance amplitude

$$A_r = \frac{F_{ex}/m}{2\gamma\sqrt{(\omega_0^2 - \gamma^2)}}$$

FHM with Mechanical System (General Approach): Alternative Method



The quality factor is defined as

$$Q \equiv \frac{\omega_r}{\Delta\omega} = \frac{f_r}{\Delta f}$$

where Δf is the half width maximum.

FHM in Electrical System

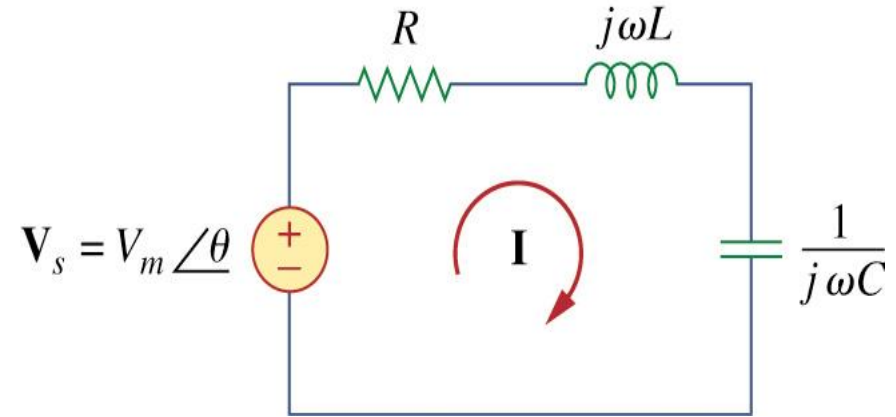
[Resonance in Electric Circuit]:

Introduction to Resonance in Electric Circuit:

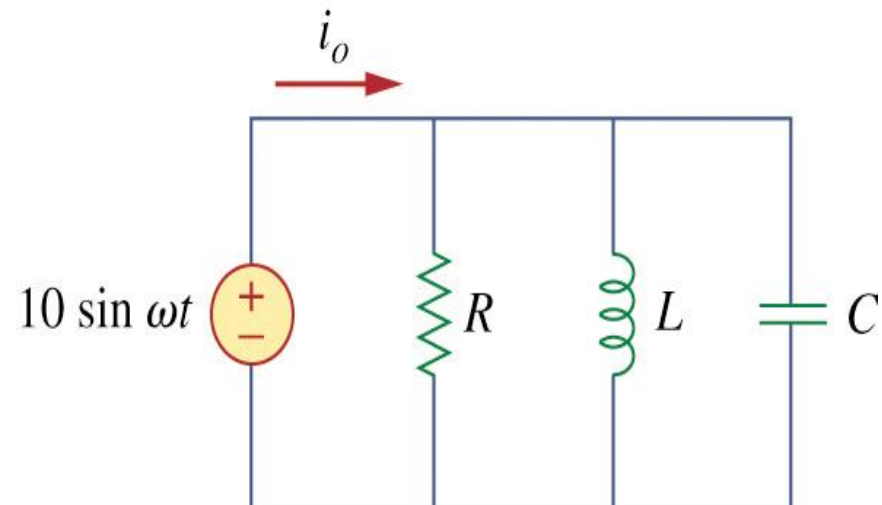
- Resonance is a condition in an RLC circuit in which the capacitive and reactive reactance are equal in magnitude, thereby resulting in a purely resistive impedance.
- Resonance circuits are useful for constructing filters and used in many application.

Resonance in Electric Circuit

Series Resonance Circuit



Parallel Resonance Circuit



Series Resonance

Total impedance of series RLC Circuit is

$$Z_{\text{Total}} = R + jX_L - jX_C$$

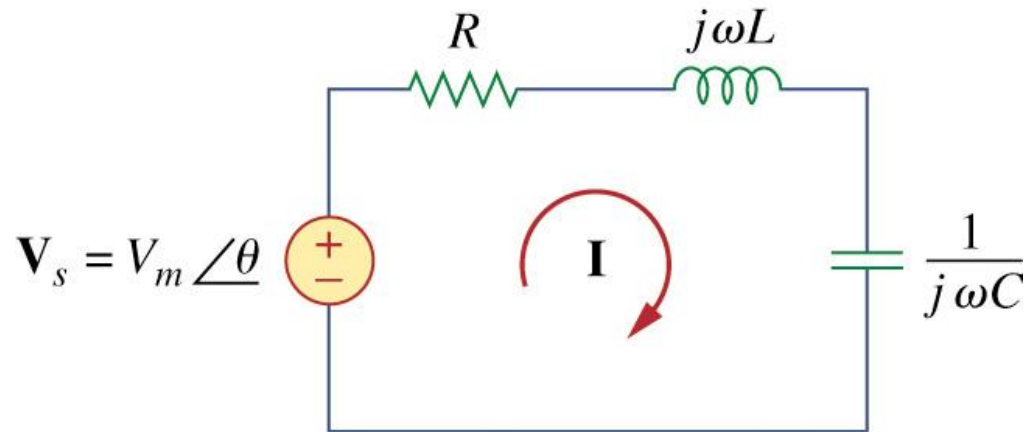
$$Z_{\text{Total}} = R + j(X_L - X_C)$$

At resonance

$$X_L = X_C$$

The impedance now reduce to

$$Z_{\text{Total}} = R$$



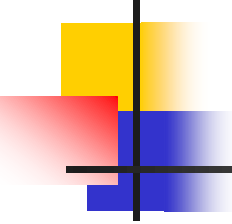
The current at resonance

$$I_m = \frac{V_s}{Z_{\text{Total}}} = \frac{V_m}{R}$$

Resonance in Electric Circuit

At Resonance

- At resonance, the impedance consists only resistive component R .
- The value of current will be maximum since the total impedance is minimum.
- The voltage and current are in phase.
- Maximum power occurs at resonance since the power factor is unity.



Resonance Frequency

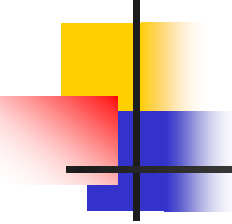
Resonance frequency is the frequency where the condition of resonance occur.

Also known as center frequency.

Resonance frequency

$$\omega_o = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

$$f_o = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$



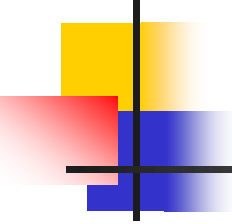
Half-power Frequency

Half-power frequencies is the frequency when the magnitude of the output voltage or current is decrease by the factor of $1 / \sqrt{2}$ from its maximum value.

Also known as cutoff frequencies.

$$\omega_1 = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)} \text{ rad/s}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)} \text{ rad/s}$$



Bandwidth, β

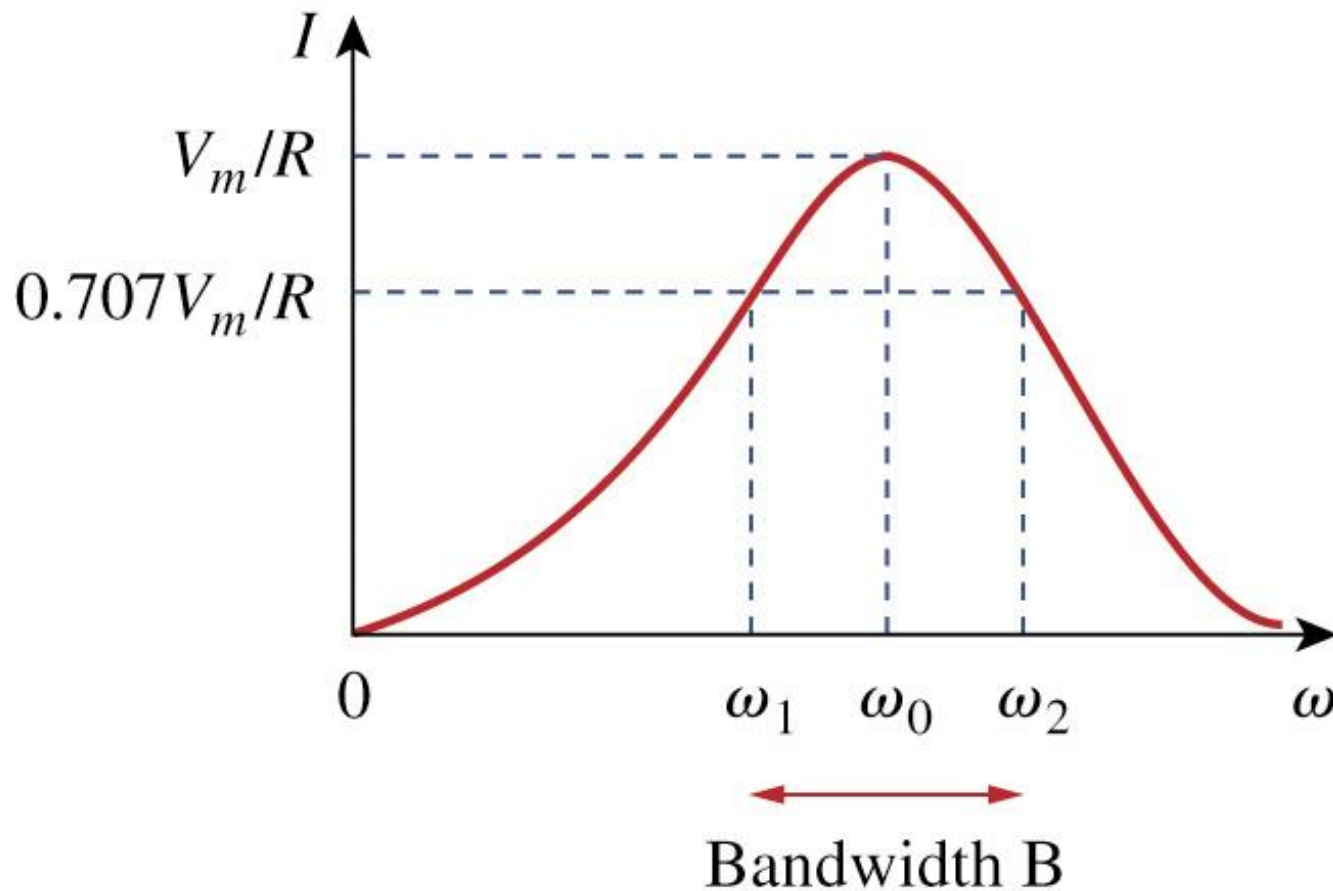
Bandwidth, β is define as the difference between the two half power frequencies.

The width of the response curve is determine by the bandwidth.

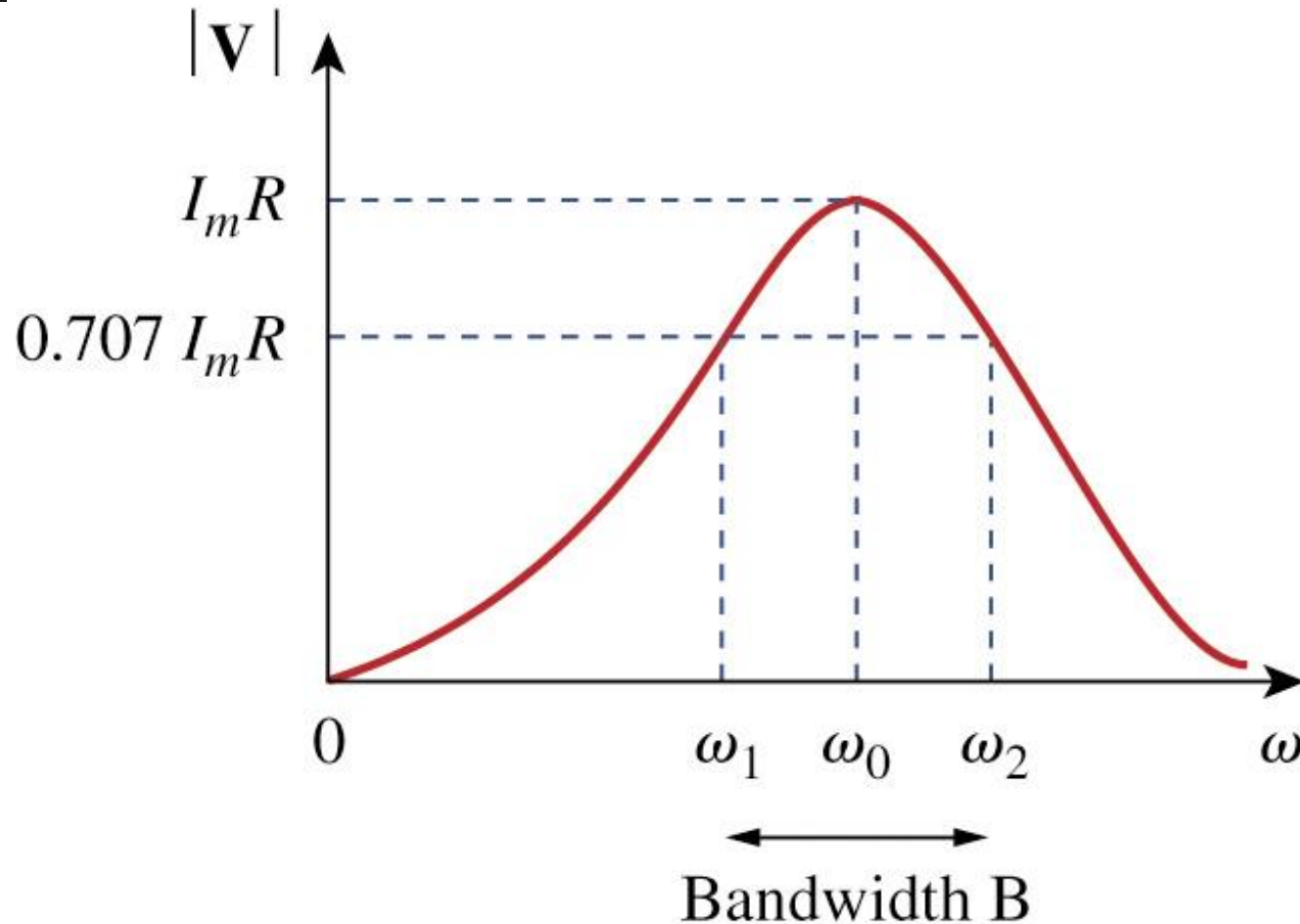
$$\beta = (\omega_2 - \omega_1) \text{rad/s}$$

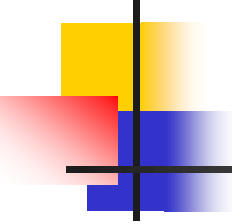
$$\beta = \frac{R}{L} \text{rad/s}$$

Current Response Curve



Voltage Response Curve





Quality Factor (Q-Factor)

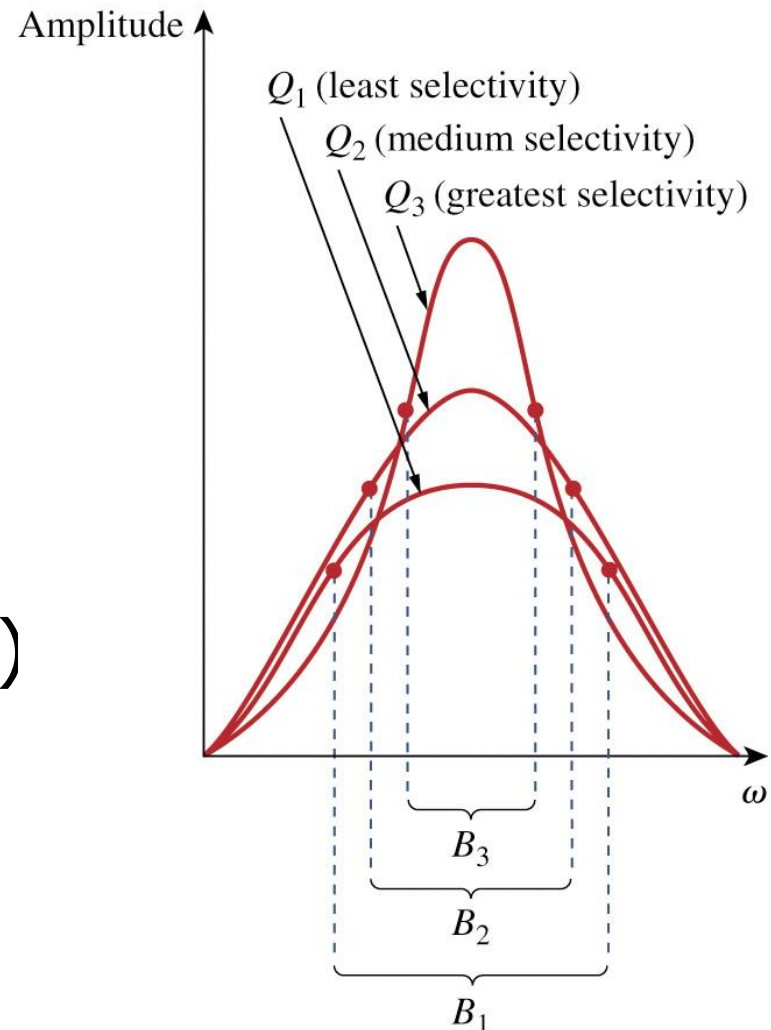
The ratio of resonance frequency to the bandwidth

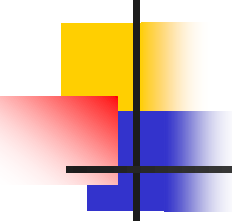
$$Q = \frac{\omega_o}{\beta} = \frac{\omega_o L}{R}$$

The “sharpness” of response curve could be measured by the quality factor, Q.

Q-Factor Vs Bandwidth

- Higher value of Q , smaller the bandwidth. (Higher the selectivity)
- Lower value of Q larger the bandwidth. (Lower the selectivity)



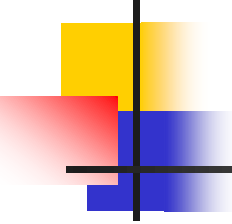


High-Q

It is to be a high-Q circuit when its quality factor is equal or greater than 10.

For a high-Q circuit ($Q \gg 10$), the half-power frequencies are, for all practical purposes, symmetrical around the resonant frequency and can be approximated as

$$\omega_1 \cong \omega_o - \frac{\beta}{2} \qquad \omega_2 \cong \omega_o + \frac{\beta}{2}$$



Maximum Power Dissipated

The average power dissipated by the RLC circuit is

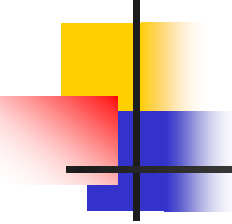
$$P(\omega_o) = \frac{1}{2} I^2 R$$

The maximum power dissipated at resonance where

$$I = \frac{V_m}{R}$$

Thus maximum power dissipated is

$$P(\omega_o) = \frac{1}{2} \frac{V_m^2}{R}$$



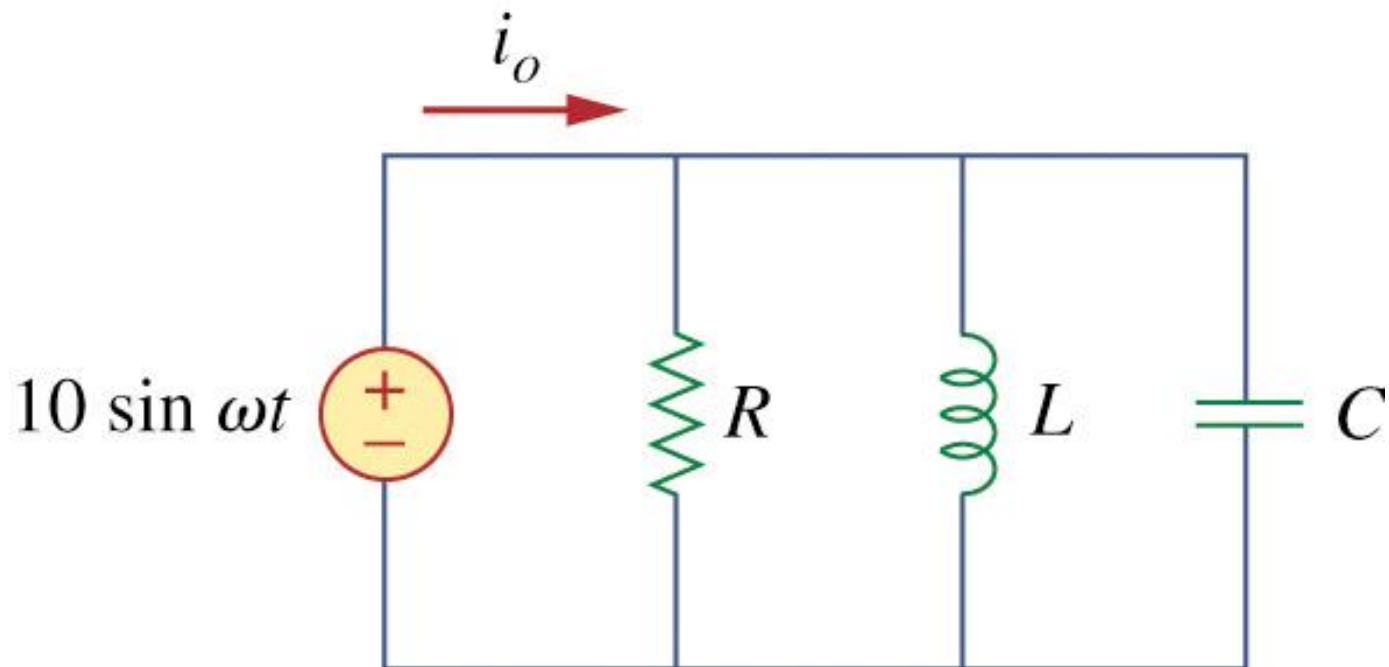
Power Dissipated at ω_1 and ω_2

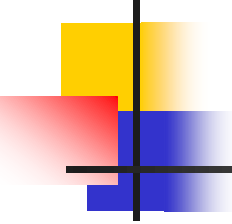
At certain frequencies, where $\omega = \omega_1$ and ω_2 , the dissipated power is half of maximum power

$$P(\omega_1) = P(\omega_2) = \frac{1}{2} \frac{(V_m/\sqrt{2})^2}{R} = \frac{V_m^2}{4R}$$

Hence, ω_1 and ω_2 are called half-power frequencies.

Parallel Resonance





Parallel Resonance

The total admittance

$$Y_{\text{Total}} = Y_1 + Y_2 + Y_3$$

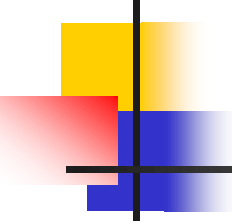
$$Y_{\text{Total}} = \frac{1}{R} + \frac{1}{(j\omega L)} + \frac{1}{(-j/\omega C)}$$

$$Y_{\text{Total}} = \frac{1}{R} + \frac{-j}{\omega L} + j\omega C$$

$$Y_{\text{Total}} = \frac{1}{R} + j(\omega C - 1/\omega L)$$

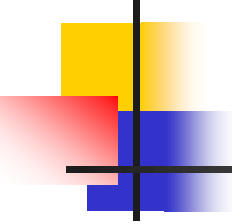
Resonance occur when

$$\omega C = \frac{1}{\omega L}$$



At Resonance

- At resonance, the impedance consists only conductance G .
- The value of current will be minimum since the total admittance is minimum.
- The voltage and current are in phase.



Parameters in Parallel Circuit

Parallel resonant circuit has same parameters as the series resonant circuit.

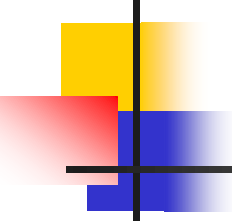
Resonance frequency

$$\omega_o = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

Half-power frequencies

$$\omega_1 = \frac{-1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \left(\frac{1}{LC}\right)} \text{ rad/s}$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \left(\frac{1}{LC}\right)} \text{ rad/s}$$



Parameters in Parallel Circuit

Bandwidth

$$\beta = \omega_2 - \omega_1 = \frac{1}{RC}$$

Quality Factor

$$Q = \frac{\omega_o}{\beta} = \omega_o RC$$

Resonance in Electric Circuit: Series Resonance

is shown in Fig. 14.22; the plot only shows the symmetry illustrated in this graph when the frequency axis is a logarithm. The average power dissipated by the RLC circuit is

$$P(\omega) = \frac{1}{2} I^2 R \quad (14.29)$$

The highest power dissipated occurs at resonance, when $I = V_m/R$, so that

$$P(\omega_0) = \frac{1}{2} \frac{V_m^2}{R} \quad (14.30)$$

At certain frequencies $\omega = \omega_1, \omega_2$, the dissipated power is half the maximum value; that is,

$$P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R} \quad (14.31)$$

Hence, ω_1 and ω_2 are called the *half-power frequencies*.

The half-power frequencies are obtained by setting Z equal to $\sqrt{2}R$, and writing

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R \quad (14.32)$$

Solving for ω , we obtain

$$\begin{aligned} \omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ \omega_2 &= \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \end{aligned} \quad (14.33)$$

We can relate the half-power frequencies with the resonant frequency. From Eqs. (14.26) and (14.33),

$$\omega_0 = \sqrt{\omega_1 \omega_2} \quad (14.34)$$

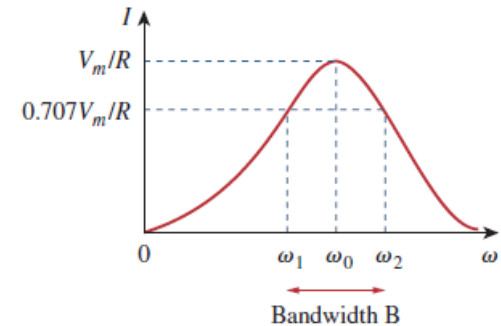


Figure 14.22

The current amplitude versus frequency for the series resonant circuit of Fig. 14.21.

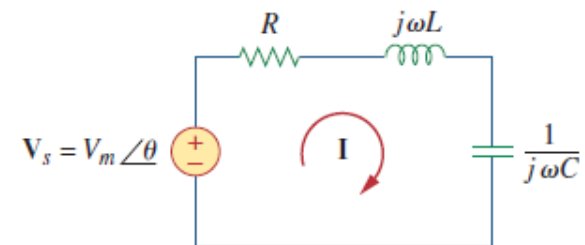


Figure 14.21

The series resonant circuit.

Resonance in Electric Circuit: Series Resonance

Although the height of the curve in Fig. 14.22 is determined by R , the width of the curve depends on other factors. The width of the response curve depends on the *bandwidth* B , which is defined as the difference between the two half-power frequencies,

$$B = \omega_2 - \omega_1 \quad (14.35)$$

This definition of bandwidth is just one of several that are commonly used. Strictly speaking, B in Eq. (14.35) is a half-power bandwidth, because it is the width of the frequency band between the half-power frequencies.

The “sharpness” of the resonance in a resonant circuit is measured quantitatively by the *quality factor* Q . At resonance, the reactive energy

The quality factor is a measure of the selectivity (or “sharpness” of resonance) of the circuit.

Resonance in Electric Circuit: Series Resonance

in the circuit oscillates between the inductor and the capacitor. The quality factor relates the maximum or peak energy stored to the energy dissipated in the circuit per cycle of oscillation:

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}} \quad (14.36)$$

It is also regarded as a measure of the energy storage property of a circuit in relation to its energy dissipation property. In the series RLC circuit, the peak energy stored is $\frac{1}{2}LI^2$, while the energy dissipated in one period is $\frac{1}{2}(I^2R)(1/f_0)$. Hence,

$$Q = 2\pi \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R(1/f_0)} = \frac{2\pi f_0 L}{R} \quad (14.37)$$

or

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} \quad (14.38)$$

Notice that the quality factor is dimensionless. The relationship between the bandwidth B and the quality factor Q is obtained by substituting Eq. (14.33) into Eq. (14.35) and utilizing Eq. (14.38).

$$B = \frac{R}{L} = \frac{\omega_0}{Q} \quad (14.39)$$

or $B = \omega_0^2 CR$. Thus

The **quality factor** of a resonant circuit is the ratio of its resonant frequency to its bandwidth.

Keep in mind that Eqs. (14.33), (14.38), and (14.39) only apply to a series RLC circuit.

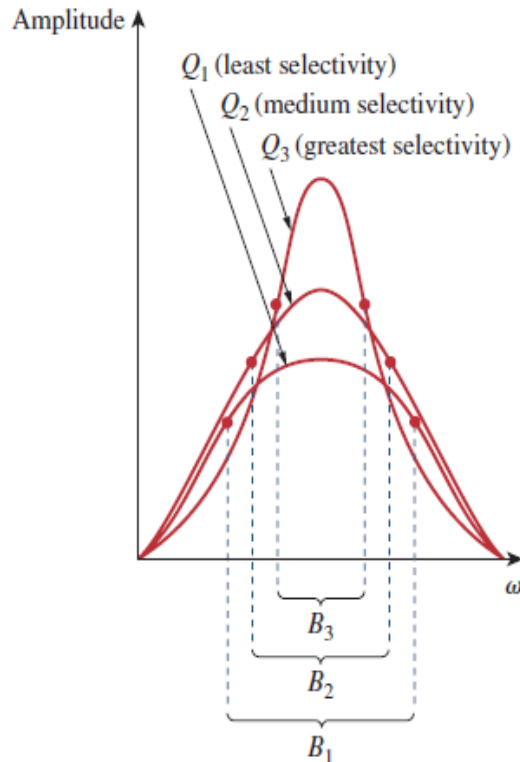


Figure 14.23

The higher the circuit Q , the smaller the bandwidth.

Resonance in Electric Circuit: Series Resonance

A resonant circuit is designed to operate at or near its resonant frequency. It is said to be a *high- Q circuit* when its quality factor is equal to or greater than 10. For high- Q circuits ($Q \geq 10$), the half-power frequencies are, for all practical purposes, symmetrical around the resonant frequency and can be approximated as

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \quad \omega_2 \simeq \omega_0 + \frac{B}{2} \quad (14.40)$$

High- Q circuits are used often in communications networks.

Resonance in Electric Circuit: Series Resonance

Example-22:

In the circuit of Fig. 14.24, $R = 2\ \Omega$, $L = 1\ \text{mH}$, and $C = 0.4\ \mu\text{F}$.

(a) Find the resonant frequency and the half-power frequencies. (b) Calculate the quality factor and bandwidth. (c) Determine the amplitude of the current at ω_0 , ω_1 , and ω_2 .

Solution:

(a) The resonant frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 0.4 \times 10^{-6}}} = 50\ \text{krad/s}$$

■ **METHOD 1** The lower half-power frequency is

$$\begin{aligned}\omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ &= -\frac{2}{2 \times 10^{-3}} + \sqrt{(10^3)^2 + (50 \times 10^3)^2} \\ &= -1 + \sqrt{1 + 2500}\ \text{krad/s} = 49\ \text{krad/s}\end{aligned}$$

Similarly, the upper half-power frequency is

$$\omega_2 = 1 + \sqrt{1 + 2500}\ \text{krad/s} = 51\ \text{krad/s}$$

(b) The bandwidth is

$$B = \omega_2 - \omega_1 = 2\ \text{krad/s}$$

or

$$B = \frac{R}{L} = \frac{2}{10^{-3}} = 2\ \text{krad/s}$$

The quality factor is

$$Q = \frac{\omega_0}{B} = \frac{50}{2} = 25$$

Example 14.7

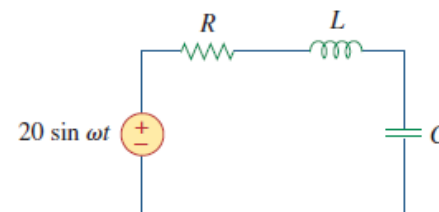


Figure 14.24
For Example 14.7.

Resonance in Electric Circuit: Series Resonance

■ **METHOD 2** Alternatively, we could find

$$Q = \frac{\omega_0 L}{R} = \frac{50 \times 10^3 \times 10^{-3}}{2} = 25$$

From Q , we find

$$B = \frac{\omega_0}{Q} = \frac{50 \times 10^3}{25} = 2 \text{ krad/s}$$

Since $Q > 10$, this is a high- Q circuit and we can obtain the half-power frequencies as

$$\omega_1 = \omega_0 - \frac{B}{2} = 50 - 1 = 49 \text{ krad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 50 + 1 = 51 \text{ krad/s}$$

as obtained earlier.

(c) At $\omega = \omega_0$,

$$I = \frac{V_m}{R} = \frac{20}{2} = 10 \text{ A}$$

At $\omega = \omega_1, \omega_2$,

$$I = \frac{V_m}{\sqrt{2}R} = \frac{10}{\sqrt{2}} = 7.071 \text{ A}$$

Resonance in Electric Circuit: Parallel Resonance

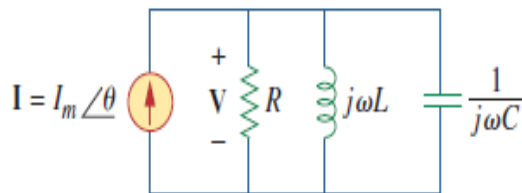


Figure 14.25

The parallel resonant circuit.

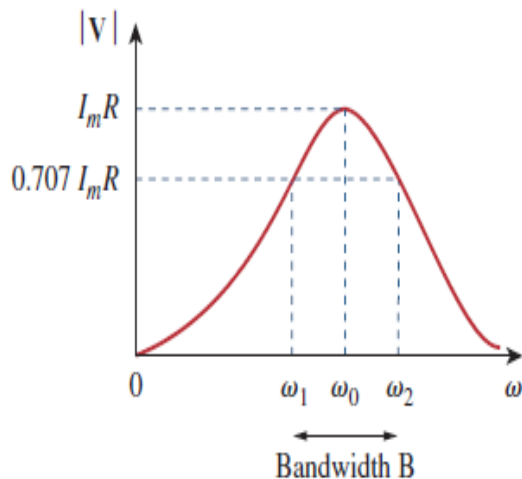


Figure 14.26

The current amplitude versus frequency for the series resonant circuit of Fig. 14.25.

14.6 Parallel Resonance

The parallel *RLC* circuit in Fig. 14.25 is the dual of the series *RLC* circuit. So we will avoid needless repetition. The admittance is

$$Y = H(\omega) = \frac{I}{V} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L} \quad (14.41)$$

or

$$Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right) \quad (14.42)$$

Resonance occurs when the imaginary part of *Y* is zero,

$$\omega C - \frac{1}{\omega L} = 0 \quad (14.43)$$

or

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s} \quad (14.44)$$

Resonance in Electric Circuit: Parallel Resonance

expressions for the series circuit with $1/R$, C , and L respectively, we obtain for the parallel circuit

$$\begin{aligned}\omega_1 &= -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} \\ \omega_2 &= \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}\end{aligned}\quad (14.45)$$

$$B = \omega_2 - \omega_1 = \frac{1}{RC} \quad (14.46)$$

$$Q = \frac{\omega_0}{B} = \omega_0 RC = \frac{R}{\omega_0 L} \quad (14.47)$$

It should be noted that Eqs. (14.45) to (14.47) apply only to a parallel RLC circuit. Using Eqs. (14.45) and (14.47), we can express the half-power frequencies in terms of the quality factor. The result is

$$\omega_1 = \omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} - \frac{\omega_0}{2Q}, \quad \omega_2 = \omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} + \frac{\omega_0}{2Q} \quad (14.48)$$

Again, for high- Q circuits ($Q \geq 10$)

$$\omega_1 \approx \omega_0 - \frac{B}{2}, \quad \omega_2 \approx \omega_0 + \frac{B}{2} \quad (14.49)$$

Resonance in Electric Circuit: Parallel Resonance

TABLE 14.4

Summary of the characteristics of resonant RLC circuits.

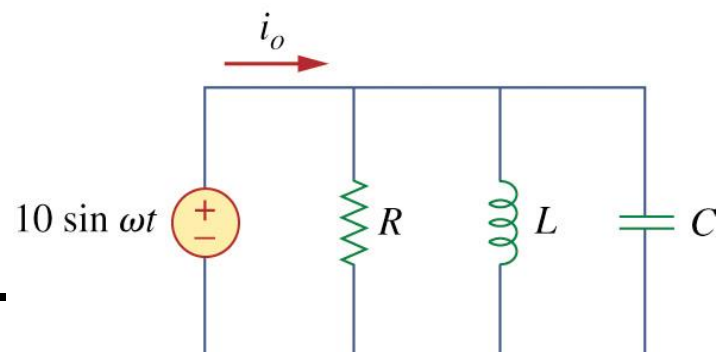
Characteristic	Series circuit	Parallel circuit
Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor, Q	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$

Resonance in Electric Circuit: Parallel Resonance

Example-23:

If $R=8\text{k}\Omega$, $L=0.2\text{mH}$ and $C=8\mu\text{F}$, calculate

- ω_o
- Q and β
- ω_1 and ω_2
- Power dissipated at ω_o , ω_1 and ω_2 .





Thank You All